UNIVERSITÉ DE GENÈVE

Archive ouverte UNIGE
https://archive-ouverte.unige.ch

Thèse 2020

Open Access

This version of the publication is provided by the author(s) and made available in accordance with the copyright holder(s).

Self-monitoring technologies to promote healthy behavior in the long term

Randriambelonoro, Mirana Michelle

How to cite

RANDRIAMBELONORO, Mirana Michelle. Self-monitoring technologies to promote healthy behavior in the long term. Doctoral Thesis, 2020. doi: 10.13097/archive-ouverte/unige:136731

This publication URL: https://archive-ouverte.unige.ch/unige:136731
Publication DOI: 10.13097/archive-ouverte/unige:136731

© This document is protected by copyright. Please refer to copyright holder(s) for terms of use.

Self-monitoring technologies to promote healthy behavior in the long term

THESIS
submitted to the
Geneva School of Economics and Management,
University of Geneva, Switzerland,

by

Mirana Randriambelonoro

Under the direction of

Prof. Dimitri Konstantas, supervisor
Prof. Antoine Geissbuhler, co-supervisor

in fulfillment of the requirements for the degree of

Docteur ès économie et management
mention Systèmes d'Information

Membres du jury de thèse :

Prof. Gilles Falquet, président du jury, Université de Genève
Prof. Alain Golay, Université de Genève
Prof. Katarzyna Wac, Université de Copenhague

Thesis no 76
Geneva, January 2020

La Faculté d'économie et de management, sur préavis du jury, a autorisé l'impression de la présente thèse, sans entendre, par-là, émettre aucune opinion sur les propositions qui s'y trouvent énoncées et qui n'engagent que la responsabilité de leur auteur.

Genève, le 31 Janvier 2020

Dean
Marcelo OLARREAGA

TABLE OF CONTENTS

Table of contents…… 3
Résumé……...… 5
Abstract……..…… 7
Acknowledgements…… 9
CHAPTER 1. GENERAL INTRODUCTION……...…..…..… 10
1.1 Background…….. 10
1.2 Thesis motivation…….… 17
1.3 Thesis objectives and research questions…….… 24
1.4 Thesis significance and intended contributions…….….. 36
1.5 Structure of the thesis document…….. 37
1.6 References…… 37

CHAPTER 2. CAN FITNESS TRACKERS HELP DIABETIC AND OBESE USERS MAKE AND SUSTAIN LIFESTYLE CHANGES?..... 50
2.1 Introduction…… 50
2.2 Related work: Empowering Users to Control Their Health…… 51
2.3 Study methodology…….. 52
2.4 Lifestyle evolution……... 53
2.5 Users' expectations and requirements……. 56
2.6 Device design recommendations…….…..... 59
2.7 Conclusions…….. 61
2.8 References……... 61

CHAPTER 3. OPPORTUNITIES AND CHALLENGES FOR SELF-MONITORING TECHNOLOGIES FOR HEALTHY AGING: AN IN-SITU STUDY…….. 63
3.1 Introduction……... 63
3.2 Related works…… 64
3.3 Methodology……..... 64
3.4 Findings……. 66
3.5 Discussions…… 70
3.6 Conclusions……... 72
3.7 Acknowledgements……... 72
3.8 References…… 72

CHAPTER 4. SOCIAL INCENTIVES IN PERVASIVE FITNESS APPS FOR OBESE AND DIABETIC PATIENTS…….. 76
4.1 Introduction……... 76
4.2 HealthyTogether……..... 76
4.3 User study…….. 77
4.4 Findings……. 78
4.5 Conclusions……... 79
4.6 References…… 79

CHAPTER 5. HOSPITAL-TO-HOME TRANSITION: USING SERIOUS GAME AND ACTIVITY MONITORING TO ENGAGE ELDERLY PATIENT TO BE MORE ACTIVE ……. 81
5.1 Abstract……. 81
5.2 Introduction……... 81
5.3 Methodology……. 83
5.4 Findings…….. 86
5.5 Discussions……. 92
5.4 Conclusions…… 95
5.5 Acknowledgements…… 95
5.6 References…….…. 95

CHAPTER 6. COMPUTER-AIDED PHYSICAL REHABILITATION OF OLDER PEOPLE: A PILOT NON-INFERIORITY RANDOMIZED CLINICAL TRIAL……..... 100
6.1 Abstract…… 100
6.2 Introduction…….. 101
6.3 Methods……..... 101
6.4 Results…….. 104
6.5 Discussions……... 108
6.6 Conclusions……..... 109
6.7 Appendix…….. 109
6.7 References…… 110

CHAPTER 7. MIRANABOT: A CONVERSATIONAL AGENT TO PROMOTE HEALTHY EATING……. 114
7.1 Introduction…….. 114
7.2 Methodology……..... 115
7.3 Findings…… 115
7.4 MiranaBot: Behavior change theory and techniques……...…… 117
7.5 MiranaBot: Components and functionalities……... 118
7.6 Conclusions…….. 120
7.7 References…… 120

CHAPTER 8. GENERAL DISCUSSION……. 123
8.1 Summary of main findings and contributions……..…… 123
8.2 Intervention prescription….…….…… 125
8.3 Limitation and future work.…….…… 126
8.3 References….……... 127

APPENDIX…… 128

RESUME

Le monde d'aujourd'hui doit faire face à deux fléaux majeurs qui sont les maladies non transmissibles et le vieillissement de la population. Parmi les principaux facteurs de risque de maladies non transmissibles figurent les habitudes de vie que nous pouvons influencer, notamment, l'inactivité physique et une mauvaise alimentation. D'autre part, à mesure que les personnes vieillissent, l'incidence des maladies chroniques qui les empêchent de demeurer autonomes augmente également. Le taux d'hospitalisation est alors trois fois plus fréquent dans la population vieillissante que dans la population jeune. Il y a donc un intérêt croissant à renforcer les capacités physiques et fonctionnelles des personnes âgées afin de favoriser leur autonomie. Cependant, bien qu'il ait été démontré que s'engager dans des comportements sains s'avère être bénéfique pour la santé et le bien-être de l'individu, le défi principal est de motiver l'adoption et l'engagement à long terme dans de tels comportements.

La présente recherche examine les technologies d'auto-monitorage visant à promouvoir des comportements sains à long terme. Bien que les technologies d'auto-monitorage se révèlent très prometteuses pour induire le changement de comportement, suivre uniquement l'évolution de ses données de santé n'est pas suffisant pour soutenir la motivation et l'engagement dans les habitudes saines, en particulier lorsqu'il s'agit de patients. Un grand nombre de techniques de changement de comportement sont en effet incorporées dans les technologies d'auto-monitorage existantes. Cela donne une indication de la capacité de ces systèmes à modifier les comportements individuels, mais l'efficacité à long terme de ces outils pour améliorer les habitudes alimentaires, l'activité physique et les comportements sédentaires reste inconnue. De plus, ces applications sont souvent généralisées au public et ne répondent pas nécessairement aux conditions, aux préférences et aux besoins des individus liés à leur pathologie particulière. Enfin, l'adoption et l'acceptabilité de ces technologies par différentes populations spécifiques restent sous-étudiées.

Cette thèse porte sur l'efficacité des technologies d'auto-monitorage pour promouvoir un changement positif à long terme des comportements modifiables, principalement en ce qui concerne l'activité physique et la nutrition. D'une part, elle met en lumière les possibilités et les limites des applications basées sur l'auto-monitorage, la gamification, les interactions sociales et les agents conversationnels. D'autre part, elle vise à fournir des recommandations pour concevoir ces technologies d'une manière efficace pour aider les patients atteints de maladies chroniques et les personnes âgées à construire une habitude saine.

Dans un premier temps, l'acceptation et l'adoption des appareils de monitorage de l'activité physique par les patients diabétiques et obèses font l'objet d'une étude longitudinale. Les résultats sont présentés sous formes de recommandations pour créer des systèmes de monitorages de santé suffisamment attrayants pour provoquer des changements à long terme chez les patients atteints de maladies chroniques. Dans un second temps, les attitudes des aînés à l'égard des technologies d'auto-monitorage de l'activité physique sont recherchées. Il en résulte un ensemble d'opportunités et de challenges qui permettrait de concevoir des technologies d'auto-monitorage qui leur permettraient de vieillir en bonne santé. Dans un troisième temps, l'impact des incitatifs sociaux pour motiver les patients obèses et diabétiques à faire de l'activité physique est étudié. Les résultats donnent un aperçu du rôle de la

coopération et de la compétition pour renforcer leur engagement à être physiquement actifs. Dans un quatrièmement temps, les effets d'une rééducation par le jeu pour inciter les patients âgés à être plus actifs font l'objet de recherches. Les résultats présentent différentes recommandations sur la conception d'un jeu sérieux pour la rééducation des personnes âgées. Dans un cinquièmement temps, l'efficacité clinique de cette réadaptation fondée sur le jeu sérieux est évaluée au moyen d'un essai clinique randomisé. Pour finir, un prototype d'agent conversationnel visant à promouvoir une alimentation saine est développé à partir de l'état de l'art, de sessions cocréation avec divers intervenants et des différents résultats présentés dans la thèse.

Dans l'ensemble, les travaux menés dans le cadre de cette thèse offrent de nouvelles perspectives sur la conception de technologies d'auto-monitorage pour les patients atteints de maladies chroniques et les personnes âgées. Les contributions de recherche sont d'intérêt pour le domaine de l'Interaction Homme-Machine, la Santé Numérique, l'Intelligence Artificielle conversationnelle, les Sciences du Comportement, la Nutrition et les Communautés de Réadaptation Physique en général.

ABSTRACT

Nowadays, the world is facing two major issues: Non-Communicable Diseases (NCDs) and ageing population. Among the main risk factors of NCDs are lifestyle habits that we can influence such as inactivity and poor diet. Furthermore, as people age, the incidence of chronic conditions preventing them to remain independent is also increasing. Hospitalization rate is then three times more frequent in the ageing population than in young age. There is therefore a growing interest in strengthening the physical and functional capacities of elderly hospitalized patients to promote their independence. However, although committing in healthy behaviors has been shown to be highly beneficial for individual's health and well-being, the challenge remains in motivating the adoption and the long-term engagement in such behaviors.

This research examines self-monitoring technologies to promote healthy behaviors in the long-term. Although self-monitoring technologies shows great promises to promote behavior change, tracking health data is not enough to sustain the motivation and engagement, especially in the patient domain. Many behaviors change techniques are indeed incorporated in existing self-monitoring technologies. This offers an indication of these systems' ability to alter individuals' behaviors, but the long-term efficacy of these systems to alter exercise, sleep, and sedentary behaviors remains unknown. Besides, these applications and the behavior theory behind are often generalized to the public and does not necessarily answer individuals' preferences and needs related to their specific pathology. Finally, the feasibility and the acceptability of these technologies across different specific population remain understudied.

This thesis focuses on the efficiency of self-monitoring technologies to promote positive change in the long-term on modifiable behaviors, mainly regarding physical activity and nutrition. It sheds light on the opportunities and the limitations of self-monitoring, gamified, social and conversational applications and intends to provide guidelines for designing these technologies for specific population, namely: chronically ill and elderly patients.

First, acceptance and adoption of fitness trackers by diabetic and obese patients are investigated. The findings are presented in a set of guidelines designing wearable health monitors that are engaging enough to effect long-term change for chronically-ill patients. Second, elderlies' attitudes towards self-monitoring technologies for physical activity are questioned. The outcomes are a set of opportunities and challenges for designing self-monitoring technologies for healthy ageing. Third, effect of social incentives to motivate physical activity on obese and diabetic patients are explored. The results give insights into the role of cooperation and competition to enhance their engagement in being physically active. Fourth, effects of a gamified rehabilitation to engage elderly patients with musculoskeletal issues to be more active are researched. The findings outline different recommendations on designing serious game for elderly rehabilitation. Fifth, the clinical efficacy of this serious game-based rehabilitation is evaluated through a non-inferiority randomized control trial. Finally, a prototype of a conversational agent to promote healthy eating is developed from state of the art, co-designing sessions with various stakeholders and the different findings outlined through the thesis.

Overall, the work conducted within this dissertation offers new perspectives on the design of self-monitoring technologies for elderly and chronically-ill patients. It makes several research contributions that are of interest to the greater Human Computer Interaction, Digital Health, Conversational AI, Behavioral sciences, Nutrition and Physical Rehabilitation communities.

ACKNOWLEDGEMENTS

"The more grateful you are… the more you will have to be grateful for." This citation of Zig Ziglar has always been guiding my life. I would like to acknowledge all the people who were part of this thesis journey and who made it possible.

First, I want to express my deep gratitude to God, who has always blessed my life. Thank you for bringing me this far. You could have chosen anyone but me, as far as I can see I'm nothing special, but You saw something that I could never see, so I am here today. Thank you.

Second, I want to thank my supervisors, Prof. Antoine Geissbuhler and Prof. Dimitri Konstantas. Antoine… that's maybe the first time I am calling you by your first name but since this page is all about sincere appreciation, let's go for it. So, Antoine, thank you for being much more than a supervisor for me. I learned so many things by working with you and what I got after 4 years is much more than a PhD degree. I was given plenty of opportunities to grow by collaborating with others, interacting with top-level researchers, conducting projects, leading teams, organizing conferences, also letting me learn by myself how challenging bringing innovation can be and why it is often necessary to look at the bigger picture. But what I appreciated most is the human dimension you bring to the work, the research and the team. So, thank you very much. Dimitri… the first time I met you, I could feel your dynamism so much that I was very motivated myself from the very beginning. Those multiple hours where we sat around thinking about how to conduct this thesis won't be forgotten for sure. Thank you for all the opportunities you have given me as well and for being sometimes overconfident about my ability. This encouraged me a lot. I am grateful to you for constantly reminding me that I could do anything and especially for thinking ahead about my future. All in all, I was blessed to have both of you as my supervisors.

I also want to thank all the jury members for agreeing to participate in this journey and for the time they dedicated to read and improve this manuscript.

Third, most of these researches would not have been possible without the financial support from REACH, a H2020 project. I am thankful to all the consortium members, the doctors, the professors, the physios, the nurse, the nutritionists who shared their expertise along the way and collaborated with me through the different projects. Than you as well to all the participants/patients who agreed to take part in this journey, this thesis wouldn't have been possible without you.

Fourth, I wouldn't forget my appreciation to all my current and former colleagues, especially Caroline, Gilles and Damien, who were always there for me not only as a colleague but as a real friend who truly cared for me. This journey was definitely so enjoyable thanks to you.

Finally, thank you to all my friends who always encouraged me and put me back on track when I wanted to give up. I'm also thankful to have a loving family who always supports me even if they are far away. I'm grateful to my dad who has always believed in me, my mother who always knows how to make me laugh and my brother who cares about me. I want to address a special thought my husband Aina, who never failed to be there for me. Thank you for doing what you do, thank you for being what you are, thank you for living.

CHAPTER 1. GENERAL INTRODUCTION

1.1. Background

1.1.1. Target population

Human behaviors account for 64% of the risk associated with preventable premature death worldwide [1]. Every year, 41 million people die due to Non-Communicable Diseases (NCDs), which represents 71% of all the deaths in the world. In Switzerland, 9 out of the top 10 causes of death and disability combined are due to NCDs, such as low back pain, ischemic heart disease, Alzheimer's disease, diabetes, headache disorders, lung cancer, stroke, COPD and neck pain [2]. Among the main risk factors of NCDs are lifestyle habits that we can influence such as smoking, drinking, drug abuse, as well as inactivity and poor diet. In response to the epidemic of diabetes in the past few decades, WHO has actively promoted healthy living through exercise and diet [3], which is less painful than surgical interventions such as gastric bypass surgery and less invasive than prescriptions that can have significant and sometimes dangerous side effects [4]. However, diet alone which often involves restricting intake can lead to a feeling of deprivation [5]. Research on diabetes and obesity has shown that physical activity not only helps prevent diabetes and obesity but also assists in treatment. Combined with a healthy diet, regular physical activity at a certain level can control blood glucose, regulate blood pressure, prevent cardiovascular events, and elevate mood [6].

In addition, the population is getting older at an unprecedented pace worldwide. There is therefore an important need to provide a healthy ageing environment. Rather than the absence of disease, “healthy ageing” is defined as a process that enables older people to continue to do the things that are important for them such as performing activity of daily living, maintaining social contact, and conserving dignity [7, 8, 9]. However, as people age, the incidence of chronic conditions preventing them to remain independent is also increasing. Sensory impairments, neck and back pain, pulmonary disease, depressive disorders, falls, diabetes, dementia and osteoarthritis are known to be the greatest cause of disability in elderly [7]. Hospitalization rate is therefore three times more frequent in the ageing population than in young age [10]. In geriatric care units, hospital-to-home transition is increasingly recognized as a critical period in the patient care, during which different incidents can occur and induce frequent re-hospitalizations. There is therefore a growing interest in strengthening the physical and functional capacities of elderly hospitalized patients to promote their independence. Being more physically active is bound to reinforce older people’s ability to perform activity of daily living and remain longer at home.

However, although committing in healthy behaviors has been shown to be highly beneficial for individual’s health and well-being, the challenge remains in motivating the adoption and the long-term engagement in such behaviors. In this thesis, the focus is on population with **specific needs** and **deeper motivation for change**, namely, obese, diabetic, and elderly patients.

1.1.2. Behavior change

Behavior is defined as "an organism or a system's external reactions to various stimuli and its environment" [11, 12, 13]. Skiing during winter or eating cheeseburger after school are examples of behaviors. Using measurable terms to define behavior, such as frequency, rate, duration, magnitude and latency, allows one to quantify behavior [14].

The American Psychological Association [15], defines behavior change as "the systematic use of principles of learning to increase the frequency of desired behaviors and/or decrease the frequency of problem behaviors". The nature of targeted problem behaviors can differ from person to person, and between different intervention programs. For this thesis, the focus is on increasing physical activity and improving nutrition. Multiple researches have been conducted to identify motivational strategies for behaviour change, especially in the nutrition and physical activity domain [16].

1.1.2.1. Behavior change theories

There are many models and theories on behavior change, some focusing on parts of the behavior change process [17], while others are more holistic, aiming to include all factors that can influence behavior change [18]. Most of the models have a psychological or psychosocial view on behavior change [19]. There are also specific frameworks that have been focusing on designing products or solutions [20]. Designing effective behavior change intervention requires to identify the target behavior and understand this target behavior in the user's context. The process should follow an effective, rigorous and standardized behavior change model. The idea here is not to present an exhaustive list of all the existing theories, but to introduce the ones that are more relevant to our context and our target behavior.

a. Hedonic Principle

The subject of behavior change dates back to Greek antiquity, who acknowledged the Hedonic Principle. People are intrinsically motivated to move away from painful and towards hedonic, or pleasurable, experiences. Later, this fundamental principle was formalized into a psychological theory [21, 22], which was expanded to describe how pleasure is associated with actions which support survival, and pain associated with the opposite actions. Hull [23] built on these theories to create reward learning theories. From these initial fundamental attempts at understanding the human motivation and how this relates to desired behavior change, psychology has evolved and developed much since those early theories.

b. Regulatory Focus Theory

More recently, the Regulatory Focus Theory (RFT) considers that the way in which people move towards pleasure and avoid pain, changes depending on the needs they are trying to satisfy, based on their individual's self-regulation focus, either promotion or prevention focus. Brockner and Higgins describe that people who are **promotion**-focused are more concerned with growth and development while people who are **prevention**-focused are driven by security. The RFT "distinguishes between promotion-focused motivation and prevention-focused motivation" [24], but does not absolutely point to either a promotion focus incentive system or a prevention focused incentive system being more effective to motivate the

individual. The RFT has the potential to support designers to understand what qualities motivate an individual user once it has been established whether they are accurately more promotion or prevention motivated [25]. However only understanding what motivates an individual is not sufficient to understand what is needed to achieve behavior change.

c. Self-Determination Theory

Motivation can be either **intrinsic**, inherently resulting from an individual's values and feelings or **extrinsic**, inherently imposed upon the user. Created by Deci and Ryan [26], self-determination theory describes motivation personality and optimal functioning and it addresses various kinds of motivation. The SDT describes three different kinds of motivation respective of three innate needs people have. These three needs are **competence**, referring to the need to have control over one's situation, **relatedness** referring to the need people have to feel an emotional bond with others and **autonomy** referring to the need for free will. Deci and Ryan describe two kinds of motivation. The first is autonomous motivation, which involves internal motivation to do a task which is intrinsically satisfying. The second kind of motivation is led by external consequences, either punishment or reward, which involves the individual feeling of appreciation or rejection due to this action.

d. Fogg's model

B.J. Fogg created a more comprehensive overview of the process of the different elements required to enable behavior change then that of the regular focus theory and expands our understanding of different kinds of motivation as compared to the self-determination theory. Fogg's behavior model (FBM) for behavior change combines the individual's intrinsic qualities of motivation and ability with the external factor of a well-timed trigger [20]. Fogg's behavior model describes three factors necessary for behavior change; **motivation**, **ability** and **triggers**. Fogg suggests that one can be motivated; firstly, by the **avoidance of pain** and **pursuit of pleasure**, secondly by **seeking hope** and **avoiding fear** and lastly, by **avoiding social rejection** and **seeking social acceptance**. Fogg's model can also be used to analyze why certain attempts and directed behavior change failed or succeeded. However, though this model addresses how triggers need to be designed appropriately to the level of motivation and ability of the individual at that time, it does not give us much insight about how this motivation and ability might change over time.

e. Trans-theoretical model

The trans-theoretical model (TTM) of behavior change describes six phases of change through which people move sequentially before and during the behavior change process [27]. The first stage the TTM talks about is the **Precontemplation** stage describing individuals who do not intend to change their behavior. Several explanations are given for the reason an individual might be in the precontemplation stage including that the individual might be under-informed or might have tried and failed so many times that they have given up trying. **Contemplation** is the second stage of change in which an individual has decided to change in the future but has not yet undertaken any real steps to change their behavior. Prochaska and Velicer explain that individuals in this stage weigh the pains from changing with the gains from their improved behavior and therefore can often get stuck in this stage for long periods

of time. When I want to lose weight but always find a reason why now is not a good week to be on a diet because it's Thanksgiving or because it's the week in which we celebrate my co-worker's birthday, I am experiencing behavioral procrastination. Individuals move on from Contemplation to **Preparation** when they decide to change in the immediate future and have a plan of action to do so. The next stage is called the **Action** stage. The trans-theoretical model only defines a modification in behavior to be action when professional would agree that this behavior change is significant enough to reduce risk. In the **Maintenance** stage, individuals usually do not rely on change processes like they do in action, however they are working to resist a relapse of their old behavior. This stage of maintenance can last anywhere from one month to five years. Finally, when the individual does not run the risk of regression, the return to an earlier stage of change, then and only then has the individual reached the **Termination** stage. But not many people ever truly make it to this stage and maintenance becomes the enduring end stage of the journey where many still struggle with their former addictions for a long time.

f. COM-B model

According to the COM-B system, a behavior change can only occur from the interaction between the three necessary conditions: **Capability**, **Opportunity**, and **Motivation** [28]. Capability involves the physical and psychological ability to enact the behavior. Opportunity is driven by the physical and the social environment that enable the behavior. Motivation includes the reflective and automatic mechanisms that activate or inhibit the behavior. This is joining Kahneman's behavior model [29] based on two modes of thought "System 1" and "System 2", where "System 1" indicates the fast brain (**automatic, intuitive**) and "System 2" represents the slower mind (**analytical, reflective**) when making a decision. In short, for a behavior to occur, first, the user should be able to perform it; second, the desired behavior should be triggered adequately and third, the user should be motivated enact accordingly.

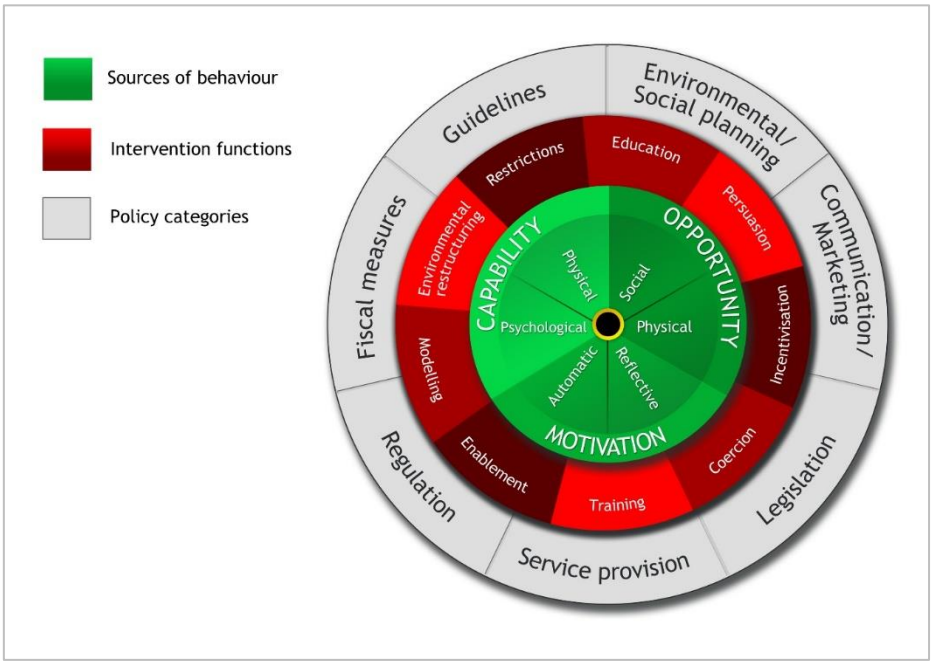


Figure 1. Behaviour Change Wheel

1.1.2.2. Behavior Change Techniques (BCTs)

Behavior Change Techniques are then drawn from these theories to produce different behavior change strategies. Michie et al. [30] developed the CALO-RE taxonomy, a set of strategies specifically used for research focusing on physical activity and nutrition intervention. 93 BCTs grouped in 19 categories are proposed: goals and planning, feedback and monitoring, social support, shaping knowledge, natural consequences, comparison of behavior, associations, repetition and substitution, comparison of outcomes, reward and threat, regulation, antecedents, identity, scheduled consequences, self-belief and covert learning.

To guide the intervention design, Michie et al., synthetized their findings into the Behavior Change Wheel (Figure 1), a framework with three main layers: sources of behavior, intervention functions and policy categories. The first layer is based on the COM-B model. The second layer consists of nine intervention functions, which are education, persuasion, incentivisation, coercion, training, enablement, modelling, environmental restructuring, and restrictions. The choice of the right behaviour change strategy is then based on the COM-B analysis and could be one or a combination of these intervention functions. Depending on the need and the context, several motivational strategies can be derived from these intervention functions. To understand the concept better, these nine intervention functions are summarized in Table 1, with corresponding examples according to our context (motivating physical activity).

Intervention function	Definition	Examples (Motivating physical activity)	Technologies examples
Education	Increasing knowledge or understanding	Providing information to promote exercise/ asking them to wear a device to make them aware of their daily activity.	Conversational agent, Self-monitoring technologies
Persuasion	Using communication to induce positive or negative feelings or stimulate action	Using game or metaphor to motivate increases in physical activity.	Conversational agent, Self-monitoring technologies
Incentivisation	Creating expectation of reward	Using prizes (real, virtual) to induce a better and a longer physical performance.	Gamification
Coercion	Creating expectation of punishment or cost	Some application proposes to do a bet on a physical challenge to motivate physical exercise.	Conversational agent, Self-monitoring technologies
Training	Imparting skills	Training with a virtual coach to do regular exercise or rehabilitation exercise.	Conversational agent
Restriction	Using rules that limit engagement in the target/competing/supporting behaviour	Limit the access to the TV if the user is too sedentary.	Conversational agent, Self-monitoring technologies
Environmental restructruring	Changing the physical or social context	Create more opportunity to exercise at home (adding more stairs, putting things higher step by step for an arm rehabilitation exercise).	Social incentives-based technologies
Modelling	Providing an	Involving the user in a social network where	Social incentives-

	example for people to aspire to or to imitate	they can share their goals and their experience. Induce competition and collaboration.	based technologies, Conversational agent
Enablement	Increasing means/reducing barriers to increase capability or opportunity	Prostheses to promote physical activity. Safe equipment for rehabilitation at home.	Self-monitoring technologies, Conversational agent

Table 1. Intervention functions and examples

Interventions based on behavior change theory have demonstrated a positive effect on patient health behavior [31, 32, 33]. Sawesi and al. [34] showed a significant relationship between theory-based health behavior change intervention and patient engagement. Unfortunately, only less than half of the existing studies referenced specific behavior theories in their research.

Our aim is then to make sure that the different motivational strategies discussed in this thesis are closely linked to the theory of behavior change.

1.1.3. Long-term behavior changes

1.1.3.1.Behavior change, a complex process

Significant behavior change is linked to many numbers of factors making it an even more complex process. First, changing physical activity or nutrition behavior often requires restructuring deep down routines already adopted by the users in their daily life. If someone wants to run during lunch time, he or she would need to find an alternative to the occasional lunch meeting, find a place to shower, get a locker for additional clothes and opt for light snacks after workouts. In that sense, it is not only a matter of exercising more but reorganizing the user's interaction with his daily environment as well as his priority. Researchers have shown that most of time, the main causes of relapse are linked to social and environmental factors [35, 36, 37]. Second, unexpected circumstances such as being sick or moving to another place may also impact user's motivation to pursue the change in their behaviors. Third, the lack of immediate impact of the change being implemented makes it often less attractive than the immediate rewards the users could get at the moment the decision has to be taken [38]. Taking the first bus seems more valued than walking home to prevent diabetes after a long and tiring day at work. Finally, relapses are frequently associated to the limitation of user self-control and self-regulation. Emotional circumstances [39,40], feeling obliged to perform the behavior [41] or falling into the old behavior even once [35] may easily break the different strategies adopted to change the behavior.

1.1.3.2.Behavior change, a long-term process

If initiating a change in the behavior represents a challenge in itself, maintaining this change over a longer period is even more difficult. Behavior change is a long-term process. Repeated performance is often required to reach a desired health outcome. For example, eating vegetables once will not achieve the same health benefits as eating them regularly over an

extended period. Efficient behavior change interventions are the ones targeting a long-term adoption leading to the maintenance of the new behavior [42].
Depending on the target behavior, maintenance may vary from months to years. Prochaska et al. [43] shared that smoking relapse rate can go from 47% to 7% after a year to 5 years of abstinence respectively. Among patients who followed an intensive 3-month cardiac rehabilitation program following a heart attack, 6 months after the program end, more than half of patients stopped to exercise regularly [36, 44]. Researches involving supervised exercise showed improved step count after 3 months, but a decrease of physical activity level 3 months later [45]. In their systematic review, Fjeldsoe et al. [46] intended to investigate the maintenance of physical activity and dietary behavior change by only considering studies reporting at least a 3 months follow-up after the intervention. Their findings showed, first, that maintenance of behavior change is not often reported (35%, 157 studies). Approximately one third of physical activity and/or dietary intervention trials published since 2000 reported on maintenance of behavioral outcomes, and this proportion fell to less than one fifth, when only randomized controlled trials were considered. Second, dietary behavior interventions or combined interventions (diet and exercise) achieved maintenance more often than physical activity interventions alone. Third, interventions conducted over a longer period (6 months), including face-to-face contact, based on multiple strategies and using follow-up prompts were more likely to achieve maintenance. In fact, in their review, Morris et al [45] reported that participation in physical activity at 12 months were observed after tailored counseling strategies.

1.1.3.3.Behavior change and habit formation

It has been argued that repeating the same behavior during a certain period would make it gradually routinized and would lead to habit formation [42]. However, whether forming habits offers a truly long-term change in behavior is still unclear. Habit is formed when there is sufficient exposure to the cue to stimulate the associated behavior unconsciously [47, 48, 49]. Lally et al. [50] stated that habit is learned though context-based repetition and follow an asymptotic growth curve. In their study, involving 96 participants, instructed to perform a certain activity in response to a certain cue (for example “going for a walk after breakfast”), they demonstrated an asymptotic evolution of the behavior where stability was reached at a median of 66 days. Similar results touching upon this asymptotic growth were found by Fournier et al. [51] who conducted a study of adoption of a novel stretching behavior were users reached a plateau at a median of 106 days. These studies reveal that habit development is not linear. The early phase depicting sharpest gains in automaticity represents a critical period during which people require most support to sustain motivation before the action becomes automatic [52]. Habit formation should be considered as a part broader sets of techniques, combining context dependent repetition with strategies that reinforce motivation, increase self-regulation and facilitate early phase repetition. Motivation and action control are pre-requisites for repetition [42]. Strategies such as self-monitoring, goal-setting, prompting cues, planning or rehearsing are suggested to optimize this critical period.

1.1.3.4.Long-term use of intervention technology

With the recent development in persuasive technology, questions arise regarding the necessity to keep using the technological solution to maintain a certain behavior. A study investigating the long-term engagement with a mobile self-management system for people with type 2 diabetes [53], observed a significant decreasing usage trend among 10 out of their 12 participants. Reduced motivation to continue using the application was explained by a sense of mastery over diabetes after a certain period and experiences of different problems through the app. In some cases, technology may serve as a gateway [54] to form a new habit or bring a certain education. Fritz et al. [55] conducted a study with 30 participants who used wearable activity monitors between 3 and 54 months. Despite the consequent number of people who eventually stopped using the device, their findings suggest that some segment of the population continued to benefit from them even after many months or years of use, by continuing to walk the stairs or inserting walking opportunity in their daily life. On the other side, researches also demonstrated correlation between active usage of intervention technology and physical activity behavior change. Cadmus-Bertram et al. [56] showed that participant wearing a monitoring device 10 hours or more per day for 4 months showed a significant improvement of their daily steps at week 3, with only 8% decline at week 16. Participants appreciated the self-regulation of their personal goal but still relied upon recording and viewing their activity data and sharing data on social networks. Both approaches are then interesting to consider when designing behavior change interventions.

1.1.3.5.Long-term approach addressed in this thesis

We have seen that addressing long-term approach of behavior change requires the integration of a well-designed follow-up in the study protocol. Our first dissertation is then based on a longitudinal study involving a 6-month follow-up to measure the maintenance of physical activity behavior following a self-monitoring intervention. We also identified the early phase of adoption of the new behavior as a critical phase to maintain the motivation and the change in the long-term. Most of our dissertation are then also focusing on uncovering the potential problems and understanding technology usage and adoption during early phase (4 weeks or 6 weeks), in order to shed light on the opportunity of the intervention in different populations. Taking into account the difficulty to conduct a long-term study with a high number of participants and following the suggestion of Klansja et al. [57] on evaluating technologies for health behavior change in the context of Human Computer Interaction research, we specifically valued a tailored evaluation of the different behavior strategies used in our intervention.

1.2. Thesis motivation

1.2.1. Self-monitoring technologies: promises

Nelson and Hayes [58] define self-monitoring as “occurring when an individual first self-assesses whether or not a target behavior has occurred, and then self-records the occurrence, frequency, duration, or so on of the target behavior”. For example, dietary self-monitoring would involve a systematic self-observation and recording of parameters related to food intake along with the context where the event occurs. Individuals are often asked to keep a detailed written journal about the type of food, the quantity, if they ate alone or not, if they ate

outside or not, if they cooked or not. The goal is to get information not only on the behavior itself but also on how it can be changed by identifying the context, and the place and the time the undesired behavior most often occurs [59]. Being aware of one's own behavior is sometimes all that is required for the behavior to change. Self-monitoring has been shown to have robust behavior change effects [60]. Today's technological advancements provide a growing opportunity for individuals to self-monitor their behavior.

Brickwood et al. demonstrated in their systematic reviews of consumer-based wearable activity trackers the potential to objectively monitor and assist individuals to remain physically active. They showed a significant increase in number of steps, moderate and vigorous physical activity and energy expenditure across all studies included in their meta-analyses [61]. Because physical activity interventions often have short-term impacts, research suggests that wearable could be an efficient tool to assist health professionals in providing continuing surveillance and long-term support. When combined with telephone counseling, these improved pedometers have been shown to have the same potential of increasing physical activity in middle-aged and older adults [62]. Additionally, dietary self-monitoring has been constantly associated with weight loss. Carter et al. [63] developed an electronic food diary and demonstrated the relation between improved weight loss and the length and frequency of use of the app. Patel et al. [64] compared self-monitoring strategies for weight loss in smartphone apps and concluded that using tailored goals and a commercial mobile app could lead to clinically significant weight loss. Digital standalone health treatments may be a feasible alternative for people seeking a less intensive approach.

Wearables have evolved from standard pedometers that count steps, to more sophisticated tools that give feedback, prompt reminders and allow to define personal goals. Today, they include a wide range of behavior change techniques typically used in clinical behavioral interventions. In their review, Lyons et al. [65] identified social support, social comparison, prompts/cues, rewards, and a focus on past success in more than half of the systems they analyzed. The majority of techniques involved self-monitoring and self-regulation methods, mostly associated to enhanced physical activity in adults and elderlies [66]. Ernsting et al. [67] conducted a population-based survey on health apps in Germany and found significant associations between planning and the health behavior physical activity, between feedback or monitoring and physical activity, and between feedback or monitoring and adherence to the doctor's advice. Self-monitoring technologies then show a great potential to be used as a medium by which these interventions could be translated for widespread use.

Research suggest that being able to integrate and control their health data drives individuals towards mHealth technology. Whelan et al. [68] demonstrated significant brain activation of people receiving personalized glucose feedback. However, to achieve successful meaningful health behavior change, collaboration models in health care are needed to support motivation and commitment [69]. Furthermore, fair and simple user experience, self-regulation, relatedness, adaptive exercise plans and just-in-time support are thought to be essential to sustain engagement over time [70].

1.2.2. Self-monitoring technologies: limitations

Although self-monitoring technologies show great promises to promote behavior change, tracking health data is not enough to sustain the motivation and engagement, especially in the patient domain [69]. Despite the growing numbers of healthcare apps on the market (47,878 on Apple App Store [71], 40,596 in Google Play [72]), little is known about the content of the feedback given, and how individual monitors may differ from one another. Furthermore, if it is true that keeping track of dietary behavior could result in weight loss, there is limited evidence of how frequently people need to monitor their diet for ideal change in their body mass [64]. Also, further research is needed to determine whether there are participant features that would reliably predict those most likely to monitor their diet on a regular basis.

A large number of behavior change techniques are indeed incorporated in existing self-monitoring technologies. This offers an indication of these systems' ability to alter individuals' behaviors, but the long-term efficacy of these systems to alter exercise, sleep, and sedentary behaviors remains unknown [73]. Techniques related to planning and providing instructions are rare. Besides, these applications and the behavior theory behind are often generalized to the public and does not necessarily answer individuals' preferences and needs related to their specific pathology. Also, rather than tracking physical activity, there is a greater need for the development of tools to self-monitor sedentary time [60]. Finally, the feasibility and the acceptability of these technologies across different specific population remain understudied.

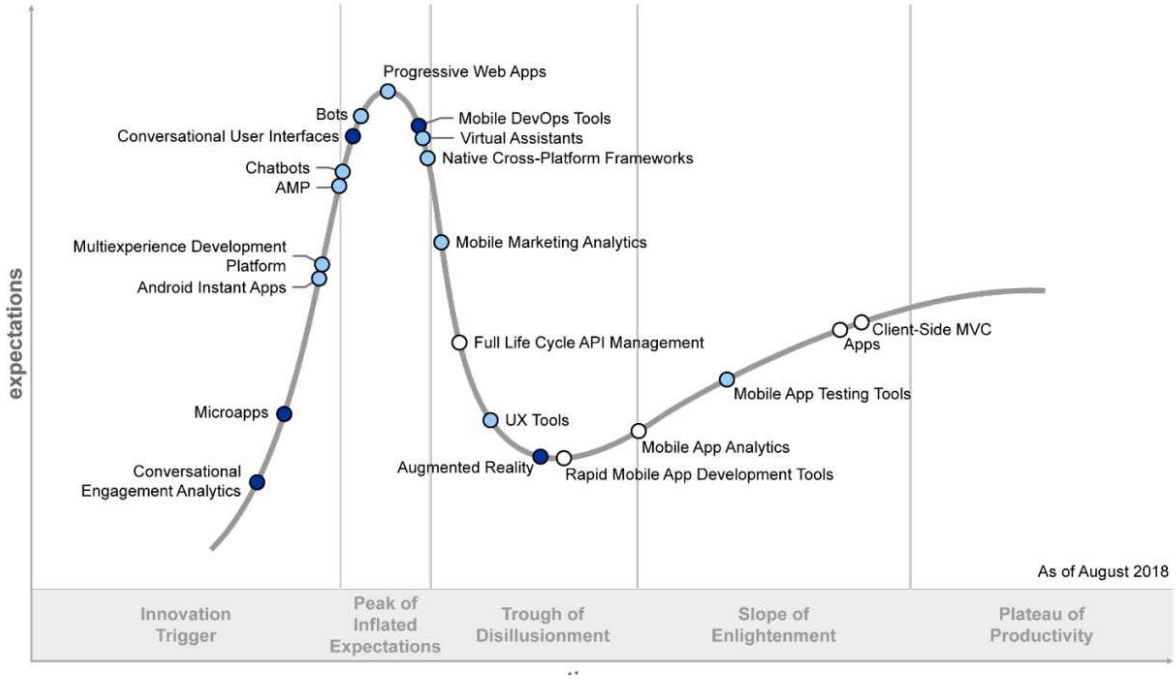


Figure 2: Gartner Hype Cycle for Mobile Apps and Multi-experience Development

In general, smartphone apps are known to have low engagement and lower re-use rates. The Gartner hype cycle [74] (Figure 2) shows how innovation trigger could lead to a peak of inflated expectations to end through a phase of disillusionment. Pokemon Go is a good example of interesting application that has swiped the world and became the most downloaded app in history [75]. Known for its complex intervention including virtual,

physical, and social dimensions, the game produced increased socialization and increased visits to public parks, museums, and historical sites. People who met the recommended (150 minutes or more per week) activity levels, have increased by 44% after playing the game [76]. However, Pokemon Go was also identified as potentially harmful [77]. Pokemon Go main motivations were based on outdoor activity, nostalgia and boredom [78], but after a certain period the engagement towards the game slowly faded. Therefore, despite the positive effect in the beginning, limited information is available about how self-monitoring technologies can sustain health behavior change and be integrated into health care. There are age-related, socioeconomic-related, literacy-related, and health-related disparities in the use of mobile technologies [67]. Health app use may reflect a user's motivation to change or maintain health behaviors and consider the needs of older people, people with low health literacy, and chronic conditions.

1.2.3. Social incentives-based technologies: promises

Gillin and Gillin [79] define social interaction as "the mutual or reciprocal influence, resulting in the modification of behavior, exerted through social contact and communication which, in turn, are established by inter-stimulation and response." Social interaction provides a number of benefits in domains such as reducing stress level, improving people's mood, or increasing life expectancy [80]. Belonging to a society and interacting with others are proved to influence positively elderly's quality of life [81].

Moreover, social incentives have been shown to have a significant impact in motivating people to reach a certain behavior. Researches in psychology and neuroscience have demonstrated that people easily mirror others behavior [82] and could be influenced by what others are doing. We thus observed many studies [83, 84, 85] that involves sharing goal and achievement on social platform to motivate people to engage in healthy behavior like doing physical exercise or eating healthy meals. They could receive social support in different format such as instrumental support, informational support, emotional support, and appraisal support [86]. Besides, most of the existing applications to promote physical activity involve a way to foster competition and challenges [87].

Salmenius-Suominen et al. [88] investigated social support through a visual food diary and showed an increase of 55% of vegetables intake and a decrease of 39% of sweets and chocolate intake after one-month intervention. Their participants reported receiving support and encouragement from their peers and enjoyed the feeling of belonging to a group. From users' perspectives, social component that may influence their engagement are social comparison, similarity and familiarity between users, and participation from other users in the network [89]. Furthermore, automation and personalization improve the delivery of individual and social elements.

Social media is a powerful tool that allows users to communicate or work together and is a social interaction mechanism for a wide range of individuals. Maher et al. [90] demonstrated significant short-term changes in physical activity associated to an intervention combining online social network and pedometers. Personal stories of health behavior change are known to affect participant attitudes, such as self-efficacy. A pilot study investigated the use of an

automated indexing of internet stories and found a significantly greater increase in self-efficacy for weight loss in the participants who received the intervention [91]. Other systematic reviews [92, 93] explored the efficiency of health behavior change interventions that uses social media and found that most of the studies reported significant improvements in outcomes related to the change of behavior. They identified “interactions with others”, “more available, shared, and tailored information”, “increased accessibility and widening access to health information”, “peer/social/emotional support”, “public health surveillance”, and “potential to influence health policy”; as the six potential key benefits in the use of social media for health.

1.2.4. Social incentives-based technologies: limitations

Despite the potential benefits mentioned previously, it is still unclear how online social networks may best be harnessed to achieve health behavior change. The same systematic review [92] also reported a low engagement and fidelity in the social media usage for changing behavior. Further studies are required to determine whether ongoing behavioral changes can be maintained and how online social networks are to be exploited to accomplish mass dissemination [94]. Furthermore, little is known about users’ acceptability and long-term engagement with social network-based interventions. Although, self-regulatory techniques and social factors are important to consider when designing a physical activity intervention, Ly Tong et al. [89] identified technological limitations and one-size-fits-all approaches as potential barriers to long-term usage. After a five months follow-up, no significant changes in physical activity, quality of life or mental health were found in the study conducted by Maher et al. [90] where they combined a social network and a pedometer.

Interaction through social media indeed allows health communication among the general public, patients, and health professionals. However, Moorhead et al. [93] shed light on twelve limitations of social network usage in their systematic review, primarily consisting of quality concerns and lack of reliability, confidentiality, and privacy. There is a need to explore potential mechanisms for monitoring and enhancing the quality and reliability of health communication using social media. This channel has the advantage to be open and easily accessible, but the quality and reliability of the information exchanged must be monitored, and confidentiality and privacy of users must be maintained. The need for health professional moderators, how often and in which format their presence is required calls for deeper investigations.

Although we know that social interaction shows numerous benefits in improving behavior, the type of social intervention adapted to a specific population is understudied. Social comparison and social collaboration may have different or complementary effect on the desired changes. Future research should also explore how to address challenges faced by physically inactive people to provide tailored advices. Users’ perspectives should be explored to learn on factors that might influence their motivation with the intervention [95].

1.2.5. Gamified technologies: promises

Gamification which is defined as “the application of game principle into non-game context” [96, 97] is being widely used in various domain such as work, marketing, education and health. The goal is to induce user engagement in performing certain actions. A plethora of research investigates the use of gamification to motivate physical exercise among people with chronic disease, cardiovascular risks and the ageing population [98].

Serious video games show promises in helping obese children adopt healthy diet and become more physically active [99]. Accessibility and significant interest in video games played a major role in their motivation. Common interests have been found in smokers who used a smoking cessation mobile phone game. They appreciated the advice and the distraction coming with the intervention [100] and expressed an intention to use it again and to recommend it to their friends. In their longitudinal study, El-Hilly et al. [101] found that game engagement was linked to intrinsic motivation and behavioral control. Gamification then shows a potential to be a low-cost and effective solution to support smoking behavior change. Furthermore, the emergence of Kinect-based technologies paves the way to the development of exergames (exercise-based videogames), for example for elderly people who are subject to frequent falls. In their study, testing serious game, Meekes et al. [102] found that social interaction was an important extrinsic motivator that increased the intrinsic motivation to adhere to the exergame program. Elderlies enjoyed playing the game and were more active and socially confident.

Gamification principles are closely related to behavior change theories and techniques that have been proven to work in improving health outcomes. Promising evidence suggests that gamification works [103, 104]. Researchers have been actively designing persuasive technologies that motivate moderate physical activities. To motivate people to walk more, activity monitoring devices are coupled with a virtual rewards system and allow users to collect badges and points [105]. Another common area of research is the use of metaphor to help people visualize in a playful and easy way their progress towards their goal. Many studies have been trying to map the number of steps achieved to the growth of virtual character like animals [106] and plants [107]. In addition to that, we observe a real interest in using serious game to engage people in performing physical exercise. Playing games that involve subtle physical movement are widely explored [108] not only for moderate activity like walking or running but also for rehabilitation exercise. Immersing the user into a virtual world does not only procure a feeling of pleasantness but also has the potential to increase user self-efficacy and confidence in doing the required exercise.

1.2.6. Gamified technologies: limitations

While gamification is promising and has proven efficient in many areas, critical issues remain as to how this technique can be used to modify health behavior. Conditions for game engagement such as purpose, user alignment, and utility need to be defined [101]. In addition, the relation between gamification elements and behavioral theory deserve a deeper investigation. Despite the proliferation of gamification in health apps, the majority are not following guidelines and theory related to behavior change, which can potentially impact the

efficacy of the intervention [104]. In a randomized control trial comparing the efficacy of educational video game versus traditional educational apps at improving doctor's decision making, small benefits were observed [109]. Real-world efficacy of gamified intervention needs further investigations.

Despite the positive echo associated with gamification, assuming that it is a magic solution to shape how people think and behave would be too far-fetched. To be effective, gamified technology must exceed other design patterns in terms of its capacity to impact the views, attitudes, or behaviors of individuals [103]. Gamified application such as Pokemon Go [78], Ingress [110] and ZombieRun [111] had substantial hype effect but were not a vector of long-term change in physical activity. How gamification may exceed short-term novelty effect remains unclear and requires further research. Furthermore, a review of stress management app [112] suggests that gamifications techniques are still scarce in this field. Guidance for the development of mHealth gamified intervention is therefore necessary in different health domain, assuming requirements may differ between various pathology. Finally, as they are mostly associated with children and youngsters, how these gamified techniques are accepted and adopted by the ageing population is still understudied.

1.2.7. Conversational agent: promises

A Conversational Agent (CA) or a chatbot is a software that interacts with users by simulating a human conversation through text or voice. An Embodied Conversational Agent (ECA) comes with an avatar that displays many of the same properties as humans in face-to-face conversation and can produce and respond to verbal and nonverbal communication [113]. With the acceleration of the development in natural language understanding, conversational agents have gained popularity over the past years. The applications range from treating anxiety [114], depression [115], loneliness [116] to relieving chronic pain [117] and other psychological issues [118]. It has also been used to enhance medication adherence [119] and to facilitate health information searching [120]. Research suggests that conversational agent shows promising results in engaging patient through Cognitive Behavioral Therapy in addition to being cost-effective and accessible [114, 121]. In a study testing the efficiency of artificial intelligence agent for mental health issues [118], patients who used the system more frequently showed significant improvement in symptoms of depression and found the system helpful and encouraging. Empathy and relational behavior are therefore a significant research themes in dialog systems for behavior change, with many pilot studies showing a preference for those features.

Conversational agent capabilities could allow a number of interesting features such as adaptive personalized messages related to the context, rich and interactive education, and support for self-monitoring of behavior change progress [122]. Tanana et al. [123] developed a patient-like conversational agent to train therapists basic counselling skills and found that participants in their ClientBot demonstrated deeper reflection capabilities, during the training, due to the real-time feedback. Additionally, a scoping review done by Scholten et al. [124] suggests that embodied conversational agents can motivate and engage individuals in the domain of learning and behavior change. Chatbots also demonstrated positive effect on

treatment self-monitoring, and medication adherence of patients with breast cancer [125]. Patients were satisfied and were willing to recommend the bot to their friends. Common positive results were found when testing bedside avatars to reduce the rate of falls among elderly inpatients fighting with delirium and loneliness [126]. Over three months study period, the rate of falls decreased by 86%.

In the nutrition domain, conversational agents allow to collect user data in an easy and user-friendly manner. Researchers at MIT developed a Web-based prototype of a speech-controlled nutrition-logging system which converts the entry spoken by the users into calories intake [127]. Researchers at the University of Applied Sciences in Western Switzerland worked on a chatbot that helps people reduce their meat and increase their fruits and vegetables consumption [128]. Users were able to set nutrition goals themselves and had a follow-up with the system every day. Although, only 11% could reach their objectives, more than half of the participants showed positive changes in their nutritional habits.

1.2.8. Conversational agent: limitations

As described previously, the application of conversational agents ranges from simple strengthening of social behaviors through emotional expressions to advanced multimodal conversational systems. However, this domain is still in its nascent age. Successful outcomes are mainly based on pilot studies and evidence on their clinical application remains low [115]. A survey investigating physicians' perceptions of chatbots identified their perceived benefits of the systems such as scheduling doctor appointments, locating health clinics, or providing medication information. The same study pointed out doctor's doubt regarding the system efficiency in caring for patients' needs, in displaying human emotion, and in providing detailed diagnosis and treatment without knowing the patient personal context [129]. Bickmore et al. [130], questioned the safety issue for using conversational assistants, such as Siri, Alexa, and Google Assistant to be used as portals for medical services. The ethical implications of these intelligent agents also raised concerns regarding the need of concrete ethical guidance while designing these systems [131].

Furthermore, conversational agents are still unable to convey empathy in a nuanced manner during a real-world conversation. Most of them still have difficulties in assessing user's emotional state while engaging in the discussion [132, 133]. Further research and collaboration between different stakeholders in various domain are needed for a conversational agent to conduct deeper and more complex interaction. There is also a lack of evidence investigating the acceptance and the efficiency of the intelligent agent in transdisciplinary area such as physical activity and nutrition. How these systems may be adapted and tailored to different population with specific needs remains understudied.

1.3. Thesis objectives and research questions

1.3.1. Thesis objectives

This dissertation investigates the efficiency of self-monitoring technologies to promote positive change in the long-term on modifiable behaviors, mainly regarding physical activity

and nutrition. It sheds light on the opportunities and the limitations of self-monitoring, gamified, social and conversational applications and intends to provide guidelines for designing these technologies for specific population, namely: chronically ill and elderly patients. First, acceptance and adoption of fitness trackers by diabetic and obese patients are investigated. The findings are presented in a set of guidelines designing wearable health monitors that are engaging enough to effect long-term change for chronically-ill patients. Second, elderlies' attitudes towards self-monitoring technologies for physical activity are questioned. The outcomes are a set of opportunities and challenges for designing self-monitoring technologies for healthy ageing. Third, effect of social incentives to motivate physical activity on obese and diabetic patients are explored. The results give insights into the role of cooperation and competition to enhance their engagement in being physically active. Fourth, effects of a gamified rehabilitation to engage elderly patients with musculoskeletal issues to be more active are researched. The findings outline different recommendations on designing serious game for elderly rehabilitation. Fifth, the clinical efficacy of this serious game-based rehabilitation is evaluated through a non-inferiority randomized control trial. Finally, a prototype of a conversational agent to promote healthy eating is developed from state of the art, co-designing sessions with various stakeholders and the different findings outlined through the thesis.

Overall, the work conducted within this dissertation offers new perspectives on the design of self-monitoring technologies for elderly and chronically-ill patients. It makes several research contributions that are of interest to the greater Human Computer Interaction, Digital Health, Conversational AI, Behavioral sciences, Nutrition and Physical Rehabilitation communities.

1.3.2. Thesis structure and research questions

This thesis is based on five original published articles and a conference poster presentation, structured in four parts (Figure 3), outlined below:

Part 1 - Acceptance and adoption of self-monitoring technologies: the first part of this thesis is an attempt to understand how chronically-ill and elderly patients accept and adopt self-monitoring technologies. Little work has been done to investigate how these technologies influence users' behavior and environment or how they can aid our target population to change their lifestyle over the long term. Our aim in this part, is first to shed lights on their attitudes towards the tools, second to give insights on how they would integrate them in their daily life and third to suggest items to consider while designing self-monitoring solution for these population. The following questions are investigated:

RQ1: How self-monitoring technologies are accepted and adopted by chronically-ill patients and engage them in healthy lifestyle?

RQ2: What are the opportunities and challenges for the elderly to be monitored and how would they integrate the system in their daily life?

Part 2 - Social incentives to motivate chronically-ill patients: the second part of this thesis investigates how social incentives work for individuals with obesity and diabetes. Social incentives such as cooperation and competition are found to motivate users in persuasive fitness applications but there is a lack of literature regarding the potentials of these social concepts to engage chronically-ill patients to be more physically active. Our goal in this part, is first to understand in what aspects collaboration and respectively competition could contribute in improving obese and diabetic patients' behavior. The second goal is to compare those two aspects to identify correct recommendations for designing social intervention for this population. The following question is investigated:

RQ3: How does collaboration and competition affect the motivation of chronically-ill patients to exercise more?

Part 3 - Gamification for elderly rehabilitation: the third part of this thesis explores the efficacy of a gamified rehabilitation intervention in the treatment of elderly patients and in motivating them to be more active. The hospital-to-home transition is increasingly recognized as a critical period in the patient care, during which different incidents can occur and induce frequent re-hospitalization. There is therefore a growing interest in strengthening the physical and functional capacities of hospitalized elderly patients to prevent re-hospitalization. Our objective in this part, is first to enlighten the community on the impact of gamified rehabilitation on elderly motivation and activity while hospitalized and second to investigate the clinical efficacy of this solution. The following questions are investigated:

RQ4: How does serious game-based rehabilitation impact elderly patients' motivation and activity at the hospital?

RQ5: How efficient is the gamified rehabilitation compared to the standard rehabilitation treatment?

Part 4 - Prototype: Conversational agent for chronically-ill and elderly patients to promote healthy eating: the last part intends to propose a conversational agent designed through co-creation sessions, the findings from the previous part of thesis and the state of the art. Despite the extensive research and the large number of existing nutrition applications, food recommender systems are still facing many challenges in terms of nutrition habits tracking and delivery of the proper recommendations. Our aim in this part, is first to shed light on the chronically-ill and elderly patients' requirements and needs for self-monitoring technology to promote healthy eating and second, to propose a prototype using a conversational agent. The following question is addressed:

RQ6: What are the needs and requirements for self-monitoring technology to promote healthy eating for chronically-ill patients?

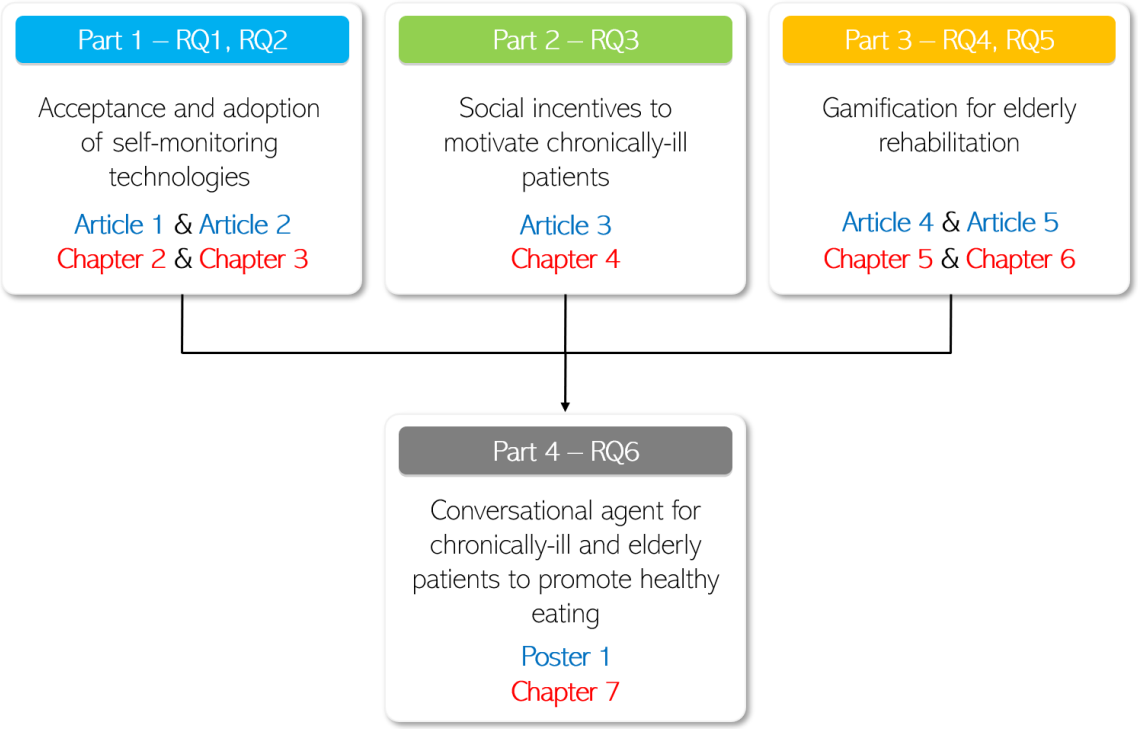


Figure 3: Schematic overview of the thesis structure

	Article 1	Article 2	Article 3	Article 4	Article 5	Poster 1
Chapter	2	3	4	5	6	7
Research focus	Acceptance and adoption of self-monitoring technologies (chronically ill patients)	Acceptance and adoption of self-monitoring technologies (elderly patients)	Social incentives to motivate chronically-ill patients	Gamification for elderly rehabilitation: engagement and motivation	Gamification for elderly rehabilitation: study of efficacy	Conversational agent for chronically-ill patients to promote healthy eating
Paper title	Can Fitness Trackers Help Diabetic and Obese Users Make and Sustain Lifestyle Changes?	Opportunities and Challenges for Self-monitoring Technologies for Healthy Aging: An In-situ Study.	Social Incentives in Pervasive Fitness Apps for Obese and Diabetic Patients.	Computer-aided physical rehabilitation of older people: a pilot non-inferiority randomized clinical trial.	Hospital-to-home transition: using serious game and activity monitoring to engage elderly patient to be more active.	MiranaBot: A conversational agent to promote healthy eating.
Authors	M. Randriambelonoro, Y. Chen and P. Pu.	M. Randriambelonoro, Y. Chen, O. Yuruten and P. Pu.	Y. Chen*, M. Randriambelonoro*, A. Geissbuhler and P. Pu.	M. Randriambelonoro, C. Perrin, A. Blocquet, D. Kozak, F. Geissbuhler, C. Graf et al.	M. Randriambelonoro, C. Perrin, G. Carmona, E. Frangos, F. Herrmann, A. Geissbuhler, C. Graf.	M. Randriambelonoro.
Methodology	**Longitudinal qualitative study** *Participants:* 18 *Age:* 36 – 73 *Duration:* 7 months	**Qualitative study** *Participants*: 20 *Age:* 65 – 89 *Duration:* 6 weeks *Condition:* senior	**Qualitative study** *Participants:* 16 *Age:* 36 – 73 *Duration:* 8 weeks *Condition:* obese	**Qualitative study** *Participants:* 57 *Age:* 81.5 (mean) *Duration:* 3 weeks *Condition:* elderly	**Multicenter non-inferiority randomized clinical trial** *Participants:* 57	**Qualitative study (observation, co-design)** *Participants:* 13 *Duration:* 1 week

	Condition: obese and diabetic	people living at home	and diabetic	patients with musculoskeletal issues	*Age:* 81.5 (mean) *Duration:* 6 weeks *Condition:* elderly patients with musculoskeletal issues	*Condition:* obese and diabetic
Intervention functions (Michie et al.)	Education Persuasion Incentivisation	Education Persuasion Incentivisation	Education Persuasion Incentivisation Modelling Environmental restructruring	Education Persuasion Incentivisation Training Enablement	Education Persuasion Incentivisation Training Enablement	Education Persuasion Training Modelling Enablement
Scientific contribution type	Empirical findings and design guidelines	Empirical findings	Empirical findings and design guidelines	Empirical findings and design guidelines	Clinical validity	Conversational agent prototype, Empirical findings and design guidelines
Published in peer-reviewed conference / Journal	**Journal** IEEE Computer Magazine. SI: March 2017 Quality-of-Life Technologies **Impact Factor: 3.564**	**Journal** Gerontechnology 16(2), Nov 2017	**Conference** In Proceedings of the 19th ACM Conference Companion on Computer Supported Cooperative Work & Social Computing (CSCW'16 Companion). Feb.	**Journal** Journal of Population Ageing. 2019	**Journal** Archive of Physical Medicine and Rehabilitation **5-Year Impact Factor: 3.618**	**Poster** REACH exhibition. Lyngby. 2019.

			2016. **Acceptance rate: 25%**			
Status	Published	Published	Published	Submitted	To be submitted	Presented
Author contributions	Contribution of PhD candidate: 90% Y.C and P.P: conceived the study M.R: conducted the experiment M.R: analyzed the data with Y.C M.R: wrote the paper	Contribution of PhD candidate: 95% M.R: conceived and designed the study. M.R: conducted the experiment M.R: analyzed the data M.R: wrote the paper	Contribution of PhD candidate: 85% Y.C and P.P: conceived the study M.R: conducted the experiment M.R: analyzed the data M.R: wrote the paper with Y.C.	Contribution of PhD candidate: 95% M.R: conceived and designed the study. M.R, A.B: conducted the experiment M.R: analyzed the data M.R: wrote the paper C.P, C.G., A.G: reviewed the paper	Contribution of PhD candidate: 90% M.R: conceived and designed the study. M.R: conducted the experiment M.R, F.H, CG: analyzed the data M.R: wrote the paper C.P, C.G., A.G: reviewed the paper	Contribution of PhD candidate: 100% M.R: conceived and designed the study. M.R: conducted the experiment M.R: analyzed the data M.R: developed the prototype

Table 2. Synoptic table of dissertation articles

1.3.3. Scope of the scientific articles

This thesis is based on five original scientific articles and one poster, which investigate the research questions defined previously through theoretical and experimental work. Table 2 gives an overview of the scope, methodology, and contributions of each of the articles. This section introduces them in more details.

1.3.3.1. Dissertation Article 1: RQ1

Article 1 presents the acceptance and adoption of fitness trackers by diabetic and obese patients and how it can influence them in the long term. The article begins with a brief introduction to the current research on pervasive fitness sensors and applications and outlines the need for a deeper understanding of the motivational techniques for using this technology. A state of the art focused on "Empowering Users to Control Their Health" is then addressed to discuss proactive decision making and motivation challenges. Later, the paper presents a longitudinal study (7-months) involving 18 individuals ages 36 to 73 - all of whom were formally diagnosed with diabetes, obesity, or both. Each participant was given the Fitbit One fitness tracker (www.fitbit.com/us/one) and was interviewed before, during, and after the study to identify what they expected from the tracker, how the tracker influenced their lifestyle, and what was required for the technology to change their long-term health decisions. After one month, changes in activity awareness, motivation to participate in activity, and a stronger sense of social connection were noted. After seven months, the 16 users still wearing the device reported visible health and lifestyle changes. At this point, physical activity was translating to visible results such as weight loss, which reinforced activity awareness, although some users reported losing the device or forgetting to wear it. Social connection also remained. Finally, a set of requirements were identified, which evolved into recommendations for the design of wearable health trackers that aim to effect long-term lifestyle changes in their users' behavior.

Type	Longitudinal qualitative study
Participants	18 obese and diabetic patients
Duration	7 months
Intervention	Wearable fitness tracker with its native application
Research question	How self-monitoring technologies are accepted and adopted by chronically-ill patients and engage them in healthy lifestyle?
Objectives	Understand acceptance and adoption in the long term
Outcomes	Guidelines to design wearable health monitors that are engaging enough to effect long-term change

Table 3. Article 1 scope and objectives

To sum-up the findings, diabetic and obese users given a fitness tracker for seven months identified the device's playfulness, practicality, persuasiveness, personalization, and privacy as the top motivators for its use. These results provide insights into designing wearable health monitors that are engaging enough to effect long-term change. The healthcare community is already envisioning technology as a driver of behavioral change. A well-designed activity monitor can do much more than record a user's glucose level and weight; through

personalization and pervasiveness, it can help users manage their lifestyle to incorporate healthier choices. This paper also showed that users' expectations of the device change with continued use. Thus, in addition to tailoring the device for the user and the user's environment and making it entertaining and easy to use, designers should consider ways that the device can adapt to changing expectations across periods of use. Table 3 summarizes Article 1 scope and objectives.

1.3.3.2. Dissertation Article 2: RQ2

Article 2 presents elderlies' attitudes towards self-monitoring technologies for physical activity and how they would integrate it in their daily life. The paper presents a qualitative study, carried out within the scope of the REACH project, with the aim to learn the older adults' behavior by collecting physical activity and health related data in order to provide personalized health recommendation to them. For this purpose, an ethnographic study for data collection was conducted to get insights on older adults' readiness, willingness, and challenges to adopt pervasive sensors and applications for healthy ageing. Despite the ongoing works on older adults' adoption of persuasive technology for behavior change, the barriers to senior's acceptance of these systems still need further investigations. The purpose of this study is an attempt to gather more insight on older adults' opportunities and challenges in adopting such technologies, for designers and engineers to propose personalized and adapted interventions that will induce steady changes in older adults' daily life. In this paper, we present the findings of an experiment with 20 older adults using pervasive sensors for 6 weeks at their homes. In particular, we present their attitudes towards physical activities and self-monitoring technologies before and after the actual usage. We then discuss the opportunities and challenges for designing self-monitoring technologies for healthy aging.

Type	Qualitative study
Participants	20 senior people living at home
Duration	6 weeks
Intervention	Wearable fitness tracker with its native application
Research question	What are the opportunities and challenges for the elderly to be monitored and how would they integrate the system in their daily life?
Objectives	Understand attitudes and adoption of self-monitoring technologies
Outcomes	A set of opportunities and challenges for designing self-monitoring technologies for healthy ageing

Table 4. Article 2 scope and objectives

Before using the devices, our intent was to understand older adults' attitudes towards increasing physical activity as well as their readiness towards tracking technologies. The main barriers of being physically active were the absence of motivation, the incapacity due to health condition and the lack of perceived usefulness of physical exercise. Furthermore, although half of the participants were not technology-oriented, personal interest and enthusiasm driven by family towards technology were observed. After 6 weeks of usage, we discovered changes in behavior and usage intention which allowed us to identify opportunities and challenges for the older adults to adopt sensors and application for health and activity management. This study showed the potential of acceptance and adoption of simple and

manageable technology for behavior change. As for usage intention, some older adults start to be eager to learn new tools as long as they are able to do it, which is more and more influenced by family from younger generation. However, designers and engineers should consider older adults' reluctance when introducing novelty in their daily life, as well as their need for social interaction and their need to remain in control of any system given to them. Table 4. summarizes Article 2 scope and objectives.

1.3.3.3. Dissertation Article 3: RQ3

Article 3 explores the effect of social incentives to motivate physical activity on obese and diabetic patients and gives us insights into the role of cooperation and competition to enhance their engagement in being physically active. A mobile fitness application named HealthyTogether was used as an experimental platform, which allows dyads to achieve fitness goals together, and compete in an online community. The paper presents a four-week study with 16 obese and diabetic patients who used HealthyTogether to exercise with a buddy. Our aim is to compare obese and diabetic patients' fitness performance when they were connected with buddies and communities, and when they exercised alone. The recruited patients participated in our longitudinal study conducted before, in which each individual used a tracker to monitor physical activities. We retrieved participants' Fitbit data during their first four-week Fitbit usage as the baseline and that during their four-week HealthyTogether usage as the intervention.

Results show that participants exercised more with social incentives compared with their baseline. Collaborating with buddies to compete in a community was reported as motivating for dyads exercising with strong ties. Social interactions could be demotivating between dyads who did not know each other well. Finally, it is crucial to consider patients' technical literacy when designing behavior-changing technologies. Table 5. summarizes Article 3 scope and objectives.

Type	Qualitative study
Participants	16 obese and diabetic patients
Duration	8 weeks (4 weeks baseline – 4 weeks intervention)
Intervention	Wearable fitness tracker and HealthyTogether application
Research question	How does collaboration and competition affect the motivation of chronically-ill patients to exercise more?
Objectives	Explore the effect of social incentives for engagement
Outcomes	Insights into the role of cooperation and competition

Table 5. Article 3 scope and objectives

1.3.3.4. Dissertation Article 4: RQ4

Article 4 presents a qualitative analysis of the effect of a serious game-based rehabilitation combined with an activity monitoring to engage elderly patients to be more active. The paper presents a multicenter non-inferiority randomized control trial with 57 elderly inpatients, carried out within the scope of the REACH project, with the aim to understand how the use of

a gamified rehabilitation equipment adds value to the continuity of patient care, specifically in terms of engagement and motivation to be more active during the hospital stay. The intervention consists of a Kinect-based virtual rehabilitation system (ActivLife) coupled with a rehabilitation machine. Elderly patients' activities were monitored via an ankle wearable tracker (Stepwatch). Semi-structured interviews were conducted to capture the patients' attitudes towards the rehabilitation process, their motivation for physical activity as well as their readiness for innovation technology.

Participants admitted feeling bored at the hospital and only following along the care process routine. Enthusiasm towards the progress brought by technological solution were observed. At the end of the hospital stay, the serious game-based rehabilitation received positive feedback. Patients felt more active and observed significant improvement in their general condition. The analysis suggests that patients following the gamified rehabilitation were higher motivated. Different recommendations on designing serious game for elderly rehabilitation were suggested, namely, playfulness, personalization, performance feedback and safety. Table 6. summarizes Article 4 scope and objectives.

Type	Qualitative study
Participants	57 elderly patients with musculoskeletal issues
Duration	6 weeks (3 weeks hospital)
Intervention	Gamified rehabilitation equipment & wearable tracker
Research question	How does serious game-based rehabilitation impact elderly patients' motivation and activity at the hospital?
Objectives	Understand attitudes and adoption of serious game-based rehabilitation
Outcomes	A set of opportunities and challenges / different recommendations on designing serious game-based rehabilitation intervention for healthy ageing

Table 6. Article 4 scope and objectives

1.3.3.5. Dissertation Article 5: RQ5

Article 5 explores the clinical efficacy of a serious game-based rehabilitation in the treatment of elderly patients with musculoskeletal issues. The paper presents a multicenter non-inferiority randomized control trial, carried out within the scope of the REACH project, with the aim to compare a gamified rehabilitation device with the standard of care in terms of efficacy in helping elderly patients to regain their functional capacities. The intervention consists of a Kinect-based virtual rehabilitation system (ActivLife) coupled with a rehabilitation machine. Elderly patients' activities were monitored via an ankle wearable tracker (Stepwatch). Our hypothesis is that the patient in the intervention would be as functional as the patients in the control group in terms of strength, speed and balance. Trial methodology including design, eligibility criteria, settings and location, sample size, intervention and assessment are described in the first part of the paper. 57 patients aged 67-95 years were randomly assigned in the intervention group (n=35) of using the gamified rehabilitation equipment 3 times a week or to a control group (n=22) of usual standard care. Outcome measures included SPPB, IHGS, FIM and number of steps. Comparisons of the groups at baseline showed no evidence of differences between the groups in any of the measures.

Findings show no significant differences were found between the control group and the intervention group for the primary outcome (SPPB: balance, gait speed, chair stand) or any of the secondary outcomes (IHGS, FIM and steps), which demonstrates the non-inferiority effect of the serious game-based intervention. In addition, patients who received the intervention demonstrated engagement in the rehabilitation process and motivation in being more active. Serious game-based rehabilitation could be an effective alternative for elderly patients to regain their functional capacities. Table 7. summarizes Article 3 scope and objectives.

Type	Multicenter non-inferiority randomized clinical trial
Participants	57 elderly patients with musculoskeletal issues
Duration	6 weeks (3 weeks hospital – 3 weeks home)
Intervention	Gamified rehabilitation equipment & wearable tracker
Research question	How efficient is the gamified rehabilitation compared to the standard rehabilitation treatment?
Objectives	Explore clinical efficacy
Outcomes	Insights into efficacy and the benefits of the intervention / complexities associated with adopting new technologies in clinical practice

Table 7. Article 5 scope and objectives

1.3.3.6. Dissertation Poster 1: RQ6

Poster 1 presents a prototype of a conversational agent to promote healthy eating, developed from state of the art, co-designing sessions with various stakeholders and the different findings outlined through the thesis. The dissertation begins with a brief introduction to the current research on food recommender systems, the recent development in natural language understanding. And outlines the need for a deeper understanding of the motivational techniques for using this technology. We then describe a one-week qualitative study with 13 patients and health professionals who helped us identify the main barriers as well as the needs and requirements for adopting a healthy nutrition behavior.

Type	Qualitative study (Observational study / Participative design)
Participants	13 participants
Duration	1 week
Intervention	Conversational agent
Research question	What are the needs and requirements for self-monitoring technology to promote healthy eating for chronically-ill patients?
Objectives	Develop a conversational agent to promote healthy eating
Outcomes	Prototype of a personalized nutrition assistant / sets of needs and requirements

Table 8. Article 5 scope and objectives

Lack of regularity, lack of variety, false belief and hunger unawareness were identified as the main barriers to healthy nutrition; whereas monitoring, education, empowerment and practicality were identified as the main needs and requirements for nutrition behavior change. These findings allowed us to suggest the appropriate BCTs to be used in our systems, namely,

self-monitoring, personalized visual feedback, goal setting, self-awareness and personalized education. Finally, these BCTs were translated into a set of components and functionalities that built up to construct our final solution: MiranaBot. Table 8. summarizes Article 3 scope and objectives.

1.4. Thesis significance and intended contributions

The body of research conducted within this dissertation resulted in several research contributions that are of interest to the greater Human Computer Interaction, Digital Health, Conversational AI, Behavioral sciences, Nutrition and Physical Rehabilitation communities. Drawing from experiences, first, with designing, implementing, conducting and evaluating qualitative and clinical experiments involving self-monitoring technologies; second, with theoretical investigations on behavioral theory and strategies, third on designing and developing and evaluating digital health applications, I provide a set of guidelines, recommendations and insights to help future designers of self-monitoring technologies for chronically-ill and elderly patients. The specific intended contributions of this dissertation are summarized as follows:

Acceptance and adoption of self-monitoring technologies: qualitative studies with chronically-ill patients (18 participants, 7 months) and elderly living at home (20 participants, 6 weeks), using a wearable tracker were conducted. The first contribution gives insights on understand acceptance and adoption in the long term of self-monitoring technologies by chronically-ill patients. The second contribution emphasizes the evolution of this population's expectations of the device over the long term. The third contribution consists of a set of guidelines to design wearable health monitors for obese and diabetic patient that are engaging enough to effect long-term change. The fourth contribution shares older adults' attitudes towards increasing physical activity as well as their readiness towards tracking technologies. The last contribution in this section presents a set of opportunities and challenges for designing self-monitoring technologies for healthy ageing.

Social incentives to motivate chronically-ill patients: a qualitative study with chronically-ill patients (16 participants, 8 weeks) was conducted. The main contribution of this part are insights into the role of cooperation and competition in engaging obese and diabetic patient to be more active.

Gamification for elderly rehabilitation: a non-inferiority 6-week randomized clinical trial with 57 elderly patients with musculoskeletal issues was conducted. The first contribution is the clinical validity of a serious gamed-based as an alternative for the rehabilitation of patient with musculoskeletal issues. The second contribution shed lights on the benefits of this intervention into engaging and motivating the elderly patient at the hospital and at home. The third contribution shares the complexities associated with adopting new technologies in clinical practice.

Conversational agent for chronically-ill and elderly patients to promote healthy eating: observational study and co-designing session with 13 participants over a week was conducted.

The contribution consists of a set of needs and requirements for self-monitoring technology to promote healthy eating for chronically-ill patients. The conception and development methodology are also shared.

1.5. Structure of the thesis document

This document is organized into eight chapters including this one. As this dissertation is based on five articles and one poster, the following chapters two to seven are each devoted to one article and poster. Each article is self-contained; however, the general background of some articles may overlap. After that, chapter eight summarizes the general findings and contributions of this dissertation and presents several design guidelines towards the design of self-monitoring technologies for chronically-ill and elderly patients to promote healthy behavior in the long term. The chapter concludes with limitations of this thesis and future work.

1.6. References

[1] Institute for Health Metrics and Evaluation. (2019). Data Visualizations. http://www.healthdata.org/results/data-visualizations Accessed 25 May 2019.

[2] Institute for Health Metrics and Evaluation. (2019). Switzerland. http://www.healthdata.org/Switzerland Accessed 25 May 2019.

[3] World Health Organization. (2011). Global Status Report on Noncommunicable Diseases. http://www.who.int/nmh /publications/ncd_report2010/en Accessed 25 May 2019.

[4] Björklund, P., Fändriks, L. (2019). The pros and cons of gastric bypass surgery – The role of the Roux-limb. *Best Practice & Research Clinical Gastroenterology.* 101638. ISSN 1521-6918. https://doi.org/10.1016/j.bpg.2019.101638

[5] Polivy, J. (1996). Psychological Consequences of Food Restriction. *Journal of the American Dietetic Association.* Volume 96, Issue 6. 589-592. ISSN 0002-8223. https://doi.org/10.1016/S0002-8223(96)00161-7

[6] Pan, X-R., Li, G-W., Hu, Y-H., Wang, J-X., Yang, W-Y., An, Z-X., et al. (1997). Effects of diet and exercise in preventing NIDDM in people with impaired glucose tolerance. The Da Qing IGT and Diabetes Study. *Diabetes Care*; 20: 537-544. https://doi.org/10.2337/diacare.20.4.537

[7] World Health Organization. 10 facts on ageing and health. (2017). https://www.who.int/features/factfiles/ageing/en/ Accessed 02 September 2019.

[8] Wiles, J.L., Leibing, A., Guberman, N., Reeve, J., Allen, R.E. (2012). The meaning of "aging in place" to older people. *Gerontologist*. Jun;52(3):357-366. https://doi.org/10.1093/geront/gnr098

[9] Bacsu, J.R., Jeffery, B., Johnson, S., Martz, D., Novik, N., & Abonyi, S. (2012). Healthy aging in place: supporting rural seniors' health needs. *Online J Rural Nurs Health Care.* 12(2):77-87. https://doi.org/10.14574/ojrnhc.v12i2.52

[10] Mattison, M. Hospital management of older adults. (2019). https://www.uptodate.com/contents/hospital-management-of-older-adults/print Accessed 02 September 2019.

[11] Minton, E.A., & Khale, L.R. (2014). Belief Systems, Religion, and Behavioral Economics: Marketing in Multicultural Environments. *New York: Business Expert Press LLC*. ISBN 978-1-60649-704-3.

[12] Bergner, R.M. (2011). What is behavior? And so what? *New Ideas in Psychology*, Volume 29, Issue 2. Pages 147-155. ISSN 0732-118X. https://doi.org/10.1016/j.newideapsych.2010.08.001

[13] Levitis, D., William Z. L., Glenn F. (2009). Behavioral biologists do not agree on what constitutes behaviour. *Animal Behaviour.* 78: 103–10. https://doi.org/10.1016/j.anbehav.2009.03.018

[14] Nock, M. K., & Kurtz, S. M. (2005). Direct behavioral observation in school settings: Bringing science to practice. *Cognitive and Behavioral Practice*, 12, 359–370. https://doi.org/10.1016/S1077-7229(05)80058-6

[15] Zimbardo & Weber. (1994). Psychology Third Edition. *HarperCollins College.* ISBN-13: 978-0065013405.

[16] Duff, O.M., Walsh, D.M., Furlong, B.A., O'Connor, N.E., Moran, K.A., Woods, C.B. (2017). Behavior Change Techniques in Physical Activity eHealth Interventions for People With Cardiovascular Disease: Systematic Review. *J Med Internet Res*;19(8):e281. https://doi.org/10.2196/jmir.7782

[17] Prochaska, J. & Velicer, W. (1997). The Transtheoretical Model of Health Behavior Change. *American journal of health promotion*. 12. 38-48. https://doi.org/10.4278/0890-1171-12.1.38

[18] Michie, S., van Stralen, M. M., & West, R. (2011). The behaviour change wheel: a new method for characterising and designing behaviour change interventions. *Implementation science*. 6, 42. https://doi.org/10.1186/1748-5908-6-42

[19] Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes.*Volume 50, Issue 2, 179-211, ISSN 0749-5978. https://doi.org/10.1016/0749-5978(91)90020-T

[20] Fogg, B.J. (2009). A behavior model for persuasive design. *Proceedings of the 4th International Conference on Persuasive Technology.* https://doi.org/10.1145/1541948.1541999

[21] Bozarth, M.A. (1994). Pleasure systems in the brain. In *Warburton D.M. (ed.), Pleasure: The politics and the reality*. 5-14.

[22] Berridge, K. C., & Kringelbach, M. L. (2015). Pleasure systems in the brain. *Neuron*, 86(3), 646–664. https://doi.org/10.1016/j.neuron.2015.02.018

[23] Marks, L. E. (2011). Chapter 2: A Brief History of Sensation and Reward, Neurobiology of Sensation and Reward. *CRC Press Taylor & Francis*.

[24] Brockner, J., & Higgins, E.T. (2001). Regulatory Focus Theory: Implications for the study of emotions at work. *Organizational Behavior and Human Decision Process*, 61 (1), 35-66. http://dx.doi.org/10.1006/obhd.2001.2972

[25] Higgins, E. (1997). Beyond pleasure and pain. *American Psychologist*, 52(12), 1280–1300. https://doi.org/10.1037//0003-066X.52.12.1280

[26] Ryan, R.M., & Deci, E.L. (2000). Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being. American Psychologist. 55. 68-78. https://doi.org/10.1037/0003-066X.55.1.68

[27] Prochaska, J.O., Velicer, W.F. (1997). The transtheoretical model of health behavior change. *Am J Health Promot*. 38-48. https://doi.org/10.4278/0890-1171-12.1.38

[28] Michie, S. (2012). Which behaviour change approach should I choose? An introduction to the Behaviour Change Wheel. *University College London, UK*. https://docplayer.net/104989-Which-behaviour-change-approach-should-i-choose-an-introduction-to-the-behaviour-change-wheel-susan-michie.html Accessed 02 September 2019.

[29] Kahneman, D. (2011). Thinking, fast and slow. *New York : Farrar, Straus And Giroux*. https://doi.org/10.1007/s00362-013-0533-y

[30] Michie, S., Ashford, S., Sniehotta, F., Dombrowski, S.U., Bishop, A. & French, D.P. (2011). A refined taxonomy of behaviour change techniques to help people change their physical activity and healthy eating behaviours: The CALO-RE taxonomy. *Psychology & Health*, 26:11, 1479-1498. https://doi.org/10.1080/08870446.2010.540664

[31] Duff, O.M., Walsh, D.M., Furlong, B.A., O'Connor, N.E., Moran, K.A., & Woods, C.B. (2017). Behavior Change Techniques in Physical Activity eHealth Interventions for People With Cardiovascular Disease: Systematic Review. *J Med Internet Res.* 19(8):e281. https://doi.org/10.2196/jmir.7782

[32] Payne, H.E., Moxley, V.B., & MacDonald, E. (2015). Health Behavior Theory in Physical Activity Game Apps: A Content Analysis. *JMIR Serious Games*. 3(2):e4. https://doi.org/10.2196/games.4187

[33] Hoj, T.H., Covey, E.L., Jones, A.C., Haines, A.C., Hall, P.C., Crookston, B.T., et al. (2017). How Do Apps Work? An Analysis of Physical Activity App Users' Perceptions of Behavior Change Mechanisms. *JMIR Mhealth Uhealth.* 5(8):e114. https://doi.org/10.2196/mhealth.7206

[34] Sawesi, S., Rashrash, M., Phalakornkule, K., Carpenter, J.S., Jones, J.F. (2016). The Impact of Information Technology on Patient Engagement and Health Behavior Change: A Systematic Review of the Literature. *JMIR Med Inform.* 4(1):e1. https://doi.org/10.2196/medinform.4514

[35] Marlatt, G.A. and George, W.H. (1998). Relapse prevention and the maintenance of optimal health. In S.A. Shumaker, E.B. Schron, J.K. Ockene and W.L. McBee, eds., The handbook of health behavior change. Springer, New York, 1998, 33-58.

[36] Moore, S.M., Dolansky, M.A., Ruland, C.M., Pashkow, F.J., & Blackburn, G.G. (2003). Predictors of women's exercise maintenance after cardiac rehabilitation. *Journal of Cardiopulmonary Rehabilitation.* 23(1): 40-49. https://doi.org/10.1097/00008483-200301000-00008

[37] Witkiewitz, K. and Marlatt, G.A. (2004). Relapse prevention for alcohol and drug problems: that was Zen, this is Tao. *The American Psychologist.* 59(4), 224-235. https://doi.org/10.1037/0003-066X.59.4.224

[38] Logue, A.W. Self-control and health behavior. (2000). In Bickel, W.K. & Vuchinich, R.E. eds., Reframing health behavior change with behavioral economics. *Lawrence Erlbaum Associates*, Inc., Mahwah, NJ, 2000, 167-192.

[39] Muraven, M. & Baumeister, R.F. (2000). Self-regulation and depletion of limited resources: Does self-control resemble a muscle? *Psychological Bulletin.* 126(2). 247-259. https://doi.org/10.1037/0033-2909.126.2.247

[40] Muraven, M., Tice, D.M., & Baumeister, R.F. (1998). Self-control as limited resource: Regulatory depletion patterns. *Journal of Personality and Social Psychology.* 74(3), 774-789. https://doi.org/10.1037//0022-3514.74.3.774

[41] Deci, E.L. & Ryan, R.M. (2000). The "What" and "Why" of Goal Pursuits: Human Needs and the Self-Determination of Behavior. *Psychological Inquiry: An International Journal for the Advancement of Psychological Theory.* 11(4), 227-268. http://dx.doi.org/10.1207/S15327965PLI1104_01

[42] Gardner, B. & Rebar, A. (2019). Habit Formation and Behavior Change. *Psychology*. http://dx.doi.org/10.1093/obo/9780199828340-0232

[43] Prochaska, J.O., Johnson, S., & Lee, P. (1998). The transtheoretical model of behavior change. In S.A. Shumaker, E.B. Schron, J.K. Ockene & W.L. McBee, eds., *The handbook of health behavior change.* Springer, New York, 59-84.

[44] Radtke, K.L. (1989). Exercise compliance in cardiac rehabilitation. *Rehabilitation Nursing: The Official Journal of the Association of Rehabilitation Nurses*. 14(4), 182-186, 195. https://doi.org/10.1002/j.2048-7940.1989.tb01092.x

[45] Morris, J. & Macgillivray, S. & MacFarlane, S. (2014). Interventions to Promote Long-Term Participation in Physical Activity After Stroke: A Systematic Review of the Literature. *Archives of Physical Medicine and Rehabilitation.* 95. https://doi.org/10.1016/j.apmr.2013.12.016

[46] Fjeldsoe, B., Neuhaus, M., Winkler, E., & Eakin, E. (2011). Systematic review of maintenance of behavior change following physical activity and dietary interventions. *Health Psychology*. 30(1), 99–109. https://doi.org/10.1037/a0021974

[47] Gardner, B. (2015). A review and analysis of the use of "habit" in understanding, predicting and influencing health-related behavior. *Health Psychology Review*. 9(3), 277–295. https://doi.org/10.1080/17437199.2013.876238

[48] Neal, D. T., Wood, W., Labrecque, J. S., & Lally, P. (2012). How do habits guide behavior? Perceived and actual triggers of habits in daily life. *Journal of Experimental Social Psychology*. 48(2), 492–498. https://doi.org/10.1016/j.jesp.2011.10.011

[49] Wood, W., Labrecque, J. S., Lin, P.-Y., & Rünger, D. (2014). Habits in dual process models. In J. W. Sherman, B. Gawronski, & Y. Trope (Eds.), Dual process theories of the social mind (pp. 371–385). New York, NY: Guilford Press.

[50] Lally, P., Van Jaarsveld, C. H., Potts, H. W., & Wardle, J. (2010). How are habits formed: Modelling habit formation in the real world. *European Journal of Social Psychology*. 40(6), 998–1009. http://dx.doi.org/10.1002/ejsp.674

[51] Fournier, M., d'Arripe-Longueville, F., Rovere, C., Easthope, C. S., Schwabe, L., El Methni,J., et al. (2017). Effects of circadian cortisol on the development of a health habit. *Health Psychology.* 36(11), 1059–1064. https://doi.org/10.1037/hea0000510

[52] Gardner, B., Abraham, C., Lally, P., & de Bruijn, G.-J. (2012). Towards parsimony in habit measurement: Testing the convergent and predictive validity of an automaticity subscale of the self-report habit index. *International Journal of Behavioral Nutrition and Physical Activity*. 9(1), 102–113. https://psycnet.apa.org/doi/10.1186/1479-5868-9-102

[53] Tatara, N., Arsand, E., Skrøvseth, S. O., & Hartvigsen, G. (2013). Long-term engagement with a mobile self-management system for people with type 2 diabetes. *JMIR mHealth and uHealth.* 1(1). https://doi.org/10.2196/mhealth.2432

[54] Schwanda, V., Ibara, S., Reynolds, S., & Cosley, D. (2011). Side Effects and 'Gateway' Tools: Advocating a Broader Look at Evaluating Persuasive Systems. *In Proc. CHI'11*. 487-496. https://doi.org/10.1145/2556288.2557383

[55] Fritz, T., Huang, E.M., Murphy, G.C., & Zimmermann, T. (2014). Persuasive technology in the real world: a study of long-term use of activity sensing devices for fitness. *In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14).* 487-496. https://doi.org/10.1145/2556288.2557383

[56] Cadmus-Bertram, L., Marcus, B.H., Patterson, R.E., Parker, B.A., & Morey, B.L. (2015). Use of the Fitbit to Measure Adherence to a Physical Activity Intervention Among Overweight or Obese, Postmenopausal Women: Self-Monitoring Trajectory During 16 Weeks. *JMIR Mhealth Uhealth*. 3(4):e96. https://doi.org/10.2196/mhealth.4229

[57] Klasnja, P., Consolvo, S. & Pratt, W. (2011). How to evaluate technologies for health behavior change in HCI research. *Conference on Human Factors in Computing Systems - Proceedings.* 3063-3072. https://doi.org/10.1145/1978942.1979396

[58] Nelson, R. O., & Hayes, S. C. (1981). Theoretical Explanations for Reactivity in Self-Monitoring. *Behavior Modification*. 5(1), 3–14. https://doi.org/10.1177/014544558151001

[59] Moore, B.A & Adams, A. (2008). Exercise as an adjunctive evidence-based treatment. In Evidence-Based Adjunctive Treatments. *Practical Resources for the Mental Health Professional.* 161-175. https://doi.org/10.1016/B978-012088520-6.50009-3

[60] Sanders, J.P., Loveday, A., Pearson, N., Edwardson, C., Yates, T., Biddle, S.J. et al. (2016). Devices for Self-Monitoring Sedentary Time or Physical Activity: A Scoping Review. *J Med Internet Res.* 18(5):e90. https://doi.org/10.2196/jmir.5373

[61] Brickwood, K.J., Watson, G., O'Brien, J., Williams, A.D. (2019). Consumer-Based Wearable Activity Trackers Increase Physical Activity Participation: Systematic Review and Meta-Analysis. *JMIR Mhealth Uhealth.* 7(4):e11819. https://doi.org/10.2196/11819

[62] Lyons, E.J., Swartz, M.C., Lewis, Z.H., Martinez, E., & Jennings, K. (2017). Feasibility and Acceptability of a Wearable Technology Physical Activity Intervention With Telephone Counseling for Mid-Aged and Older Adults: A Randomized Controlled Pilot Trial. *JMIR Mhealth Uhealth.* 5(3):e28. https://doi.org/10.2196/mhealth.6967

[63] Carter, M.C., Burley, V.J., Cade, J.E. (2017). Weight Loss Associated With Different Patterns of Self-Monitoring Using the Mobile Phone App My Meal Mate. *JMIR Mhealth Uhealth.* 5(2):e8. https://doi.org/10.2196/mhealth.4520

[64] Patel, M.L., Hopkins, C.M., Brooks, T.L., & Bennett, G.G. (2019). Comparing Self-Monitoring Strategies for Weight Loss in a Smartphone App: Randomized Controlled Trial. *JMIR Mhealth Uhealth.* 7(2):e12209. https://doi.org/10.2196/12209

[65] Lyons, E., Lewis, Z., Mayrsohn, B. & Rowland, J. (2014). Behavior Change Techniques Implemented in Electronic Lifestyle Activity Monitors: A Systematic Content Analysis. *Journal of medical Internet research.* 16. e192. https://doi.org/10.2196/jmir.3469

[66] Mercer, K., Li, M., Giangregorio, L., Burns, C., & Grindrod, K. (2016). Behavior Change Techniques Present in Wearable Activity Trackers: A Critical Analysis. *JMIR Mhealth Uhealth.* 4(2):e40. https://doi.org/10.2196/mhealth.4461

[67] Ernsting, C., Dombrowski, S., Šteko, M., O'Sullivan, J., Kanzler, M., Kuhlmey, et al. (2017). Using Smartphones and Health Apps to Change and Manage Health Behaviors: A Population-Based Survey. *Journal of Medical Internet Research.* 19. e101. https://doi.org/10.2196/jmir.6838

[68] Whelan, M., Morgan, P., Sherar, L., Kingsnorth, A., Magistro, D. & Esliger, D. (2017). Brain Activation in Response to Personalized Behavioral and Physiological Feedback From Self-Monitoring Technology: Pilot Study. *Journal of Medical Internet Research.* 19. e384. https://doi.org/10.2196/jmir.8890

[69] Miyamoto, S., Henderson, S., Young, H., Pande, A. & Han, J. (2016). Tracking Health Data Is Not Enough: A Qualitative Exploration of the Role of Healthcare Partnerships and mHealth Technology to Promote Physical Activity and to Sustain Behavior Change. *JMIR mHealth and uHealth.* 4. e5. https://doi.org/10.2196/mhealth.4814

[70] Baretta, D., Perski, O. & Steca, P. (2019). Exploring users experiences of the uptake and adoption of physical activity apps: a longitudinal qualitative study. *JMIR Mhealth Uhealth.* 7(2):e11636. https://doi.org/10.2196/preprints.11636

[71] Mikulic, M. (2019). Number of mHealth apps available in the Apple App Store from 1st quarter 2015 to 3rd quarter 2019. https://www.statista.com/statistics/779910/health-apps-available-ios-worldwide Accessed on 15 September 2019.

[72] Mikulic, M. (2019). Number of mHealth apps available in the Apple App Store from 1st quarter 2015 to 3rd quarter 2019. https://www.statista.com/statistics/779919/health-apps-available-google-play-worldwide Accessed on 15 September 2019.

[73] Duncan, M., Murawski, B., Short, C.E., Rebar, A.L., Schoeppe, S., Alley, S. et al. (2017). Activity Trackers Implement Different Behavior Change Techniques for Activity, Sleep, and Sedentary Behaviors. *Interact J Med Res.* 6(2):e13. https://doi.org/10.2196/ijmr.6685

[74] Gartner Hype Cycle. https://www.gartner.com/en/research/methodologies/gartner-hype-cycle Accessed on 15 September 2019.

[75] Clark, A. M., & Clark, M. T. G. (2016). Pokémon Go and Research: Qualitative, Mixed Methods Research, and the Supercomplexity of Interventions. *International Journal of Qualitative Methods.* https://doi.org/10.1177/1609406916667765

[76] Wagner-Greene, V.R., Wotring, A.J., Castor, T., Kruger, J., Mortemore, S. et al. (2017). Pokémon GO: Healthy or Harmful? *American Journal of Public Health*. Vol. 107, N° 1, 35-36. https://doi.org/10.2105/AJPH.2016.303548

[77] Colley, A., Thebault-Spieker, J., Yilun Lin, A., Degraen, D., Fischman, B., Häkkilä, J. et al. (2017). The Geography of Pokémon GO: Beneficial and Problematic Effects on Places and Movement. *In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17).* 1179-1192. https://doi.org/10.1145/3025453.3025495

[78] Zsila, A., Orosz, G., Bőthe, B., Tóth-Király, I., Király, O., Griffiths, M. et al. (2018). An empirical study on the motivations underlying augmented reality games: The case of Pokémon Go during and after Pokémon fever. *Personality and Individual Differences.* Vol 133. 56-66. https://doi.org/10.1016/j.paid.2017.06.024

[79] Gillin, J.L. & Gillin, J.P. (1948). Cultural Sociology. https://archive.org/details/in.ernet.dli.2015.186912/page/n2 Accessed on 16 September 2019.

[80] Sasidharan. V., Payne, P., Orsega-Smith, E. & Godbey, G. (2006). Older adults' physical activity participation and perceptions of wellbeing: Examining the role of social support for leisure. *Managing Leisure.* 11(3):164-185. http://dx.doi.org/10.1080/13606710600715242

[81] Grosinger, J., Vetere, F., & Fitzpatrick, G. (2012). Agile life: addressing knowledge and social motivations for active aging. *Proceedings of the 24th Australian Computer-Human Interaction Conference.* 162-165. https://doi.org/10.1145/2414536.2414566

[82] Iacoboni, M. (2009). Imitation, Empathy, and Mirror Neurons. *Annual Review of Psychology.* 60:653-70. https://doi.org/10.1146/annurev.psych.60.110707.163604

[83] Roelof A. J. de Vries, Truong, K.P., Kwint, S., Drossaert, C.H.C, & Evers, V. (2016). Crowd-Designed Motivation: Motivational Messages for Exercise Adherence Based on Behavior Change Theory. *In Proc. CHI 2016.* 297-308. https://doi.org/10.1145/2858036.2858229

[84] Keyani, P., Hsieh, G., Mutlu, B., Easterday, M., & Forlizzi, J. (2005). DanceAlong: Supporting Positive Social Exchange and Exercise for the Elderly Through Dance. In Proc. Conference on Human Factors in Computing Systems. 1541-1544. https://doi.org/10.1145/1056808.1056961

[85] Brox, E., Luque, L.F., Evertsen, G.J., & Hernandez, J.E.G. (2011). Exergames for elderly: Social exergames to persuade seniors to increase physical activity. Pervasive Computing Technologies for Healthcare (PervasiveHealth). 546-549. https://doi.org/10.4108/icst.pervasivehealth.2011.246049

[86] Uchino, B. (2004). Social Support and Physical Health: Understanding the Health Consequences of Relationships. *New Haven, CT: Yale University Press.* https://doi.org/10.12987/yale/9780300102185.001.0001

[87] Pyae A., Luimula M., & Smed J. (2016) Pre-studies on Using Digital Games for the Elderly's Physical Activities. In: Li H., Nykänen P., Suomi R., Wickramasinghe N., Widén

G., Zhan M. (eds) Building Sustainable Health Ecosystems. *WIS 2016. Communications in Computer and Information Science.* vol 636. https://doi.org/10.1007/978-3-319-44672-1_8

[88] Salmenius-Suominen, H., Lehtovirta, M., Vepsäläinen, H., Konttinen, H. & Erkkola, M. (2016). Visual Food Diary for Social Support, Dietary Changes and Weight Loss. *Iproceedings.* 2. e38. https://doi.org/10.2196/iproc.6135

[89] Tong, H.L., Coiera, E., & Laranjo, L. (2018). Using a Mobile Social Networking App to Promote Physical Activity: A Qualitative Study of Users' Perspectives. *J Med Internet Res*. 20(12):e11439. https://doi.org/10.2196/11439

[90] Maher, C., Ferguson, M., Vandelanotte, C., Plotnikoff, R., De Bourdeaudhuij, I,, Thomas S et al. (2015). A Web-Based, Social Networking Physical Activity Intervention for Insufficiently Active Adults Delivered via Facebook App: Randomized Controlled Trial. *J Med Internet Res.* 17(7):e174. https://doi.org/10.2196/jmir.4086

[91] Manuvinakurike, R., Velice,r W.F., & Bickmore, T.W. (2014). Automated Indexing of Internet Stories for Health Behavior Change: Weight Loss Attitude Pilot Study. *J Med Internet Res.* 16(12):e285. https://doi.org/10.2196/jmir.3702

[92] Maher, C.A., Lewis, L.K., Ferrar, K., Marshall, S., De Bourdeaudhuij, I., & Vandelanotte, C. (2014). Are Health Behavior Change Interventions That Use Online Social Networks Effective? A Systematic Review. J Med Internet Res. 16(2):e40. https://doi.org/10.2196/jmir.2952

[93] Moorhead, S.A., Hazlett, D.E., Harrison, L., Carroll, J.K., Irwin, A., & Hoving, C. (2013). A New Dimension of Health Care: Systematic Review of the Uses, Benefits, and Limitations of Social Media for Health Communication. *J Med Internet Res.* 15(4):e85. https://doi.org/10.2196/jmir.1933

[94] Leahey, T. & Rosen, J. (2014). DietBet: A Web-Based Program that Uses Social Gaming and Financial Incentives to Promote Weight Loss. JMIR Serious Games. 2(1):e2. https://doi.org/10.2196/games.2987

[95] Tong, H.L., Coiera, E., Tong, W., Wang, Y., Quiroz, J.C., Martin, P., et al. (2019). Efficacy of a Mobile Social Networking Intervention in Promoting Physical Activity: Quasi-Experimental Study. *JMIR Mhealth Uhealth.* 7(3):e12181. https://doi.org/10.2196/12181

[96] Huotari, K., & Hamari, J. (2012). Defining Gamification – A Service Marketing Perspective. *Proceedings of the 16th International Academic MindTrek Conference 2012*. 17-22. https://doi.org/10.1145/2393132.2393137

[97] Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness: Defining "gamification". *Proceedings of the 15th International Academic MindTrek Conference.* 9-15. https://doi.org/10.1145/2181037.2181040

[98] Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does Gamification Work? A Literature Review of Empirical Studies on Gamification. *47th Hawaii International Conference on System Sciences (HICSS)*. https://doi.org/10.1109/HICSS.2014.377

[99] Thompson D. (2014). What serious video games can offer child obesity prevention. *JMIR serious games.* 2(2), e8. https://doi.org/10.2196/games.3480

[100] Edwards, E.A., Caton, H., Lumsden, J., Rivas, C., Steed, L., Pirunsarn, Y. et al. (2018). Creating a Theoretically Grounded, Gamified Health App: Lessons From Developing the Cigbreak Smoking Cessation Mobile Phone Game. *JMIR Serious Games.* 6(4):e10252. https://doi.org/10.2196/10252

[101] El-Hilly, A.A., Iqbal, S.S., Ahmed, M., Sherwani, Y., Muntasir, M., Siddiqui, S. et al. (2016). Game On? Smoking Cessation Through the Gamification of mHealth: A Longitudinal Qualitative Study. *JMIR Serious Games.* 4(2):e18. https://doi.org/10.2196/games.5678

[102] Meekes, W. & Stanmore, E.K. (2017). Motivational Determinants of Exergame Participation for Older People in Assisted Living Facilities: Mixed-Methods Study. *J Med Internet Res.* 19(7):e238. https://doi.org/10.2196/jmir.6841

[103] Cugelman B. (2013). Gamification: what it is and why it matters to digital health behavior change developers. *JMIR serious games.* 1(1), e3. https://doi.org/10.2196/games.3139

[104] Lister, C., West, J.H., Cannon, B., Sax, T., & Brodegard, D. (2014). Just a Fad? Gamification in Health and Fitness Apps. *JMIR Serious Games.* 2(2):e9. https://doi.org/10.2196/games.3413

[105] Skriloff, S.J.G, Gonzalez, D.C., Christensen, K.C., Bentley, L.J., & Mortensen, C.V. (2016). FitPlay Games: Increasing Exercise Motivation Through Asynchronous Social Gaming. CHI Extended Abstracts. 164-167. https://doi.org/10.1145/2851581.2890367

[106] Lin, J.J., Mamykina, L., Lindtner, S., Delajoux, G., & Strub, H.B. (2006). Fish'n'Steps: Encouraging Physical Activity with an Interactive Computer Game. *In Proc. Ubicomp 2006.* 261-278. https://doi.org/10.1007/11853565_16

[107] Consolvo, S., McDonald, D.W., Toscos, T., Chen, M.Y., Froehlich, J., Harrison, B. et al. (2008). Activity sensing in the wild: a field trial of UbiFit Garden. *In Proc. CHI 2008.* 1797-1806. https://doi.org/10.1145/1357054.1357335

[108] Rice, M.D., Sun, M.W.T., Ling, J.N.S, Foo, M.H.N., Janell, K., Sheng, S.L.W. et al. (2011). Evaluating Gesture-based Games with Healthy Older Adults on a Large Screen Display. *CM SIGGRAPH Symposium on Video Games.* 17-24. https://doi.org/10.1145/2018556.2018560

[109] Mohan, D., Farris, C., Fischhoff, B., Rosengart, M., Angus, D., Yealy, D. et al. (2017). Efficacy of educational video game versus traditional educational apps at improving physician decision making in trauma triage: Randomized controlled trial. *BMJ*. 359. j5416. https://doi.org/10.1136/bmj.j5416

[110] Söbke, H., Hauge, J. & Stefan, I. (2017). Prime Example Ingress Reframing the Pervasive Game Design Framework (PGDF). *International Journal of Serious Games.* 2384-8766. 4. https://doi.org/10.17083/ijsg.v4i2.182

[111] Southerton, C. (2014). 'Zombies, Run!': Rethinking Immersion in Light of Nontraditional Gaming Contexts. *In Transmedia Practice: A Collective Approach.* https://doi.org/10.1163/9781848882614_013

[112] Hoffmann, A., Christmann, C. A., & Bleser, G. (2017). Gamification in Stress Management Apps: A Critical App Review. *JMIR serious games*, 5(2), e13. https://doi.org/10.2196/games.7216

[113] Cassell, J., Sullivan, J., Prevost, S. & Churchill, E.F. (2000). Embodied Conversational Agents. https://mitpress.mit.edu/books/embodied-conversational-agents Accessed on 18 September 2019.

[114] Fitzpatrick, K.K., Darcy, A., & Vierhile, M. (2017). Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. *JMIR Ment Health.* 4(2):e19. https://doi.org/10.2196/mental.7785

[115] Provoost, S., Lau, H.M., Ruwaard, J,, Riper, H, (2017). Embodied Conversational Agents in Clinical Psychology: A Scoping Review. *J Med Internet Res.* 19(5):e151. https://doi.org/10.2196/jmir.6553

[116] Loveys, K., Fricchione, G., Kolappa, K., Sagar, M,, & Broadbent, E. (2019). Reducing Patient Loneliness With Artificial Agents: Design Insights From Evolutionary Neuropsychiatry. J Med Internet Res. 21(7):e13664. https://doi.org/10.2196/13664

[117] Gardiner, P, Negash, N.L., Shamekhi, A., Bickmore, T., Gergen-Barnett, K., Lestoquoy, A.S. et al. (2016). Utilization of an Embodied Conversational Agent in an Integrative Medical Group Visit for Patients with Chronic Pain and Depression. *Iproc.* 2(1):e6. https://doi.org/10.2196/iproc.6099

[118] Inkster, B., Sarda, S., & Subramanian, V. (2018). An Empathy-Driven, Conversational Artificial Intelligence Agent (Wysa) for Digital Mental Well-Being: Real-World Data Evaluation Mixed-Methods Study. *JMIR Mhealth Uhealth.* 6(11):e12106. https://doi.org/10.2196/12106

[119] Magnani, J. W., Schlusser, C. L., Kimani, E., Rollman, B. L., Paasche-Orlow, M. K., & Bickmore, T. W. (2017). The Atrial Fibrillation Health Literacy Information Technology System: Pilot Assessment. *JMIR cardio.* 1(2), e7. https://doi.org/10.2196/cardio.8543

[120] Bickmore, T.W., Utami, D., Matsuyama, R., & Paasche-Orlow, M.K. (2016). Improving Access to Online Health Information With Conversational Agents: A Randomized Controlled Experiment. *J Med Internet Res.* 18(1):e1. https://doi.org/10.2196/jmir.5239

[121] Fulmer, R., Joerin, A., Gentile, B., Lakerink, L., & Rauws, M. Using Psychological Artificial Intelligence (Tess) to Relieve Symptoms of Depression and Anxiety: Randomized Controlled Trial. *JMIR Ment Health.* 5(4):e64. https://doi.org/10.2196/mental.9782

[122] Kennedy, C.M., Powell, J., Payne, T.H., Ainsworth, J., Boyd, A., & Buchan, I. (2012). Active Assistance Technology for Health-Related Behavior Change: An Interdisciplinary Review. *J Med Internet Res.* 14(3):e80. https://doi.org/10.2196/jmir.1893

[123] Tanana, M.J., Soma, C.S., Srikumar, V., Atkins, D.C., & Imel, Z.E. (2019). Development and Evaluation of ClientBot: Patient-Like Conversational Agent to Train Basic Counseling Skills. *J Med Internet Res.* 21(7):e12529. https://doi.org/10.2196/12529

[124] Scholten, M.R., Kelders, S.M., & Van Gemert-Pijnen, J.E. (2017). Self-Guided Web-Based Interventions: Scoping Review on User Needs and the Potential of Embodied Conversational Agents to Address Them. *J Med Internet Res.* 19(11):e383. https://doi.org/10.2196/jmir.7351

[125] Chaix, B., Bibault, J.E., Pienkowski, A., Delamon, G., Guillemassé, A., Nectoux, P. et al. (2019). When Chatbots Meet Patients: One-Year Prospective Study of Conversations Between Patients With Breast Cancer and a Chatbot. *JMIR Cancer.* 5(1):e12856. https://doi.org/10.2196/12856

[126] Bott, N., Wexler, S., Drury, L., Pollak, C., Wang, V., Scher, K. et al. (2019). A Protocol-Driven, Digital Conversational Agent at the Hospital Bedside to Support Nurse Teams and Mitigate Risks of Hospitalization in Older Adults. *J Med Internet Res.* 21(10):e13440. https://doi.org/10.2196/13440

[127] Korpusik, M. & Glass, J. (2017). Spoken Language Understanding for a Nutrition Dialogue System . *IEEE Transactions on Audio, Speech, and Language Processing.* 1450 – 1461. https://doi.org/10.1109/TASLP.2017.2694699

[128] Casas, J., Mugellini, E., & Khaled, O.A. (2018). Food Diary Coaching Chatbot. *In Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (UbiComp '18).* 1676-1680. https://doi.org/10.1145/3267305.3274191

[129] Palanica, A., Flaschner, P., Thommandram, A., Li, M, & Fossat, Y. (2019). Physicians' Perceptions of Chatbots in Health Care: Cross-Sectional Web-Based Survey. *J Med Internet Res.* 21(4):e12887. https://doi.org/10.2196/12887

[130] Bickmore, T.W., Trinh, H., Olafsson, S., O'Leary, T.K., Asadi, R., Rickles, N.M. et al. (2018). Patient and Consumer Safety Risks When Using Conversational Assistants for Medical Information: An Observational Study of Siri, Alexa, and Google Assistant. *J Med Internet Res.* 20(9):e11510. https://doi.org/10.2196/11510

[131] Fiske, A., Henningsen, P., & Buyx, A. (2019). Your Robot Therapist Will See You Now: Ethical Implications of Embodied Artificial Intelligence in Psychiatry, Psychology, and Psychotherapy. *J Med Internet Res.* 21(5):e13216. https://doi.org/10.2196/13216

[132] Morris, R.R., Kouddous, K., Kshirsagar, R., & Schueller, S.M. (2018). Towards an Artificially Empathic Conversational Agent for Mental Health Applications: System Design and User Perceptions. J Med Internet Res. 20(6):e10148. https://doi.org/10.2196/10148

CHAPTER 2. CAN FITNESS TRACKERS HELP DIABETIC AND OBESE USERS MAKE AND SUSTAIN LIFESTYLE CHANGES?

This chapter has been published as:

Randriambelonoro, M., Chen, Y. & Pu, P. **"Can Fitness Trackers Help Diabetic and Obese Users Make and Sustain Lifestyle Changes?"** in Computer, vol. 50, no. 3, pp. 20-29, Mar. 2017. doi: 10.1109/MC.2017.92

Diabetic and obese users given a Fitbit fitness tracker for seven months identified the device's playfulness, practicality, persuasiveness, personalization, and privacy as the top motivators for its use. These results provide insights into designing wearable health monitors that are engaging enough to effect long-term change.

2.1. Introduction

According to a World Health Organization (WHO) report on the global status of noncommunicable diseases in 2010, diabetes and obesity are among the diseases most responsible for worldwide mortality.1 In response to the epidemic of diabetes in the past few decades, WHO has actively promoted healthy living through exercise and diet, which is less painful than surgical interventions such as gastric bypass surgery and less invasive than prescriptions that can have significant and sometimes dangerous side effects.

However, diet alone—which often involves restricting intake—can lead to a feeling of deprivation. Research on diabetes and obesity has shown that physical activity not only helps prevent diabetes and obesity but also assists in treatment. Combined with a healthy diet, regular physical activity at a certain level can control blood glucose, regulate blood pressure, prevent cardiovascular events, and elevate mood.

Unfortunately, developing ways to engage and motivate people to adopt a healthy behavior has proven more difficult than imagined, even with affordable wearable health trackers and creative companion applications. The state of the art "Empowering Users to Control Their Health" describes research to define device features that users find motivating enough to change their lifestyle.

Despite the extensive research on pervasive fitness sensors and applications and how diabetic and obese users can benefit from them, motivational techniques for using this technology remain largely unexplored. Research is focusing instead on aspects such as self-monitoring glucose levels,2 body weight, and food choices. Little work has been done to investigate how these technologies influence users' behavior and environment or how they can aid users with chronic diseases to change their lifestyle over the long term through solutions such as increased physical activity.

To address this lack of research, we conducted a study over seven months (December 2013 to June 2014) involving 18 individuals ages 36 to 73—all of whom were formally diagnosed with diabetes, obesity, or both. We gave each participant the Fitbit One fitness tracker

(www.fitbit.com/us/one) and interviewed them before, during, and after the study to identify what they expected from the tracker, how the tracker influenced their lifestyle, and what was required for the technology to change their long-term health decisions. The seven-month study is an extension of a previous one-month study of behavioral change.3

Our second study gave us a deeper understanding of how fitness trackers can motivate diabetic and obese users to engage in more physical activity over the long term. We identified a set of requirements, which we evolved into recommendations for the design of wearable health trackers that aim to effect long-term lifestyle changes in their users' behavior.

2.2. Related work: Empowering Users to Control Their Health

Researchers have extensively studied user needs and research prototypes for wearable devices,1,2 and they have investigated how users employ their devices and evaluate them.3 Results show that these devices have tremendous potential to equip diabetic and obese users to manage their own health. Eric Topol's book The Patient Will See You Now effectively develops the paradigm of patient empowerment in which patients are no longer followers but the main actors in managing their health.4 Studies about proactive health management systems that empower chronically ill patients with self-care are on the rise, as many individuals realize noncommunicable diseases like diabetes can be prevented and managed by adopting a healthy lifestyle.

2.2.1. Proactive decision making

Many researchers have pointed out the need for patients to actively learn about their diseases and make healthy decisions.5,6 One ethnographic study attempted to understand current diabetes management practices and identified that helping users to associate glucose levels with their lifestyle is an essential design requirement.7 In another study, researchers designed the Continuous Health Awareness Program, which lets users log glucose levels and complement them with a report of activities, diet, medication, and emotional states.5 Another group also extended the value of glucose levels and supplemented it with photographs of users' meals, exercise routines, and events that made them happy or sad, aiming to help diabetics reflect on past actions and make decisions about future ones.6

2.2.2. Motivation challenges

Motivating and engaging users with chronic diseases to actively monitor and change their lifestyle is a promising yet challenging area in patient self-management.2,5 One study showed the difficultly of introducing exercise habits into some diabetics' sedentary lifestyles, and noted that the participants relapsed in a short time.6 Another group proposed persuasive technologies to address this issue, but provided no concrete applications or empirical studies.5 Meanwhile, technologies that help users monitor their lifestyle are on the rise.7 Researchers have been actively designing technologies that motivate physical activities to prevent obesity,2 but few studies have evaluated the impact of these technologies on obese and diabetic patients. The study most relevant to our work examined the experience of normal users who had already adopted activity-monitoring devices for a long term (3 to 54 months).8 However, the study targeted the general population rather than patients.

2.3. Study methodology

We recruited our 18 users from a primary healthcare center in a large city in Switzerland. Our study was endorsed by the center's ethical committee to protect patient safety and privacy. Users included 12 diabetics, 5 of whom were also obese, and 6 obese and overweight patients. Of the 18 users, 7 were male and 11 were female.

Over the seven months, we held three interviews with each user: an initial hour-long interview before users began wearing the Fitbit One and two half-hour interviews—one a month after device use and one at the end of the study. The interviews were semi-structured in that we had prepared topics for questions from a sensor-adoption model,4 but we also encouraged users to share their expectations for the device and experience with technology in general (first interview) and then to describe their reactions to the tracker and show us how they used it (second and third interviews). At least two researchers were present for each interview: one to interact with the users and the other to observe and take notes.

We chose to use the Fitbit One tracker because it is unobtrusive and users can easily attach it to their clothing. It also collects relatively comprehensive physical activity data, including number of steps, number of stairs, calories burned, and distance traveled in kilometers. In addition, Fitbit awards badges for walking milestones. Users can also manually log their diet, weight, and glucose levels and access all their data from accounts on both the Fitbit website and mobile app. As an incentive to participate in the study, we told users they could quit the study at any time and still keep the device.

2.3.1. Interviews

The goal of the first interview was to understand users' daily life, disease management, and expectations for a fitness tracker. All but two users had smartphones, which they used mainly for calls and texting; seven had experience with pedometers; and two had previously used mobile fitness applications but had stopped employing them because of accuracy or usability problems. In the two remaining interviews, our focus was on device use. Questions were designed to assess users' experience and their perspectives on the tracker's effectiveness. The last interview was to ascertain if users had continued to employ the tracker and whether it had prompted them to change their lifestyle or environment.

2.3.2. Data analysis

We audio-recorded all interviews (total of 1,963 min), transcribed them into English, and analyzed the resulting 268,931 words according to the grounded-theory approach. After reading all the transcripts, two researchers generated a starting list of codes based on the research questions and the patterns that emerged from the data. The researchers then returned to the data and conducted a systematic axial coding to identify emergent themes. After several coding iterations, the researchers identified and categorized the themes and reached a consensus mediated by a third researcher.

2.4. Lifestyle evolution

Overall, users made positive lifestyle changes. By the end of the study, onlytwo users had stopped employing the device. As a result, the data reported for seven months is for 16 users, although we still have interview feedback from the two users who dropped out. For simplicity and anonymity, we labeled users P1 to P18.

2.4.1. Before the study

Most users entered the study physically inactive, yet were aware of the benefits of physical activity in maintaining proper glucose levels and body weight. They reported that they did not perform routine or intense physical exercises on their own, only those that the hospital organized. Half had participated in exercise programs for diabetic and obese patients organized by the hospital, usually in three one-hour sessions weekly that lasted for three months. Users reported that regular training programs had benefits such as making them feel accomplished and more confident. For example, P3 said, “With the monitoring, I had to take a pace and did everything regularly. I liked it, and I think it was fun, but it was just for three months.”

Users also liked the program because it was an external motivator to push them to exercise with professional and clinical support. However, none of them continued exercising regularly after the program ended. P5 said, “I want to do more physical exercises, but I am always postponing it. I want everything to be perfect and at a steady pace. I want to do it right.”

Some users also reported physical restrictions from their disease and its complications. Three had gastric bypass surgery, which gave rise to many side effects such as nausea, trauma, and slow recovery from anesthesia. Therefore, it seems that the users had not integrated physical exercise into their daily lives and tended to rely on professional support.

Overall, those who prioritized their time sacrificed physical activity to other duties, reporting that their busy lifestyle prevented them from exercising. P4 did not have time because she was too busy with her work: “I often drive because I’m responsible for many associations, and I feel like I’m wasting my time walking. Although it’s not far, I still drive because I always carry many files.” P18 claimed to be too busy with her family: “When I’m with my husband and my son, I can’t exercise even if I want to.”

2.4.2. After a month

We noted changes in activity awareness, motivation to participate in activity, and a stronger sense of social connection.

Activity awareness. Users initially reported that monitoring their daily steps and stairs helped them be more aware of their inactivity. P5 said, “I realized I don’t have to go through stairs at all in my everyday habits, so I started to take the stairs from the underground parking to the mall.” Thirteen participants had chosen opportunities to exercise more each day. In addition to gradually taking the elevator and escalator less often, some had deliberately created opportunities to walk more, parking their car farther away or exiting at a bus stop before their regular one.

Although these actions taken alone might seem trivial, their repetition over time shows the potential to create a more meaningful long-term change. The tracker prompted them to rethink and redefine physical activity in terms of number of steps, which they could easily integrate into their daily routine. P8, for example, was surprised by the number of steps taken while doing his weekly job of monitoring cars in a parking lot.

Motivation. As users felt the benefits of integrating physical activity on their health and wellness, they were more motivated to be active. P10 reported, "As I walk I feel like my glucose level is more stable, my health is better." P18 said, "Instead of just sitting, I do many things as long as I can increase my number of steps. After I send my son to school, I walk in the city. I have lost 1 kg since I started using Fitbit."

Social connection. Wearing the device gave users a sense of pride when showing their tracker to others. They felt they were not only following the technological trends, but were also adopting a healthy behavior in the form of more physical activity. P4 reported, "I showed it to my parents and my children. Now, I'm a cool mom."

Moreover, they were able to influence others by sharing their experience. Some users related their study participation to their diabetic association and asked for financial help to distribute it to all the members. Some participants influenced their family, friends, and colleagues to be more physically active. P3 said, "Just by seeing how inactive I am with my number of steps, my colleague who is always with me could relate to it and started to be aware of being too sedentary as well." P2 reported, "When I was with my daughter, I told her to take the stairs and not the elevator."

2.4.3. After seven months

After seven months, the 16 users still wearing the device reported visible health and lifestyle changes. At this point, physical activity was translating to visible results such as weight loss, which reinforced activity awareness, although some users reported losing the device or forgetting to wear it. Social connection also remained.

Activity awareness. Many users became aware of the benefits of physical activity in lasting change. For example, P18 lost an additional 3 kg (on top of her loss of 1 kg at the one-month milestone), and intentionally changed her habits to be consistent with an active lifestyle. This is a significant improvement for obese patients. As one researcher in a separate study noted, "the achievement of weight normalization is often unrealistic and does not have to be the ultimate goal of a weight-reduction strategy. Moderate weight loss can have substantial health benefits."5

P13, who was both obese and diabetic, reached an average of 70,000 steps per week, which he had considered challenging in the past. He proudly shared his achievements with his friends when he reached a total distance of 1,000 km of walking. P8 and P16 began participating in a Nordic Track walking program every Monday afternoon and were continuing to do so as of late January 2017.

Table 1 shows the average steps and stairs taken by the remaining 16 users at two points in the study.

Of the two users who stopped wearing the device, one became so busy that he forgot to wear the tracker after the first two weeks. The other user reported being more interested in her average caloric intake and average steps. Once she had that information, she did not want to continue wearing the device.

Social connection. Some participants reported even greater impact on their social circles. P8 got financing through his diabetes association to distribute the device to a larger population. P11, who works as a secretary, also had the opportunity to influence others. Her office was planning to buy a coffee machine because they had to walk a long distance to the cafeteria. After an extended period with the Fitbit tracker, this user realized that she had been too sedentary and began to get up more often to get water. When her coworkers and boss noticed her activity, she explained her need to exercise. Ultimately, a decision was made to refrain from buying the coffee machine and continue walking to the cafeteria. We were surprised by this spreading fitness effect: users altered not only their behavior but also influenced those around them to do the same.

TABLE 1. Average activity reported from 16 users.

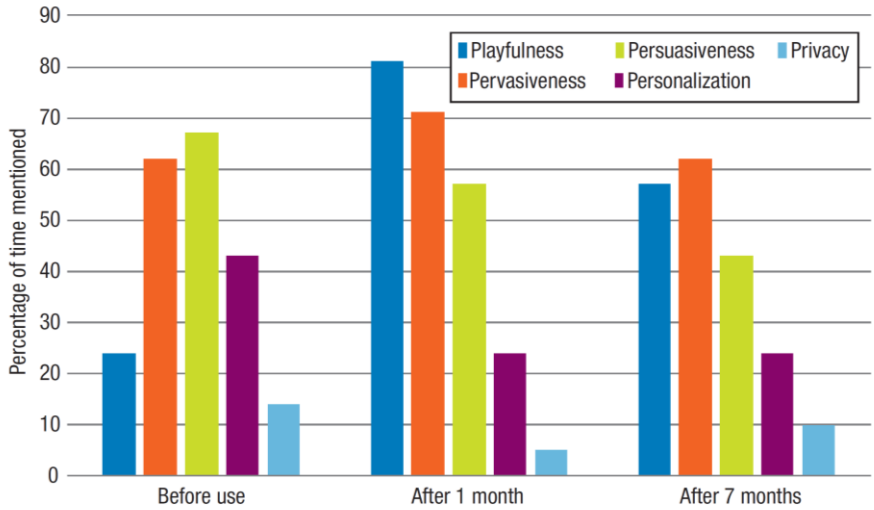
Activity	After 1 month	After 2 to 6 months
No. of steps	7,852.72	9,002.30
No. of stairs	19.23	22.06

TABLE 2. Users' requirements for activity-monitoring technologies.

Requirements	Explanations	No. of users mentioning requirement		
		Before use	After 1 month	After 7 months
Playfulness	Metaphors, fun messages, motivating messages, colors	5	15	10
Pervasiveness and practicality	Ease of integration into daily life, small, lightweight, simple information	11	13	11
Persuasiveness	Notifications, rewards, warnings, comparisons, suggestions	12	10	9
Personalization	Personalized goal, information relevant to disease, recommendations	9	5	5
Privacy	Control over data sharing	3	1	2

2.5. Users' expectations and requirements

On the basis of interviews and observations, we identified five requirements that users have for a wearable activity monitor: persuasiveness, pervasiveness and practicality, personalization, playfulness, and privacy. Table 2 summarizes these properties along with the number of participants who mentioned them. Figure 1 shows how often users mentioned each requirement as a percentage of the number of times all requirements were mentioned.



2.5.1. Before the study

As Table 2 shows, users most often cited persuasiveness followed by pervasiveness and practicality, and then by personalization. Playfulness and privacy were less desirable overall.

Persuasiveness. Most users (12) wanted a tool that nudges them to exercise. Some complained about applications that only display activity data without analyzing it or providing some user interaction. They expected a coaching function such as persuasion, reminders, rewards, warnings, comparisons with previous days, visualizations, and so on. P10 noted, "It would be nice if it could send me motivating messages from time to time. We need something that stays around and reminds us what we need to do. The problem is that we forget easily."

Pervasiveness and practicality. This requirement was easily the next in demand, with 11 users mentioning that the device should be integrated into their life seamlessly and that information should be easy to understand. Users in their 60s were less curious about new data, preferring that the technology be easy to use and practical.

Information simplicity also influenced users' willingness to adopt the technology. P16 said, "It's good to have some value that we understand easily and not just some meaningless number." Visualization techniques are a possible solution. For example, P18 expected the results to be displayed in figures that are easy to interpret.

Personalization. Nine users expected advice and information related to their disease. As P14 reported, “I want it to give me some practical advice on what I should do with my disease and not just show the numbers.”

Because diabetes and obesity represent major risk factors for a number of other chronic diseases like high blood pressure, heart disease, and high blood cholesterol, users also expected the device to give them information such as heart rate and glucose level.

Playfulness. Fewer than a third of the users (five) wanted a wearable device that is fun and entertaining. Some users were motivated by things they like to do, instead of things they have to do. Having a chronic disease already filled their life with constraints and stress, so a device that could lighten that burden would be more desirable. P2 expected some music functions to encourage physical activities, and P6 noted, “It has to be fun and playful. We already have enough problems every day, so it should be entertaining.”

Privacy. Three users were concerned about confidentiality, mainly that others could guess their activities unless they controlled data sharing. P7 said, “I am worried that someone may see what I did. I should be able to control who can see my profile and my performance.”

2.5.2. After a month

The users’ requirements ranking changed drastically a month into the study, with playfulness now being the most important requirement for continued use, followed by pervasiveness and practicality and then persuasiveness. Personalization and privacy continued to be in low demand.

Playfulness. Of the 16 remaining users, all but one considered the fun element the most important motivating property. Of these, 10 users were amused by the device’s messages. P4 said, “I love the message on the Fitbit screen. One day when I tried to hang it on my clothes, it said ‘Go for a short walk?’ and I answered, ‘No, I still have to eat.’ That made me laugh out loud.”

Eleven users liked the flower on the Fitbit screen, which shows at a glance whether the user has been active because it gets taller as the user walks. P4 said, “I look at the screen at least twice per day, in the morning and in the evening, and also when I’m back from a walk. I want to see if the flower is growing.”

Some users were entertained by the color coding of activity levels on the Fitbit dashboard. P14 said, “When I reach the daily goal, everything suddenly turns from orange to green; when my performance is bad, everything is red. The color system is really interesting.”

Pervasiveness and practicality. After wearing the device for an extended time, 13 users mentioned that Fitbit is pervasive and easy to use and that wearing it and checking steps were convenient. P4 said, “It’s handy. In the morning, I fix it to my trousers and at night I check how much I did during the day.”

Users also appreciated that the tracker is unobtrusive, being small, light, and easy to wear in a way that is not obvious and potentially embarrassing. They did not perceive wearing the device as a burden. However, the flip side of unobtrusiveness is that the device is easy to

misplace. P10 noted, “It is too small. For an old person like me, I forget everything and lose it easily. If we can call it like a phone, it would be easy to locate it. I am frustrated when I cannot find it.”

Fitbit helped users integrate physical activities into their daily life rather than having to set aside a chunk of time dedicated to physical activity. P8 reported, “I was checking 29 cars for my association the other day and had three times more steps [than usual]. Before using the sensor, I did not know that this job would lead to so much physical activity.”

Users tended to avoid functions like sleep tracking that involved more effort and were less convenient. Most did not want to wear the wristband with the sensor inside while they slept. P4 said, “I just used the sensor to get fitness data. I didn’t use the sleep tracking because I didn’t understand how to use it.” Additionally, even though the Fitbit application provides functions for logging diet, weight, glucose levels, and blood pressure, users rarely used them because of the extra effort required.

Finally, for many users, “ease of use” translated to having a minimal IT background, and even with that assumption, a few of the older users encountered technical problems while using the application. The spouse of P7, who joined the user interview, reported, “We did not know how to use the dashboard on the computer. What we saw was not the same as that on the Fitbit screen. I was frustrated with the computer, and he was frustrated with the Fitbit. We are old and it is difficult.” Some participants also encountered various technical problems but managed to use the essential functionality.

Persuasiveness. Users appreciated Fitbit’s monitoring and features that nudged them to exercise. They also wanted congratulations and rewards when they were doing well by their own standards. P2 shared, “Riding a bicycle for 30 minutes is not easy for me but when I saw that I had been active for 28 minutes, I told myself ‘I am almost there, and I won’t die if I reach 29.’ If there are some sweet messages like that, it can motivate me to stay till the end.”

Personalization. Only five users expected the device to adaptively personalize fitness goals for them. Most users expected to start with an achievable goal and then increase it steadily. However, some were overwhelmed by the Fitbit’s “demands.” P6 said, “I was disappointed because I thought every time I reach 5,000 steps, it will congratulate me. 5,000 was already very difficult, but it asks for 10,000 steps. It is almost discouraging. If the goal is increased little by little based on my performance, I will be more motivated.”

Users wanted accurate monitoring and comparisons with past performance. While accuracy and self-comparison are important for healthy individuals as well, such information is particularly valuable for patients to associate with their glucose levels and body weight and learn about health practices by themselves. For example, P9 mentioned that knowing his glucose level while exercising is important so that he can determine if he is in a hypoglycemic state and needs to eat something sweet.

Privacy. Only one participant mentioned privacy, being concerned that insurance companies might obtain her health-related data and adjust her payment on the basis of it.

2.5.3. After the study

According to the transtheoretical model of behavior change,6 users went through all the phases, from precontemplation—in which they were aware of physical activity's benefits but were not motivated to be more active—to maintenance, in which they are engaged in performing the new behavior over the long-term. Our results are consistent with other researchers who showed that even though goals and practices varied along the study, participants still derived value and motivation from the technology.7 However, these researchers did not provide concrete insights into how motivation and requirements evolve.

As Table 2 shows, our users' expectations and requirements changed across the seven months. By the end of the study, the top requirement was pervasiveness and practicality followed by playfulness and then persuasiveness. Personalization and privacy remained the least mentioned.

Because the end of the study gave us our final set of insights, we opted to focus on design implications in presenting our seven-month results.

2.6. Device design recommendations

Before wearing the Fitbit One tracker, users were more concerned about whether such technology could nudge them and provide personalized healthcare information. They also expected the device to be pervasive, practical, and easy to use. However, after wearing it for one month, users became more impressed with its playfulness and fun elements, although they still wanted it to be pervasive and practical to use. Only a few users were initially concerned about privacy, and this lack of concern persisted. This was somewhat surprising, as privacy is typically considered a sensitive topic in clinical data collection.

The shift of focus in user requirements across the study demonstrates the importance of playfulness in engaging new users and the need for pervasiveness and practicality in sustaining that use. On the basis of our findings, we derived four requirement areas with recommendations for wearable health monitor designers: persuasiveness and personalization (which we combined into one area), playfulness, pervasiveness and practicality, and privacy.

2.6.1. Persuasiveness and personalization

The goal should be to create persuasive techniques that are easy for the user to integrate into daily life. The device should consider the user's preferences and environment and give a timely recommendation based on the user's past behavior. An app that provides information at a glance can reduce the time users spend with the system and thus increase their willingness to use it over a long period.

Personalized information is also important. For users suffering from chronic conditions, a continuous update of health-related parameters is essential. Health parameters should be conveyed and displayed in a way that is easy to understand, for example, not just showing curves and graphics but also including an understandable interpretation in terms of impact on the user's health. To maximize user acceptance, goals and recommendations must be tailored to the user's needs, preferences, and abilities.

2.6.2. Playfulness

Technology should motivate users in an entertaining way, such as by using metaphors adapted to the user's preferences (such as music) to convey information. Users should be able to reflect on their behavior without feeling pressured to reach a performance goal. Another approach is to associate virtual incentives with the user's improvement to foster their extrinsic motivation to exercise more. For example, after a certain amount of walking, users could receive badges or coins that enable them to play music. The more they walk, the more they can enjoy their favorite songs.

With the increasing popularity of virtual reality, another solution might be to immerse users in a virtual world and map their achievement in that world. This could motivate users to perform beyond their daily habits. For example, the user might be an avatar in a game to find hidden images, which can be unlocked only by moving in the physical world—along the lines of Pokémon Go.

2.6.3. Pervasiveness and practicality

To keep users motivated to wear the device, it must be pervasive and practical to use. Techniques might include notifications that strike a balance between making the device unobtrusive and reminding the user to wear it. The system should be as simple as possible and minimize the need for user input, such as logging into the website or app or entering health information manually. A smartphone's locked screen could display the user's improvement automatically, for example.

Additional usability studies will help designers and engineers find the right balance between the need for unobtrusiveness and the minimum interaction that the system requires. A timely notification in the right context can help users remember the device, application, or health goals they have set. The device might suggest walking instead of driving to an event, for example, or present a social challenge that requires walking if the user is with someone. Finally, both the mode of system interaction and the information displayed must be simple and easy to understand.

2.6.4. Privacy

We believe that our users were less concerned about privacy because they did not understand the privacy risks involved in using the device. In general, any device must incorporate methods to address privacy concerns before the technology is deployed.

Managing the tradeoff between the benefits of using the system and the risks associated with sharing data will lead to fewer concerns about privacy. However, a system that ensures privacy, particularly with regard to health-related data, remains essential in engaging users. Privacy settings should be simple enough for users to easily decide what to share and with whom. For users to freely decide how much privacy they want, they need a straightforward way to modify privacy settings and easily understood information about the possible risks of sharing each data type.

2.7. Conclusion

Our longitudinal ethnographic study showed that when sufficiently motivated, diabetic and obese users can change their habits to align with a healthier lifestyle. The healthcare community is already envisioning technology as a driver of behavioral change. A well-designed activity monitor can do much more than record a user's glucose level and weight; through personalization and pervasiveness, it can help users manage their lifestyle to incorporate healthier choices. Our study also showed that users' expectations of the device change with continued use. Thus, in addition to tailoring the device for the user and the user's environment and making it entertaining and easy to use, designers should consider ways that the device can adapt to changing expectations across periods of use.

We recognize that our study has a small and narrowly focused sample size, but our aim was to gain a deeper knowledge of how these technologies are accepted and adopted by chronically ill patients, how technology can help patients engage in a healthier lifestyle, how the device's requirements change over time, and how designers can make the device more efficient.

We plan to follow up on our results in a study with a broader sample size for a longer period to quantitatively investigate the behavior and health changes for individuals with other chronic illnesses. It is also worth exploring how social and living environments affect behavior and how technology can alter those environments to promote desirable behavioral changes.

2.8. References

Paper references

1. Global Status Report on Noncommunicable Diseases—2010, World Health Org., 2011; www.who.int/nmh /publications/ncd_report2010/en.

2. P. Klasnja, S. Consolvo, and W. Pratt, "How to Evaluate Technologies for Health Behavior Change in HCI Research," Proc. ACM SIGCHI Conf. Human Factors in Computing Systems (CHI 11), 2011, pp. 3063−3072.

3. M. Randriambelonoro et al., "Exploring Physical Activity Monitoring Devices for Diabetic and Obese Patients," Adjunct Proc. ACM Int'l Joint Conf. Pervasive and Ubiquitous Computing and ACM Int'l Symp. Wearable Computers (UbiComp/ISWC 15), 2015, pp. 1003−1008.

4. R. Fensli et al., "Sensor Acceptance Model–Measuring Patient Acceptance of Wearable Sensors," Methods of Information in Medicine, vol. 47, no. 1, 2008, pp. 89−95.

5. K. Suastika, "Update in the Management of Obesity," Acta Medica Indonesiana (Indonesian J. Internal Medicine), vol. 38, no. 4, 2006, pp. 231−237.

6. J.O. Prochaska et al., "The Transtheoretical Model of Health Behavior Change" American J. Health Promotion, vol. 12, no. 1, 1997, pp. 38-48.

7. T. Fritz et al., “Persuasive Technology in the Real World: A Study of Long- Term Use of Activity Sensing Devices for Fitness,” Proc. ACM SIGCHI Conf. Human Factors in Computing Systems (CHI 14), 2014, pp. 487−496.

Related work references

1. S. Consolvo et al., “Activity Sensing in the Wild: A Field Trial of UbiFit Garden,” Proc. ACM SIGCHI Conf. Human−Computer Interaction (CHI 08), 2008, pp. 1797−1806.

2. S. Consolvo et al., “Design Requirements for Technologies that Encourage Physical Activity,” Proc. ACM SIGCHI Conf. Human Factors in Computing Systems (CHI 06), 2006, pp. 457−466.

3. J.J. Lin et al., “Fish’n’Steps: Encouraging Physical Activity with an Interactive Computer Game,” Proc. 8th Int’l Conf. Ubiquitous Computing (Ubicomp 06), 2006, pp. 261−278.

4. E.J. Topol, The Patient Will See You Now: The Future of Medicine Is in Your Hands, Basic Books, 2015.

5. L. Mamykina, E.D. Mynatt, and D. Kaufman, “Investigating Health Management Practices of Individuals with Diabetes,” Proc. ACM SIGCHI Conf. Human Factors in Computing Systems (CHI 06), 2006, pp. 927−936.

6. B.K. Smith et al., “Integrating Glucometers and Digital Photography as Experience Capture tools to Enhance Patient Understanding and Communication of Diabetes Self-Management Practices,” Personal and Ubiquitous Computing, vol. 11, no. 4, 2007, pp. 273−286.

7. Y. Chen, “Take It Personally: Accounting for Individual Difference in Designing Diabetes Management Systems,” Proc. 8th ACM Conf. Designing Interactive Systems (DIS 10), 2010, pp. 252−261.

8. T. Fritz et al., “Persuasive Technology in the Real World: A Study of Long-Term Use of Activity Sensing Devices for Fitness,” Proc. ACM SIGCHI Conf. Human Factors in Computing Systems (CHI 14), 2014, pp. 487−496.

CHAPTER 3. OPPORTUNITIES AND CHALLENGES FOR SELF-MONITORING TECHNOLOGIES FOR HEALTHY AGING: AN IN-SITU STUDY

This chapter has been published as:

Randriambelonoro, M., Chen, Y., Yürüten, O. & Pu, P. (2017). **Opportunities and challenges for self-monitoring technologies for healthy aging: An in-situ study.** Gerontechnology. 16. 173-180. Doi: 10.4017/gt.2017.16.3.006.00.

The current need to adopt a healthy lifestyle opens the way for researchers to investigate the efficiency of using technology to motivate a change towards such behavior. Senior people's technology adoption barriers have been largely investigated in previous research but remain unclear to design efficient solution to promote behavior change in the long term. In this paper, we present the findings of an experiment with 20 older adults using pervasive sensors for 6 weeks at their homes. In particular, we present their attitudes for physical activities and self-monitoring technologies before and after the actual usage. We then discuss the opportunities and challenges for designing self-monitoring technologies for healthy aging.

3.1. Introduction

Faced with the constant growth of aging population, the need to promote an environment for healthy aging is expanding1-3. Although maintaining healthy behavior has been shown to be highly beneficial for older adults' health and well-being, the challenge remains in motivating the adoption and the long-term engagement in such behavior4. There are opportunities for emerging technology to increase older adults' engagement in being physically active and managing their health. Within the European REACH (Responsive Engagement of the Elderly promoting Activity and Customized Healthcare) project5, the goal is to learn the older adults' behavior by collecting physical activity and health related data in order to provide personalized health recommendation to them. For this purpose, we conducted an ethnographic study for data collection to get insights on older adults' readiness, willingness, and challenges to adopt pervasive sensors and applications for healthy ageing.

The goal of this ethnographic study is three folded: First, we will obtain insights on older adults' attitudes towards increasing physical activity as well as their readiness towards tracking technologies. Second, we will identify senior's potential behavior changes as well as their usage intention. Third, we will shed light on the opportunities and barriers for them to be monitored and try to understand how they would integrate the system in their daily life.

We present here the findings of a 6-week study with 20 senior people using health monitoring sensors and applications in their home environment. We discuss senior individuals' readiness, willingness and challenges to accept such technology to manage their health. We then discuss their behavior change and aptitude for sustained usage at the end of the study.

3.2. Related works

A sufficient amount of moderate intensity physical activities (150 minutes) have been shown to have a substantial impact on older adults' (age > 65) health6, which would result in a stronger independence and a higher quality of life7. However, the vast majority of older population has difficulty in meeting this recommended lifestyle8.

Meanwhile, technologies for behavior change to promote healthy living are widely investigated in the research community4,9. Researchers tested different technologies including pervasive sensors10 and mobile applications11-13 to help people track their health data. A rich body of work also leveraged goal-setting, rewards, and metaphors to motivate healthy behavior change, such as promoting physical activities.

For senior adults, in particular, Schrack et al.,10 explored the efficiency of several wearable activity monitors, and Vichitvanichphong et al.,14 focused on the older adults' perceived value of assistive technologies. Several studies15-17 using pervasive sensors as intervention showed promising results in terms of health awareness and small improvement in exercise levels, but no consistent changes have been observed. Consequently, despite these ongoing works on older adults' adoption of persuasive technology for behavior change, the barriers to senior's acceptance of these systems still need further investigations. The purpose of this study is an attempt to give more insights on older adults' opportunities and challenges in adopting such technologies for designers and engineers to propose personalized and adapted interventions that will induce steady changes in older adults' daily life.

3.3. Methodology

3.3.1. Participants

We recruited 20 older adult participants from a daily care home, a primary healthcare center and a main hospital in a large city in Switzerland. As we ensured the anonymity of participants' data, we were exempted to go through the institution's Ethical Committee. Our inclusion criteria were: 65+ years old, living at home, and in need of occasional help for their daily activities. The participants include 13 females and 7 males; among them 3 couples took part in the study together. Their ages range between 65 and 89, with a mean age of 77. 6 participants live alone and the rest live with their spouses. Only a few of them still have occasional social activities. 8 of them have serious health conditions limiting them in doing physical activities. All of them have a TV and watch it or use it as background noise on purpose. Most of them use landline phones for communication. 8 of them have smartphones, and among them, four use social media applications (WhatsApp) motivated by their grandchildren; one of them uses a pedometer application introduced by her daughter and 1 uses a photo editing app. 9 participants have tablets and use them mainly for watching news, searching on internet and listening to music. 2 of them had experience using pedometers before the study but they stopped using it after a year due to device failure.

We choose to include 20 participants to make sure our findings come to theoretical saturation. In a qualitative study, saturation is reached when additional participants do not provide

additional insights. Guest et al.,18, Crouch et al.,19, and Strauss20 suggests that saturation occurs around 12 participants if the group is homogeneous. A minimum of 15 participants is then necessary to ensure saturation. In line with other qualitative studies related to our subject21-23, we decided to include 20 participants.

3.3.2. Materials and incentives

We provided each participant with a fitness tracker and a smart scale. For fitness tracker, we chose Fitbit Charge 2 tracker (https://www.fitbit. com/charge2). For smart scales, half of the participants received the Fitbit Aria scale (https:// www.fitbit.com/aria) and the other half the Withings Body Cardio scale (https://www.withings. com/ch/en/products/body-cardio). We selected our sensor through a comparison of products based on a number of criteria from Yumak et al., survey9. These criteria include safety certifications, measurement accuracies, ease of use (wearability, battery life, data storage, ease of use of the software interface), usefulness (the range of information collected via the sensor), seamlessness of data transmission, data accessibility (using public API, manual download), and affordability–see *Table 1* and *Table 2* for the summary. Participants received the tracker and the scale as incentives at the end the study.

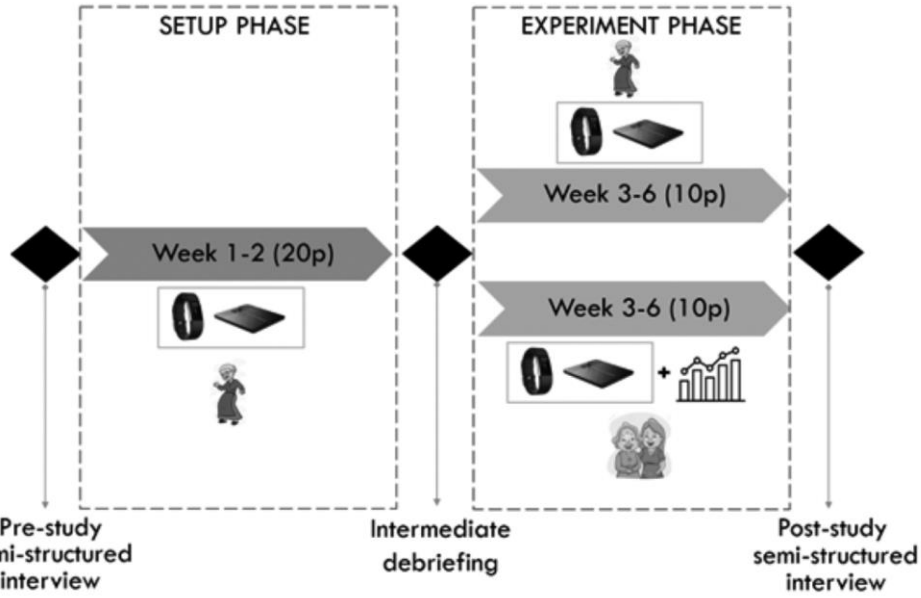


Figure 1. Setup phase and experiment phase of the study

3.3.3. Procedure

We conducted a qualitative study with participants who used the sensors at their homes for 6 weeks. Each participant went through two main phases: the setup phase and the experiment phase *(Figure 1)*. At the beginning of the setup phase, in-between the 2 phases and at the end of the experiment phase, we met the participants for individual interviews.

During the setup phase, the participants received the devices and familiarized them for 2 weeks. The goal of this phase is for them to get familiar with wearing and using the sensors and for the study conductors to identify any potential difficulties.

During the experiment phase, we randomly split the participants into two groups. The first group continued to use the devices for 4 weeks. For the second group, in addition to continuing to use the devices for 4 weeks, they also received visualizations about their activity data.

We then conducted 2 semi-structured interviews (at the beginning and at the end) and an intermediate debriefing (in-between) with each participant. We prepared the topics of interview questions by referring to the ethnographic methodologies by Sabelli et al.,24 and requirements for lifestyle management systems by Yumak et al.,9 We encouraged the users to share their experience and to show us how they used the devices at their own willingness. The interviews were conducted at the participant's home in French, which is the participants' native language. All the interviews were audio-recorded. We first debriefed them about the study and obtained their consents.

The pre-study semi-structured interview lasted around 60minutes. The goal was to understand participants' daily life, health condition and physical activities. We also aimed to know their technology usage and their expectation about technologies for health management and physical activity monitoring.

The intermediate debriefing lasted 15 minutes. The goal was to make sure that the participants could use the devices correctly and integrate them in their daily life. We started to get few feedbacks after a first usage. We then started the experiment phase and helped half of the participant visualize their activity and health parameters on the Fitbit application and the Withings application.

The post-study interview lasted around 45 minutes. The goal of this interview was to get user's feedback after interacting with the devices and to understand their acceptance and adoption of the tools, as well as the challenges they faced during the experiment.

3.3.4. Data analysis

We audio-recorded all interviews (total of 2,247 min), transcribed them into English, and analyzed the resulting 307,839 words according to the grounded-theory approach. After reading all the transcripts, we generated a starting list of codes based on the research questions and the patterns that emerged from the data. We then returned to the data and conducted a systematic axial coding to identify emergent themes. After several coding iterations, we identified and categorized the themes and reached a consensus.

3.4. Findings

In this section, we focus on the older adult people's perception on activity tracking devices before and after usage. Results on their attitudes on scales and visualization will be presented in another paper.

As our primary purpose in this paper is not to quantitatively measure the intervention effectiveness but rather to elicit users' experience in using self-monitoring technologies for

healthy aging, we do not have a control group. However, the evolution of the participants' number of steps during the experiment showed an increase of 1609 steps. We compared the mean number of steps in Week 1 (4573) and the mean number of steps in Week 6 (6182).

Device name	Accuracy	Safety	Data accessibility	Ease of use	Usefulness	Seamless data transmission	Affordability
Wahoo Tickr X	●	●	○	◉	◉	●	●
Suunto Smart Sensor	○	●	◉	○	○	◉	●
TomTom Spark3 Cardio	●	●	○	◉	●	●	◉
Mio Alpha 2	●	●	○	●	◉	◉	◉
Garmin Vivosmart HR	○	○	◉	●	◉	◉	◉
Fitbit Charge 2	◉	●	●	●	●	●	◉
Fitbit One	◉	●	●	●	◉	●	●
Hexoskin	◉	●	●	◉	●	●	○

Table 1. The summary of our sensor evaluations. Full-black circles, partially full circles, and white circles indicate high, acceptable, and low scores respectively

Device name	Accuracy	Safety	Data accessibility	Ease of use	Usefulness	Seamless data transmission	Affordability
Fitbit Aria	◉	●	●	●	◉	●	●
Withings Body Cardio	◉	●	●	●	●	●	◉

Table 2. The summary of our intelligent scale evaluations. Full-black circles, partially full circles, and white circles indicate high, acceptable, and low scores respectively

3.4.1. Before usage

3.4.1.1. Attitudes towards increasing physical activities

Regarding older adults' opinion about physical activity, different points of view emerged from the discussion. Most of them (n=14) were aware of the benefits of doing physical exercise for their health but failed to do so due to inability and lack of motivation. P2 said*: "I know it is good but I am not really motivated to walk when I am alone. When my friend was there I was more motivated but now she is sick"*. P10 mentioned: *"I really like walking but with my leg I cannot go far anymore, it is really painful so I stay home. When I am bored at home, I try to go out a bit but only to take coffee and cake downstairs, not far away".*

Two participants had been very active in sport in their younger age, but they no longer felt the need to continue anymore. P16 said: "*You know I did a lot of football before, during 30 years*

of my life. Now I think I have done enough, it's not for my age anymore. I just watch sport on TV". One participant who is very healthy doesn't feel the need to do any physical exercise. He said: *"My doctor told me to walk 3 times a week but I don't do it, because I don't feel like I need it. I don't have any condition, I am very healthy. I am not the type to go walk just for the pleasure to walk, there should be a purpose besides".*

3.4.1.2. Readiness towards tracking technologies

Here, we attempt to understand older adults' readiness to adopt such devices and integrate them in their daily life. Nowadays, technological solution is not always adapted to senior's needs and ability, which then creates a disinterest on their side25. However, some of our participants (n=4) were eager to try new tools and devices. P4 told us: *"I am very interested in technology. It's a pity that I don't have the ability to use more complex things like computer but my children offered me a tablet and I learned by myself how to use it. I spend several nights trying to understand how it works and now I manage to use it"*. P15 often spends his afternoon trying to help his friends doing things on his computer. He reported: "*As I am very interested in computer, he calls me when he has a problem so we try to solve it together".*

Five participants got interest in diverse mobile applications introduced by their family or friends. P7 proudly told us: *"let me show what I can do with my smartphone, I know how to use WhatsApp, you know, and I even edit my photo before sending it to my grandchildren"*. P3 said: *"my daughter got me interested in this health application on the phone that can count steps, so I try to do 6000 steps every day".*

One participant who was not interested in technology at all found the devices we gave them manageable and acceptable to use. P5 said: *"I am not very a technological person, I don't have computer, neither a smartphone, it is too complicated. I just have a fixed phone and a mobile phone that I barely manage to use. I am a bit afraid, but maybe I can manage to learn it if it is simple."*

Before using the devices, our intent was to understand older adults' attitudes towards increasing physical activity as well as their readiness towards tracking technologies. The main barriers of being physically active were the absence of motivation, the incapacity due to health condition and the lack of perceived usefulness of physical exercise. Furthermore, although half of the participants were not technology oriented, personal interest and enthusiasm driven by family towards technology were observed.

3.4.2. After usage

For post-usage, two participants (P13 and P20) couldn't have been reached due to the wife hospitalization, so we analyzed the data based on 18 participants.

3.4.2.1. Behavioral change

After 6 weeks' usage, 10 participants reported to us a positive change on their behavior in their everyday life. They replaced some of their sedentary activities such as reading, knitting or writing letters to a friend to going outside and walking in their neighborhood. P5 shared: *"I feel like I don't have time to do my little things anymore, because I go out to walk every day. I*

realize it takes me a lot of time to do it, but I like doing it. So, I stay up late at night to catch up with the things I didn't finish".

Even though we didn't purposely ask them to reach a specific goal, most of them fixed a personal objective themselves and acted accordingly in order to reach it. P1 said: *"I make sure I do at least 4000 steps every day, so I go out and walk more since I have the watch"*. P4 who is very motivated even walk at home to increase her number of steps. She told us: *"Now, I even do the exercise the physiotherapist asked me to do, and I look at the watch to know how much I have done".* Some participants didn't walk much more but tried to take more stairs. P15 said: *"My physiotherapist always told me that I should do at least 2 floors by stairs as I live on the 4th floor, but now I always take the stairs to come up here. I feel like I have less difficulty to breath while I do that."*

For P17, even though he was on a wheelchair, wearing the watch motivated him to move around. He said*: "now I am going in circle inside the house instead of sleeping in my wheelchair while waiting for the nurse to come".* His wife reported to us: "*as he cannot see what is displayed on the watch, he always bugs me to check how much steps he has done"*. P17 also became very dependent on the device. His wife contacted us after the watch stopped working: *"the watch didn't work for 3 days now, and my husband is sleeping a lot again. He really insisted that I contact you so that you can solve the problem because he really needs it as a motivation".*

Meanwhile, five participants reported that no changes had occurred in their daily life. They were doing the same activities as before and discovered that they could reach a certain number steps or stairs by going around at home and having their daily routine. P19 said: *"I can't say I changed my habits due to the watch but with what I do every day I could reach around 7000 steps. This is what I can do, I don't think I can do more and the watch wouldn't change that"*. P7 also reported: *"I don't feel that I am walking more but as I live in a house with stairs, I have a proof now that I am doing a lot of activity by going up and down"*.

3.4.2.2. Usage Intention

We investigated usage intention by looking at whether participants had the intention to continue using the device. Overall, 15 out of 18 participants expressed willingness to continue using the device, while 3 out of 18 participants did not plan to continue using it.

Participants were motivated to continue using the device because it was engaging, fun and allowed them to set personal goal and track their improvement. In total 15 participants kept the devices. Some of them (P5, P14, P17) even started to make plans and future goals with the tools. P5 ensured she will be able to charge it anywhere because she is planning to go on holiday in a few months. P14 wanted to know more about making a food plan through the application because he would like to lose weight. P17 who is half-paralyzed told us: *"Now I do at least 2000 steps a day but my goal is to reach at least 3000 steps in summer"*.

We were surprised by P5 behavior. As she didn't know how to use a computer, she took the time to note every parameter in a notebook for every day of the experiment, without us asking her to do that. She started to self-reflect on her personal data and asked us to explain some

parameters she couldn't understand. She was ready with a list of questions about how to use either a functionality of the devices either a data she couldn't understand. She said: *"I liked it because it makes me have a goal and walk more. Someone told me that since the study is over now, I will have more time for other activities. But I told her that even if the study finished, I will continue to walk and record everything because I like it"*.

Although some of the participants (n=5) were a bit skeptical about the efficiency of the technological solution at the beginning and wanted to return the device to us at the end of the study, 4 of them changed their mind. P16 said: *"it gives interesting information and it doesn't require so much effort from me, so I will keep it"*. P11 reported: *"at first when you came, I thought what is this thing again, but you now, I find it very funny, I get used to the little message. It's like a game. If you told me you would take it now, I would feel that something is missing"*.

However, some participants did not want to use the devices anymore because of perceived inutility, unwanted behavior related to the tools, and fear of the equipment negatively impacting their health. For example, P12 who has a lot of difficulty in walking gave us back the device at the end the study. She said: *"I knew from the beginning that it wouldn't change anything in my daily life. My problem is really that I can't do much and it is not a technology that would tell me what I have to do"*. Another participant (P2) also returned the watch but kept the scale. She didn't like the fact that she wanted to look at the watch all the time to see how much she has done. One participant (P9) dropped the study at the end of the setup phase. Due to a health problem that occurred during the experience, she preferred to stop using the devices, fearing that there would be a link to that.

3.5. Discussion

Our findings shed light on the older adults' readiness, experience, and challenges for adopting technologies for healthy aging.

3.5.1. Opportunities

Our participants had different reasons to participate in the experience. 10 thought the experience would be interesting, 6 participated due to their trust in the institution which proposed the project and 4 said they accepted to take part in the study only to help the investigator. However, in the end 15 out of 18 kept the devices and had the intention to continue using them. While designing augmented reality game for older adults, Laine et al.,26 also observed hesitation in their participants at first, but discovered later that the team competitiveness succeeded to engage them in doing the physical exercise through the game. In our case, senior's usage intention and willingness to integrate the devices in their daily life could be explained by the systems simplicity, practicality, and the possibility to customize it to their needs and ability.

Participants also demonstrated a stronger motivation when the device usage was linked to a personal goal they set. The usefulness takes an important role in user willingness to adopt such devices. Designers of systems that encourage being physically active should consider

emphasizing the added value of the technology usage. This could be done by offering timely feedback on the older adults' progress and giving easy to understand correlation between their daily activity and the variation of their personal parameters to allow a self-reflection on their behavior.

In addition, the devices really impacted some user's behavior and induced a relation of dependence. As the seniors broke their old habits to integrate a new system in their daily life, once getting used to it, it became a part of their lifestyle. They considered the device as a companion, a buddy that shows interesting information without constraining them. The need for timely information became greater as well as the need for receiving message that makes them less lonely. Considering this inter-relational aspect would benefit researchers who would want to increase long term engagement in technology usage for behavior change.

3.5.2. Challenges

Half of our participants showed interest and began to change little things in their behavior. Five participants didn't have any changes in their daily routine but used the tools only as a source of information. However, five participants still found it challenging to integrate in their daily life. The main reasons were linked to the fear of introducing novelty and breaking their old habits, the fear to be dependent of the tools, and the need of a human presence interacting with them.
The fear of novelty, introducing new things and breaking their daily routine remained a challenge for some of our participants. As people get older, they start to create a daily routine to facilitate they everyday life. In addition, although seniors are now more exposed to new technology, their acceptance of it is still difficult to achieve, mainly because of lack of adaptation of these systems to their age-related condition27 and their lack of confidence in using the tools correctly. Considering these aspects, it becomes challenging to lead the older adults into creating new habits by inserting new unfamiliar elements in their lifestyle.

One important aspect addressed was also the fear to be dependent of the tools. While some older adult people needs to rely on something due to loneliness and the need of interaction and assistance, others do not find this relation of dependence acceptable. The sensation of being able to control the systems they are using is also very important to them. Engineers should propose a system that allows the older adults to turn it off when necessary, so that they would not feel obliged to look at it every time. Furthermore, designers should focus on positive improvement to avoid senior's culpability, which will decrease their fear of being dependent of the systems.

Finally, the last challenge we observed were linked to the need of social interaction. Although they found the tools interesting, most of them still value the interaction with others. One important aspect that researchers should take into account while designing intervention to encourage physical activity, is to involve real meeting with family, friends and peers through technology. Talking to people, commenting on the news around a coffee or discovering together a new neighborhood would be an efficient motivation for seniors to go out of their home.

3.6. Conclusion

We conducted an ethnographic study for 6 weeks with 20 senior people at their home. The ultimate goal of this project is to understand whether the older adults would accept to be monitored and whether they would adopt and integrate such tools in their daily life. Although our procedure involved different sensors and setups, we mainly report in this paper older adult people's perception on activity tracking devices before and after usage. After 6 weeks of usage, we discovered changes in behavior and usage intention which allowed us to identify opportunities and challenges for the older adults to adopt sensors and application for health and activity management. This study showed the potential of acceptance and adoption of simple and manageable technology for behavior change. As for usage intention, some older adults start to be eager to learn new tools as long as they are able to do it, which is more and more influenced by family from younger generation. However, designers and engineers should consider older adults' fear introducing novelty in their daily life as well as their need for social interaction and their need to remain in control of any system given to them.

So far, this study focused mostly on the qualitative aspect, and particularly on physical activity tracking. In the future, we would focus on the impact of the scales and the data visualization presented in the procedure, go deeper in analyzing quantitatively the participants' data and identify potential guidelines for designing pervasive technology for healthy aging.

3.7. Acknowledgements

We are grateful to the anonymous reviewers for their valuable feedback. We also thank Samira Hammoudi Bouchez and Marie-Luce Guillermin Spahr for helping us recruit the participant. Finally, we thank all the participants for their time and their involvement in the study. This research has received funding from the European Union's Horizon 2020 research and innovation program under grant agreement No 690425.

3.8. References

1. Dishman E. Inventing wellness systems for aging in place. Computer, 2004;37(5):34-41; https://doi. org/10.1109/MC.2004.1297237

2. Ying AHP. Creating a supportive environment for elderly with chronic illness. Plenary session of the Healthy Ageing Convention 2001

3. Wood A, Stankovic J, Virone G, Selavo L, He Z, Cao Q, Doan T, Wu Y, Fang L, Stoleru R. Context-Aware Wireless Sensor Networks for Assisted-Living and Residential Monitoring. IEEE Network 2008;22(4); https://doi.org/10.1109/ MNET.2008.4579768

4. Araullo J, Potter LE. Promoting Physical Activity in Seniors: Future Opportunities with Emerging Technologies. Proceedings of the 2016 ACM SIGMIS Conference on Computers and People Research 2016; pp 57-64; https://doi. org/10.1145/2890602.2890616

5. Responsive Engagement of the Elderly promoting Activity and Customized Healthcare; http:// reach2020.eu; retrieved May 30th, 2017

6. Centers for Disease Control and Prevention, US Department of Health and Human Services. How much physical activity to older adults need? https://www.cdc.gov/physicalactivity/basics/older_ adults/index.htm; retrieved June 07th, 2017

7. Sims J, Hill K, Hunt S, Haralambous B. Physical activity recommendations for older Australians. Australasian Journal on Ageing 2010;29(2):81-87; https://doi.org/10.1111/j.1741-6612.2009.00388.x

8. Prohaska T, Belansky E, Belza B, Buchner D, Marshall V, McTigue K, Wilcox S. Physical activity, public health, and aging: Critical issues and research priorities. The Journals of Gerontology Series B: Psychological Sciences and Social Sciences 2006;61(5):267-273; https://doi.org/10.1093/ geronb/61.5.S267

9. Yumak Z, Pu P. Survey of Sensor-Based Personal Wellness Management Systems. BioNanoScience 2013;3(3):254-269; https://doi.org/10.1007/ s12668-013-0099-0

10. Schrack JA, Cooper R, Koster A, Shiroma EJ, Murabito JM, Rejeski WJ, Ferrucci L, Harris TB. Assessing Daily Physical Activity in Older Adults: Unraveling the Complexity of Monitors, Measures, and Methods. Journals of Gerontology Series A: Biomedical Sciences and Medical Sciences 2016;71(8):1039- 1048; https://doi.org/10.1093/gerona/glw026

11. Lin JJ, Mamykina L, Lindtner S, Delajoux G, Strub HB. Fish'n'Steps: Encouraging Physical Activity with an Interactive Computer Game UbiComp 2006: Ubiquitous Computing; pp 261-278; https:// doi.org/10.1007/11853565_16

12. Consolvo S, McDonald DW, Toscos T, CheN MY, Froehlich J, Harrison B, Klasnja P, LaMarca A, LeGrand L, Libby R, Smith I, Landay JA. Activity sensing in the wild: a field trial of UbiFit Garden. In Proceedings of the SIGCHI conference on human factors in computing systems. ACM 2008 ; pp 1797- 1806; https://doi.org/10.1145/1357054.1357335

13. Lane ND, Mohammod M, Lin M, Yang X, Lu H, Ali S, Doryab A, Berke E, Choudhury T, Campbell A. BeWell: A smartphone application to monitor, model and promote wellbeing. In 5th International ICST Conference on Pervasive Computing Technologies for Healthcare 2011;23-26; https://doi. org/10.4108/icst.pervasivehealth.2011.246161

14. Vichitvanichphong S, Talaei-Khoei A, Kerr D. Elderlys' Perception about The Value of Assistive Technologies for their Daily Living: Impacting Factors and Theoretical Support. Hawaii International Conference on System Sciences 2017; https://doi. org/10.24251/HICSS.2017.445

15. Sookhai L, Coppola JF, Gaur C. Intergenerational activity tracker program: Impact with health related outcomes on older adults. Systems, Applications and Technology Conference (LISAT), IEEE Long Island 2015; pp 1-7; https://doi.org/10.1109/ LISAT.2015.7160218

16. Lebron J, Escalante K, Coppola J, Gaur C. Activity tracker technologies for older adults: Successful adoption via intergenerational telehealth. Systems, Applications and Technology Conference (LISAT), IEEE Long Island 2015; https://doi.org/10.1109/LI-SAT.2015.7160200

17. Burke L, Lee AH, Jancey J, Xiang L, Kerr DA, Howat PA, Hills AP, Anderson AS. Physical activity and nutrition behavioral outcomes of a home-based intervention program for seniors: a randomized controlled trial. International Journal of Behavioral Nutrition and Physical Activity 2013; https://doi. org/10.1186/1479-5868-10-14

18. Guest G, Bunce A, Johnson L. How many interviews are enough? An experiment with data saturation and variability. Field Methods 2006;18(1):24; https://doi.org/10.1177/1525822X05279903

19. Crouch M, McKenzie H. The logic of small samples in interview-based qualitative research. Social Science Information 2006;45(4):18; https://doi. org/10.1177/0539018406069584

20. Corbin J, Strauss A. Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory (3rd ed.). Thousand Oaks, CA: Sage 2008; https://doi. org/10.4135/9781452230153

21. Bisafar FI, Parker AG. Confidence & Control: Examining Adolescent Preferences for Technologies that Promote Wellness. In Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW ‘16), ACM, New York, NY, USA 2016; pp 160-171; http://dx.doi.org/10.1145/2818048.2820028

22. Stawarz K, Cox AL, Blandford A. Beyond self-tracking and reminders: Designing smartphone apps that support habit formation. In Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI ‘15), ACM, New York, NY, USA 2015; pp 2653-2662; http:// dx.doi.org/10.1145/2702123.2702230

23. Schorch M, Wan L, Randall DW, Wulf V. Designing for those who are overlooked: Insider perspectives on care practices and cooperative work of elderly informal caregivers. In Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW ‘16), ACM, New York, NY, USA 2016; pp 787-799; http://dx.doi.org/10.1145/2818048.2819999

24. Sabelli AM, Kanda T, Hagita N. A conversational robot in an elderly care center: An ethnographic study. Proceedings of the 6th international conference on Human-robot interaction 2011; pp 37-44; https://doi.org/10.1145/1957656.1957669

25. Olphert W, Damodaran L, Balatsoukas P, Parkinson C. Process requirements for building sustainable digital assistive technology for older people. Journal of Assistive Technologies 2009;3(3):4-13; https://doi.org/10.1108/17549450200900019

26. Laine TH, Suk HJ. Designing Mobile Augmented Reality Exergames. Games and Culture 2015;11(5):548-580; https://doi.org/ 10.1177/1555412015572006

27. Olphert W, Damodaran L, Balatsoukas P, Parkinson C. Process requirements for building sustainable digital assistive technology for older people. Journal of Assistive Technologies 2009;3(3):4-13; https://doi.org/10.1108/17549450200900019

CHAPTER 4. SOCIAL INCENTIVES IN PERVASIVE FITNESS APPS FOR OBESE AND DIABETIC PATIENTS

This chapter has been published as:

Chen*, Y., Randriambelonoro*, M., Geissbuhler, A. & Pu, P. **Social Incentives in Pervasive Fitness Apps for Obese and Diabetic Patients.** In Proceedings of the 19th ACM Conference Companion on Computer Supported Cooperative Work & Social Computing (CSCW'16 Companion). San Francisco, USA, Feb. 2016 (*equal contribution)

Social incentives such as cooperation and competition are found to motivate users in pervasive fitness applications. This work investigates how social incentives work for individuals with obesity and diabetes. We used a mobile fitness application called HealthyTogether as an experimental platform, which allows dyads to achieve fitness goals together and compete in an online community. We conducted a four-week study with 16 obese and diabetic patients who used HealthyTogether to exercise with a buddy. Results show that participants exercised more with social incentives compared with their baseline. Collaborating with buddies to compete in a community was reported as motivating for dyads exercising with strong ties. Social interactions could be demotivating between dyads who did not know each other well. Finally, it is crucial to consider patients' technical literacy when designing behavior-changing technologies.

Author Keywords

Health; diabetes; obesity; behavior change; social interaction; mobile fitness application

4.1. Introduction

Recent decades witness the rapid development of an epidemic of diabetes and obesity mainly due to unhealthy diet and physical inactivity. A major challenge for designing lifestyle management systems for individuals with diabetes and obesity is to provide effective motivation methods for behavior change. Previous empirical studies [2,3,5,8] showed that users exercised significantly more when they walked with a buddy than when they walked alone. Clinical research also indicated promising effects of integrating peer support provided by patients with same symptoms [1,4,6]. Such support provides diabetics with emotional, appraisal and information assistance to improve their physical, mental and social wellbeing [9]. In this work, we investigate the effectiveness of social incentives for obese and diabetic patients using HealthyTogether [2] as an experimental platform.

4.2. HealthyTogether

HealthyTogether is an Android application that involves dyads to exercise and earn badges together [2]. It retrieves users' performance in walking and climbing stairs from Fitbit tracker. HealthyTogether employs three dimensions of social incentives by allowing dyads to 1) win collective badges based on their average steps and floors of the day, 2) pledge the sum of steps and floors of the day and 3) compete with other dyads in the team leaderboard. Dyads

can also interact with each other by sending cheering or taunting messages. Figure 1 shows the dyad and community interface of HealthyTogether.

4.3. User study

We deployed a four-week study to compare obese and diabetic patients' fitness performance when they were connected with buddies and communities with when they exercised alone. We recruited the patients from a major hospital in Switzerland. They have participated in a previous study [7] in which each individual used Fitbit to monitor physical activities for at least one month. Sixteen patients agreed to participate in the study, including five obese and eleven diabetics. They were five males and eleven females from five different countries (Swiss, France, Italy, Cap Vert and Madagascar). Three of them worked full time and others were retired or housewives. They were paired into eight dyads. Four dyads were strong ties (i.e., couples or friends) and four dyads were weak ties (i.e., acquaintances from this study). We provided each participant with an Android phone with a SIM card with 1Go data plan. They could keep the phones after the study as a compensation for their time.

We invited them to the hospital for an information meeting and helped them install HealthyTogether. All participants started the study in the same week and used HealthyTogether for four weeks to facilitate community competition. Finally, they returned to the hospital to attend a 30-minute post-study interview. We conducted the interviews in French, audio-recorded them, transcribed and translated into English.

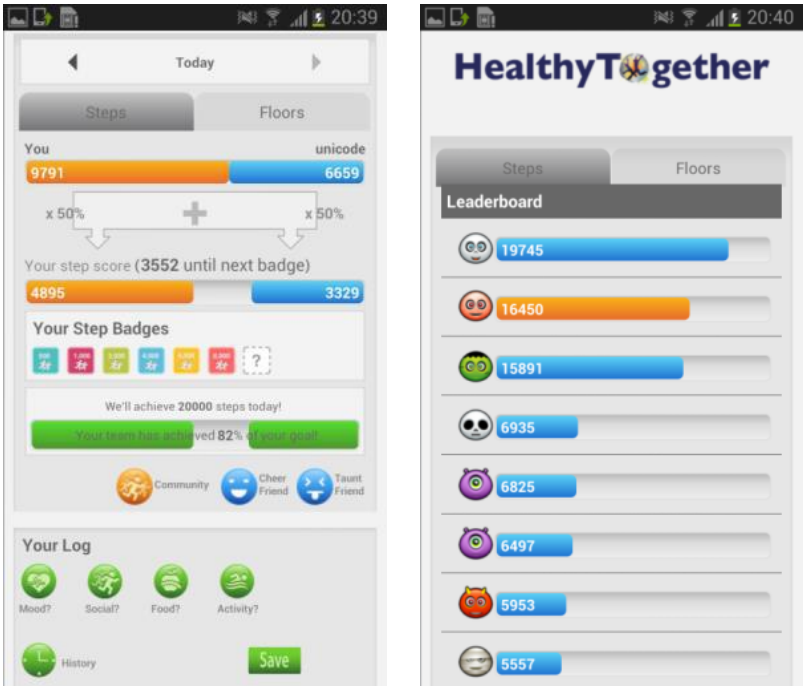


Figure 1. HealthyTogether screenshots: (a) dyad interface; (b) community interface.

4.4. Findings

We retrieved participants' Fitbit data during their first four-week Fitbit usage as the **baseline** and that during their four-week HealthyTogether usage as the **intervention**. One dyad, who did not know each other well, withdrew out from the study because of personal and privacy concerns. Another participant encountered technical issues with her Android phone, which impacted her in using HealthyTogether. So we quantitatively analyzed Fitbit data from 13 patients.

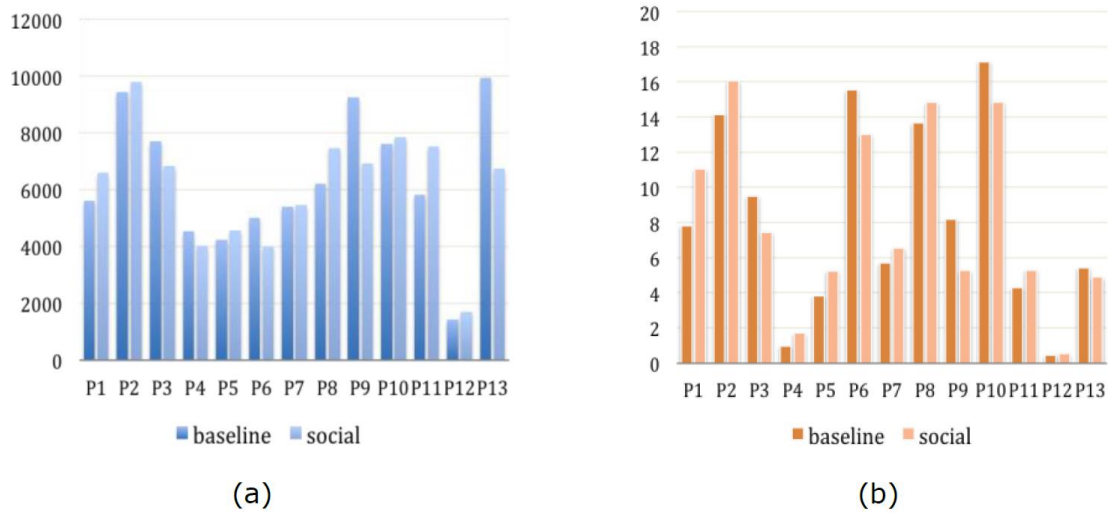


Figure 2. Average steps/day and floors/days in baseline vs. social intervention grouped by individuals: (a) Steps; (b) Floors.

To examine the intervention effects of social incentives, we conducted a Linear Mixed Model (LMM) analysis in SPSS. Daily Step count and daily floor count are the dependent variables. Intervention (before/after using HealthyTogether) is within-subject variable, and relationship type (strong tie/weak ties) is between-subject variable.

For number of daily floors, we found a tendency of increased performance for intervention: $F(1,753)=3.029$, $p = .082$, MeanBefore = 7.72, SE =5.15; MeanAfter= 9.17, SE=5.16. The interaction between intervention and relationship type is not significant. For daily steps, the number slightly increased, but the main effect of intervention is not significant $F(1,753)=1.083$, $p = .298$, MeanBefore = 6332.8, SE =187.9; MeanAfter= 6631.3, SE=216.7.

When we examined each participant, eight of them have increased their number of steps and floors from baseline. Three patients had an average increase of over 1,000 steps/day, which is considered significant improvement in health intervention literature [5]. Interview results showed that the most frequently (N = 8) mentioned motivating factor is the competition within the patient community. Participants used their ranking in the **community** as a reference for their performance. As P3 reported: "We could see what other groups are doing.

The ranking is motivating. I would do more with my buddy if we do not rank well in the community."

Participants also appreciated **collaborating** with and being accompanied by a buddy (N = 6). For P4, achieving a better score with her husband made her happy. They cheered each other up to compete with other patients. As her husband said, "I was trying my best, but it was not easy. Sometimes I cannot just take the stairs with my legs. Seeing her performance on HealthyTogether, I tried to exercise more by creating some occasion to go out and walk. Although my physical conditions did not allow me to reach my wife's level, I was trying and have contributed to a collective improvement." P6 usually sent a message to taunt his buddy when he was not moving. They participated in a group walk one hour and a half every Monday in order to achieve their common goal. P6 is happy that they could achieve the daily goal of 20,000 steps.

However, three participants who were not familiar with their buddies reported to be demotivated. Their exercise amount dropped after pairing up with a buddy. P7 mentioned that she did not know his buddy well, so she barely interacted with her buddy: "If I did it with my best friends at another period, it would have been different". His buddy, P8 said: "I hesitated in sending messages and taunting because she does not like pressure. So I just watched how she was doing."

Technical literacy is another issue reported by participants. Twelve out of the sixteen patients were above 50 years old, and some of them were not familiar with Android system. For example, P12 contacted us because after updating the phone, all the application disappeared including Fitbit and HealthyTogether. She was frustrated and re-installed all the apps. By contrast, earlier studies with university students hardly showed any technology issues even though most participants used iPhones as their primary phones [2]. While obesity and diabetes are not targeted at the elderly, the findings suggest that technologies should be designed simple and familiar for patients to minimize technical barriers.

4.5. Conclusions

This study investigates the effectiveness of social incentives for lifestyle changes for obese and diabetic patients. Sixteen patients participated in a four-week intervention. Compared with the baseline, there was a trend of improvement in daily floors when connecting with a buddy and competing in the community: 7.72 floors/day vs. 9.17 floors/day. Participants also slightly walked more (6332 steps/day vs. 6631 steps/day). Three of them have an increase of more than 1,000 steps per day. Participants found community competition and the support from strong ties motivating. However, pairing up with weak ties was reported demotivating. Finally, considering patients' technical literacy is essential in lifestyle intervention technologies. In the future, we will further validate our results and conduct the study with more patients and they will use HealthyTogether on their primary phones.

4.6. References

[1] Manuel Barrera Jr, Glasgow E. Russell, Mckay H. Garth, Boles M. Shawn, and Feil G. Edward. 2002. Do Internet-based support interventions change perceptions of social support?:

An experimental trial of approaches for supporting diabetes self-management. American journal of community psychology 30, 5: 637-654.

[2] Yu Chen and Pearl Pu. 2014. HealthyTogether: exploring social incentives for mobile fitness applications. In Proceedings of the Second International Symposium of Chinese CHI (Chinese CHI '14), 25-34.

[3] Sunny Consolvo, Katherine Everitt, Ian Smith, and James A. Landay. 2006. Design requirements for technologies that encourage physical activity. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '06), 457-466.

[4] Michele Heisler and Piette D. John. 2005. "I Help You, and You Help Me" Facilitated Telephone Peer Support Among Patients With Diabetes. The Diabetes Educator 31, 6: 869-879.

[5] Andrew D. Miller and Elizabeth D. Mynatt. 2014. StepStream: a school-based pervasive social fitness system for everyday adolescent health. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14), 2823-2832.

[6] Gillian M. Paul, Susan M. Smith, David L. Whitford, Eamon O'Shea, Fergus O'Kelly, and Tom O'Dowd. 2007. Peer support in type 2 diabetes: a randomised controlled trial in primary care with parallel economic and qualitative analyses: pilot study and protocol. BMC family practice 8, 1: 45.

[7] Mirana Randriambelonoro, Yu Chen, Antoine Geissbuhler, and Pearl Pu. 2015. Exploring physical activity monitoring devices for diabetic and obese patients. In Adjunct Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Computers (UbiComp/ISWC'15 Adjunct), 1003-1008.

[8] Tammy Toscos, Anne Faber, Kay Connelly, and Adity Mutsuddi Upoma. 2008. Encouraging physical activity in teens: Can technology help reduce barriers to physical activity in adolescent girls? In Proceedings of Second International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth'08), 218-221.

[9] Henk A. van Dam, Frans G. van der Horst, Lut Knoops, Richard M. Ryckman, Harry FJM Crebolder, and Bart HW van den Borne. 2005. Social support in diabetes: a systematic review of controlled intervention studies. Patient education and counseling 59, 1: 1-12.

CHAPTER 5. HOSPITAL-TO-HOME TRANSITION: USING SERIOUS GAME AND ACTIVITY MONITORING TO ENGAGE ELDERLY PATIENT TO BE MORE ACTIVE

This chapter has been submitted as:

Randriambelonoro, M., Perrin, C., Blocquet, A., Kozak, D., Fernandez, J.T., Marfaing, T., Bolomey, E., Benhissen, Z., Frangos, E., Geissbuhler, A., Graf, C. **Hospital-to-home transition for older patients: using serious games to improve the motivation for rehabilitation.** Journal of Population Ageing

5.1. Abstract

Background: Traditional physiotherapy is often perceived as repetitive and boring by patients, leading to issues in maintaining their interest and performing the required exercises. This paper investigates older patients' engagement and motivation for physical activity using a serious game-based rehabilitation compared to a control group using a standard care therapy, during hospitalization.

Methodology: We conducted a randomized clinical trial with 57 geriatric patients and demonstrated that the serious game-based rehabilitation was as efficient as the standard rehabilitation program in terms of improving their functional capacity. In this paper, we mainly focus on reporting qualitative analysis of the patient's behavior change thorough the intervention. The intervention lasted 3 weeks. Semi-structured interviews, including all the participants, were conducted before and after the intervention. Patients' attitudes towards rehabilitation process, physical activity and innovation technology were investigated. Motivation to be active following the intervention were explored.

Results: Participants admitted feeling bored at the hospital and only following along the care process routine. Enthusiasm towards the progress brought by technological solution was observed. At the end of the hospital stay, the serious game-based rehabilitation received positive feedback. Patients felt more active and observed significant improvement of their general condition.

Conclusion: The analysis suggests that patients participating in the gamified rehabilitation are more motivated. Several recommendations on designing serious game for elderly rehabilitation are proposed.

Keywords: serious game; elderly; rehabilitation; motivation; engagement; qualitative study

5.2.Introduction

By 2050, one in six people in the world is expected to be part of the 65+ age group (United Nations, 2019). This demographic shift towards a higher proportion of older people will increase the number of people at risk for disability (World Health Organization, 2017). Indeed, elderly people are subject to more frequent and longer hospitalisation than younger ones due to the complexity of their illness, leading to a higher risk of functional decline

(Covinsky et al. 2003, Kwok et al. 2016). A rehabilitation process is therefore necessary to restore geriatric patient's independence prior to returning home. Regular physical therapy increases muscle strength and functional capacities of hospitalized older people with musculoskeletal issues (Ling et al. 2017). However, it is often associated with frustration and poor adherence to the proposed exercise (Lang et al 2009; Burke et al 2009; Liddle et al 2007) which represents a barrier for rehabilitation.

Recent technological developments paved the way to techniques that motivate behavior change, such as gamification. Gamification, defined as “the application of game principle into non-game context” (Huotari and Hamari 2012; Deterding et al 2011), is being widely used in various domains such as work environment, marketing, education and health. The goal is to induce user engagement in performing certain actions. A plethora of research investigates the use of gamification to motivate physical exercise among people with chronic disease, cardiovascular risks and in the aging population (Hamari et al 2014). Gamification principles are closely related to behavior change theories and techniques that have been demonstrated to improve health outcomes.

Promising evidence suggests the positive impact of gamification on motivation to be active (Johnson 2016; Cugelman 2013; Lister et al 2014). Researchers are designing persuasive technologies that motivate moderate physical activities. For example, to motivate people to walk more, activity monitoring devices are coupled with a virtual reward-system and allow users to collect badges and points (Skriloff et al 2016). Another common area of research is the use of metaphors to help people visualize in a playful and easy way their progress towards their goal. Many studies have been trying to map the number of steps achieved to the growth of virtual characters such as animals (Lin et al 2006) and plants (Consolvo et al 2008). In addition, we observe a real interest in using serious game to engage people in performing physical exercise. Playing games that involve subtle physical movement are widely explored (Rice et al 2011) not only for moderate activities like walking or running but also for rehabilitation exercises. Immersing the user into a virtual world does not only procure a feeling of pleasantness but also has the potential to increase user self-efficacy and confidence in doing the required exercise.

Previous research (Ling et al 2017; Idriss et al 2017; Brox et al 2017) suggests that serious games increase motivation in rehabilitation sessions which consist of repetitive exercises. Game characteristics which generally involve cognitive and motor skills represent an effective way to retain elderly’s attention (Krichevets et al 1995) and distract them from pain and boredom (Burke et al 2009). Furthermore, the possibility to increase the game level tailored to the patient’s ability gives a sense of challenge followed by a feeling of satisfaction when achieving the task (Lumsden et al 2016). Moreover, the emergence of (Kinect-based) motion detection technologies opens the way to the development of exergames (exercise-based videogames) for elderly people who are subject to frequent falls. Meekes and Stanmore (2017) found that social interaction was an important extrinsic motivator that increased the intrinsic motivation to adhere to the exergame program. Older people enjoyed playing the

game and were more active and socially confident. This study had a small population size and was conducted in two living facilities quite different from a hospital context.

While gamification showed promising results, a systematic review of gamification usage in e-health (Sardi and al, 2017) emphasized the lack of valid empirical evidence in this field. In addition, serious game-based intervention is often associated with short-term engagement mainly based on extrinsic rewards. Conditions for game engagement such as purpose, user alignment and utility need to be defined (El-Hilly and al 2016) for gamification to reach its full potential. Finally, it remains unclear how gamification can be used to modify health behavior in a hospital setting while doing rehabilitation. Understanding the patient's context and how the gamified intervention could be aligned with their needs, during hospital stay, is therefore necessary.

This paper aims to determine whether and how the use of a gamified rehabilitation equipment adds value to the rehabilitation process, specifically in terms of engagement and motivation to increase physical activity.

5.3.Methodology

We conducted a randomized clinical trial and a qualitative study with 57 patients, 65+ years old, for 3 weeks to measure the added value of the REACH concept. REACH is an EU-funded project within the Horizon 2020 program. The project acronym stands for Responsive Engagement of the Elderly Promoting Activity and Customized Healthcare. The intervention consists of a Kinect-based virtual rehabilitation system (ActivLife) coupled with a rehabilitation machine (Fig 1). Elderly patients' activities were monitored via an ankle wearable tracker (Stepwatch) (Fig 1). The study has been approved by the Institution's Ethical Committee (No 2018-01516).

5.3.1. Participants recruitment

We recruited the participants from two urban rehabilitation hospitals in a large city in Switzerland. Patients were randomly assigned in the Intervention Group (N=35) of using the gamified rehabilitation equipment 3 times a week or to a Control Group (N=22) of usual rehabilitation therapies. Our inclusion criteria were: 65+ years old, hospitalized in one of the involved sites, with musculoskeletal issues (hip fracture, knee or hip prosthesis, falls and low back pain withour vertebral fracture), able to remain standing with help, and understanding the instruction.

5.3.2. Materials

Participants in the intervention group used two devices: the rehabilitation equipment (ActivLife) and the wearable sensor (Fig 1). ActivLife allowed the participants to do their exercise via a series of rehabilitation game while Stepwatch measured the evolution of their walking activity during the study.

ActivLife is an activation equipment coupled with a rehabilitation game. The games and the instructions are displayed on a screen in front the patient who is secured in a mechanical platform ActivLife. ActivLife allows to maintain a safe and upright standing position while performing balance, upper limbs and lower limbs exercises. The device is also equipped with a mechanism assisting the patient in standing up. Performing squats with the equipment has been shown to increase muscle strength on the lower parts of the body (Les et al 2017). The screen has a Kinect sensor that allows the software to determine if the patient is doing the exercise correctly. The software allows the physical therapist to program and schedule a specific treatment (series of rehabilitation games) for each patient.

Stepwatch is a wearable sensor that measures the activity of the patient in terms of number of steps, activity (low, medium, high), cadence and velocity. It is a wearable watch that does not display any information and can be worn on the ankle. Stepwatch demonstrated to capture small changes in step rate (99% accuracy (Cindy Ng et al 2012; McCullagh et al 2017)), thus it can be used to assess changes in physical activity in individuals who walk slowly or use a walking aid such as a rollator. Furthermore, it allows to collect the data locally, which ensure the patient's privacy.

Fig 1. ActivLife / Stepwatch

5.3.3. Procedure

Fig 2 provides an overview on the study procedure.

The hospital electronic medical records of all consecutive new hospitalized patients were accessed (from February to June 2019) to identify potential participants. All patients who fulfilled the eligibility criteria were screened by two medical doctors (in each of the sites) and one nurse. Two researchers then present the project and provided written information about the study to the identified subjects. If the contacted patient agreed to participate, researchers ask the patient to sign an informed consent form. Participants were then allocated randomly to

one of the two arms of the trial. The randomization was based on a single allocation ratio. Due to the intervention type, allocation was not masked to participants in the intervention and control groups or to researchers who recruited patients.

Intervention consisted in five therapies per week: three 30 minutes therapies using ActivLife and 2 standard physical therapies Participants in the control group wore the Stepwatch sensor and followed usual rehabilitation program, consisting in five standard physical therapies (standard care) per week. The rehabilitation was performed under the supervision of physical therapists. Both groups were monitored constantly with the Stepwatch sensor and participated in training sessions of 30 minutes every day.

5.3.4. Assessment

Semi-structured interviews were conducted to collect older patient's inputs at time t0 (before the intervention) and t1 (end of the hospitalization). Interviews lasted between 17 minutes to 38 minutes, with a mean of 26 minutes. Steps data collected by the sensor was coded on collection.

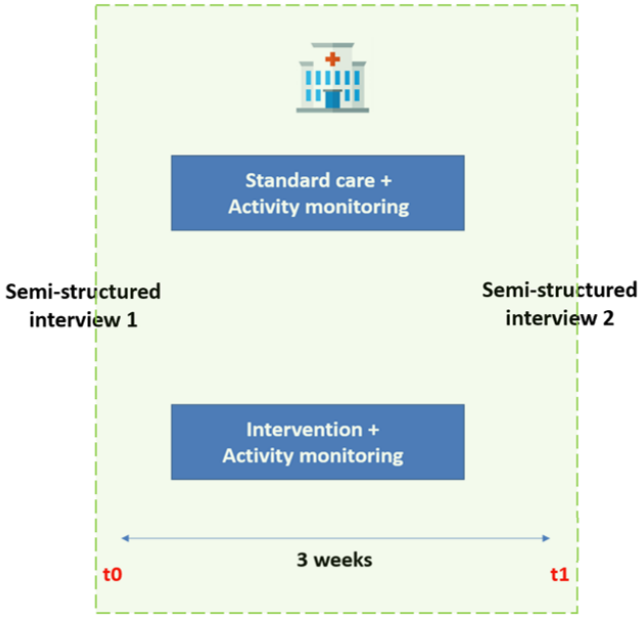


Fig 2. Research procedure

5.3.5. Data analysis

We audio-recorded all interviews (total of 3,990 min), transcribed them into English, and analyzed the resulting 546,630 words according to the grounded-theory approach. After reading all the transcripts, we generated a starting list of codes based on the research questions and the patterns that emerged from the data. We then returned to the data and conducted a systematic axial coding to identify emergent themes. After several coding iterations, we identified and categorized the themes and reached a consensus. Orange data mining software and TreeCloud.org were used to visualize interviews results and confirm the coding originally performed by the researchers.

5.4.Findings

Table 1 summarizes the attitudes before the intervention and behavior changes after the intervention.

5.4.1. Before intervention (t0)

Results from the first semi-structured interview conducted before the intervention are reported here. Participants from intervention group are described with prefix I (I1,…, I35) where as participants from control group are described with prefix C (C1,…, C22).

5.4.1.1.Population characteristics

The participants were 39 females (68%). Their age ranges between 67 and 95, with a mean age of 81.5 ± 6.8. Two thirds of the participants live alone, and the rest live with their family. Although 6 participate in gymnastics for seniors, most of them indicate moderate walking as their main physical activity. The majority (41) enjoys light activities such as watching TV, reading newspapers, knitting or playing word games as a hobby. News and animal documentaries are the most watched emission. Six of the participants fancy technology and mention it as a distraction to do video and photo editing or follow-up on previous professional activities. One participant gives courses on radio technology and telegraphy. Most of them use landline phones for communication. Twenty-three have smartphones, tablets or computers, and among them, eight use social media applications (WhatsApp□), mostly motivated by their family. Two used a pedometer before but stopped using it due to loss and two uses an application on the phone to monitor activity.

5.4.1.2.Elderly attitudes towards rehabilitation process

The geriatric rehabilitation process is often linked to complex care issues involving a panel of health professionals throughout their care path. The participants shared their view on the rehabilitation process at the hospital.

Forty-five patients (79%) reported not having any specific habits but only following hospital routine. I3 said *"It's the same every day. Days start with morning care, breakfast, toilet, another care… a little bit later there is a physiotherapy session. Not too long after that, it's lunch again. The afternoon is calm. We finish with diner around 6 pm and evening care."* C4 mentioned *"There are lots of caregivers coming and going all day. It can be tiring sometimes but that's the hospital rhythm, so we follow along."* Thirty-two patients (56%) admitted feeling bored being at the hospital. I8 said *"Days here are very monotonous. I wonder what I am doing here."* C1 reported *"We are not doing much here, I am a little bored."*

5.4.1.3.Elderly attitudes towards physical activity

Regarding elderly opinion about physical activity at the hospital, different points of view emerged from the discussion.

Thirty-five patients (61%) reported pain and tiredness after the surgery as the main barrier for being active. I6 said *"Normally, I am pushed in a wheelchair. However, once I took the rollator to the cafeteria. I shouldn't have because after that I was very tired."* C3 stated *"I can barely stand up and I find myself shaky. So, I rest regularly in the chair or take naps."* I4, I23, C6 reported having pain while trying to walk. I3 mentioned *"Sometimes, I try go outside but when I make some effort, it is painful, so I go back to my room."* Fourteen patients (24%) stated being at the hospital to rest and not to exercise. I33 said *"I am a patient, so I stay on my bed. I am here to rest."* C1 stated *"I am doing nothing here, but this rhythm is fine with me."* I24 and I27 reported sleeping a lot due to lack of energy, concentration and motivation.

Forty-one patients (72%) reported sitting or light walking in the corridor as their main activity at the hospital. C7 said *"Yesterday, for the first time, I went all the way down the hall*". I13 mentioned *"In the morning, I often sit in the hallways. I play word games or chat with the person next to me."* However, ten patients (17%) stated making the effort to go outside. I8 said *"I go outside three times a day because there's nothing to see in the corridor."* C10 mentioned *"After lunch, I go to the park which is very pleasant, then I sit there to read newspapers."* I26 reported *"As soon as I can go outside, I'm out. I have pain, but that doesn't stop me from going outside."*

5.4.1.4.Elderly technology readiness

We attempt to understand older adults' attitudes towards innovation. Today, technological solutions are not always tailored to senior citizen's requirements and capacity, which results in a lack of interest on their part (Olphert et al 2009). However, fourteen patients (25%) showed positive interest and were in favor of the progress brought by technology. C2 said *"It is so useful because you can get in touch with other people far away from you." I26* mentioned *"I am so happy to have my phone. I also have my computer here, it's in the closet. It's a good distraction."* I8 added *"It's just fantastic. If you look at scanners or MRIs, it became much easier to detect pathologies. Especially, technologies used in healthcare are really useful."* Eight participants (14%) reported actively using social media to communicate with their friends and family. C6 said *"My son wanted to teach me how to take pictures, but I didn't need him, I knew how to do it. I use my phone for calls, text messages, calendar and WhatsApp."* I2 mentioned *"I use WhatsApp and SMS to share videos and news to my family."* I3 said *"I am on Facebook. I use it a lot. I don't have any privacy concerns because I have a very limited number of friends. I only use WhatsApp to chat with the children."* Still, twelve participants (21%) reported disinterest in technology mainly due to a lack of ability. C7 said *"Yes we have internet but it's not for me. I don't know how to use it."* C4 mentioned *"It can be annoying sometimes when it does not work."* I9 stated *"In practice, it's catastrophic. I've already demolished computers and thrown them away in the trash because it did not work."*

Nine patients (16%) heard about pedometer before in different circumstances. C7 said *"I've always wanted to have a step counter when I had a dog, but it was quite expensive."* I23 reported *"Yes I know, it gives your number of steps. I have a friend who walks a lot, occasionally, he wears a pedometer."* Four participants (7%) had previous experience with sensing technology and encountered difficulties. I3 said *"I used the step counter in my phone.*

But, sometimes even if I walk, it shows 0. There is something I am doing wrong maybe." Despite, hearing about sensing technology and serious game for the first time, the majority expressed interest in trying something that could potentially help their rehabilitation. C1 said *"I don't mind using them if it's something I really need."* I4 stated *"Why not? I can be fun"* I26 said *"If it is useful, then there is no problem."*

5.4.1.5.Summary

Before using the different devices, our intent was to understand older patients' attitudes towards the rehabilitation process, their motivation for physical activity as well as their technology readiness. Table 1 summarizes the findings. Participants admitted feeling bored at the hospital and only following along the care process routine. Sitting in the hallways, reading newspapers,playing word games and light walking represents their main activity at the hospital. Pain, lack of energy and motivation after surgery were identified as the main barriers to being active. Furthermore, although some patients (N=12) reported disinterest in technology, mainly due to a lack of ability, the same proportion (N=14) showed enthusiasm and were in favor of the progress brought by technological solution. Interest in trying a tool that could potentially help their rehabilitation was observed.

Before intervention			After intervention		
Attitudes	Both group		Behavior changes	Control group	Intervention group
Attitudes towards rehabilitation	No specific habits, only following hospital routine	45 (79%)	Showed positive change in their behavior	5 (31%)	15 (60%)
	Feeling bored	32 (56%)	Still reported pain and lack of confidence	7 (44%)	3 (12%)
Attitudes towards physical activity	Sitting or light activity in the hallways as main activity	41 (72%)	Felt more active	No mention	10 (40%)
	Pain and tiredness after surgery as main barrier to be active	35 (61%)	Took part in social walking group	No mention	3 (12%)
	Consider hospital stay as a resting time	14 (24%)			
	Making slight effort to go outside	10 (17%)			
Elderly technology readiness	Positive interest	14 (24%)			
	Disinterest due to lack of ability	12 (21%)			
	Using social media	8 (14%)			
	Used/using step counter	4 (7%)			

Table 1. Attitudes before the intervention and behavior changes after the intervention

5.4.2. After intervention (t1)

Results from the second semi-structured interview conducted after the intervention (meaning at the end of hospitalization) are reported here.

	Mean (nb steps / day) during the last week at the hospital	Nb of patients with less than mean 1000 steps / day (%)
Intervention group	2375	3 (12%)
Control group	1696	7 (44%)

Table 2. Mean number of steps per day during the last week of hospitalization

During the study, sixteen participants (Intervention Group=10; Control Group=6) dropped out. Seven wanted to stop, four were transferred to another hospital, three were found not eligible, one left the hospital earlier and one died. Here, we proceed to the analysis of the remaining forty-one participants (Intervention Group =25: Control Group =16). The mean number of steps per day (Steps_Intervention_Group=2375; Steps_Control_Group=1696) (Table 2) showed that participants from the intervention group were more active during the last week of hospitalization compared to the control group. Three patients from the intervention group VS seven patients from the control group had less than a mean 1000 steps/day during the last week of hospitalization.

5.4.2.1.Acceptance and adoption

Intervention group

At the end of hospitalization, twenty-three patients (92%) from the intervention group reported positive feedback towards the gamified rehabilitation. They particularly enjoyed the playfulness as it brought back family memory. I6 said *"Frankly, I am like a kid. I like to play these games. I really enjoyed it. When I throw balls at cans, it reminds me of my childhood."* I23 mentioned *"It's nice and funny. It makes me thinks of my grandchildren when they challenge each other."* I1 said *"The games are funny. If my little ones could see that, they'd be so happy. It's like a child game."*

Eighteen patients (72%) reported the gamified intervention to be stimulating and challenging. They were curious and motivated to do more. I1 stated *"When these pawns were passing by on the screen and I could shoot them all, I was so happy. I told myself, today I did well."* I4 mentioned *"I was wondering what the next step in the game was."* I8 said *"When I take the conventional stairs, I easily stop. But here, there is a challenge. It motivates me. So, I did my best."* I3 reported *"Yes, it was a motivation, absolutely. I think I would have made less effort if I was doing the standard physiotherapy."*

Seventeen patients (68%) shared about feeling safe while doing the exercise inside the equipment. I6 said *"I did not even worry about safety, it was secure."* I24 stated *"I felt safe, there is no way I could fall when I am inside the equipment."* Few participants, who were still quite weak, felt a bit unstable inside the platform at first, but later gained confidence. I13 reported *"At the beginning, I was afraid to fall, so when I went into the equipment, I was afraid too, but later, I gained some stability, so it was fine inside the equipment as well."*

Fifteen patients (60%) acknowledged that the serious game encouraged self-learning. They felt engaged and more confident in doing the exercise. I4 said *"It started with things that seem natural, like cleaning the window. But then, I understood that I must go to the edges,*

that I should bend down and go up front. Little by little, I got caught up in the game." I18 reported *"I loved it. I even dreamt about it at night. Then, I asked the physiotherapist, if I could start cleaning my windows once I get back home, but she told me to wait."* I25 stated *"As the games progressed, I felt that all my body and my senses were stimulated."* I3 was happy to share *"I was very studious and rigorous during the exercises. According to the physiotherapist, I am the best here."*

Fig 3. Patients' feedbacks towards the gamified rehabilitation

Ten patients (40%) appreciated the intervention as a nice distraction from the hospital context. I9 said *"This made my hospital stay less boring."* I4 mentioned *"It's a very good idea because, little by little, we lose sight of our disability. It made me forget that I was at the hospital."* I1 reported *"It was a funny way to make the time pass." I19 added "It was nice. We forgot a bit about our troubles. I liked it."*

Even though they enjoyed doing the exercises, twelve patients (48%) felt exhausted after each session. I22 said *"I had a lot of fun but, after a while, I got tired of cleaning the windows with my arms."* I27 reported *"I was enjoyable but still a little physical, when you really try to do it correctly."* I7 stated *"It's good, but it's exhausting. For me, the most tiring games were the plane and the ambulance."*

Fig 3 illustrates older patients' feedbacks regarding the gamified rehabilitation.

5.4.2.2.Behavior change

Intervention group

At the end of the hospitalization, fifteen patients (60%) reported a positive change of their behavior at the hospital. They replaced some of their sedentary activities such as sleeping, sitting in the hallways or playing word games to going outside and breathing fresh air. I8 said "*Sometimes, I go out for a walk in the garden, to take as many steps as I can. It last about an hour."* I23 mentioned *"I try to move around a bit, it doesn't suit me to be seated all the time."* I28 reported *"In the afternoon, I circle the park twice. Then, I sit there and read a book. I was never in my room lately."* I3 added *"We're at the end of our stay here, so I am making some effort to move."*

Ten patients (40%) felt more active and observed significant improvement in their general condition. I22 said *"I think I have more confidence now when I walk. A little out of breath though, but I'm less tired than before when I move. I am more active, because there are things, I was afraid to do that I can do now, like walking for example."* I9 went through a rapid progression. She started from using a wheelchair, to a walking support, then two canes, and walked without any auxiliary in the end. I6 reported *"I feel like I've come a long way back. I'm completely different. I have made a lot of progress and my friends have also really seen the difference. I am more active and much more relaxed."*

I4 particularly showed a significant improvement thanks to the pedometer during his hospital stay, as he went from 3000 steps to 9000 steps in just one week. He reported *"I'm definitely more active. I feel motivated. When I go in the garden, I make sure I've done my tours, before I sit on a bench to read something. Before all of this, I didn't care about the number of tours I made. I just sat somewhere in the park. But now, my goal has changed as I have this step counter. I want to do more. It's a stimulus. I want to see how many steps I could make if I finish 7 or 8 tours of the garden. When I succeed, I'm satisfied."* I4 set a personal goal of doing at least 6 tours of the park every day. After a certain period, he was able to exercise and walk without feeling any constraint. He stated *"If I'm given a challenge, I'll play along. I tried to achieve my steps goals without sweating. I wanted to do it naturally. Now, doing 6 tours in the garden has become natural for me. Later, I think, I would make 10,000 steps without any problem, if my leg doesn't hurt."*

Three patients (12%) took part in social activity organized by the hospital motivating them to exercise more. I3 shared *"Every Tuesday and Wednesday afternoon I exercise with the walking group. The caregivers proposed me some activity, so I chose what I like."* I7 said "*I'm walking with a group of people. We exercise and walk together about one hour every week."*

However, three patients (12%) were not confident enough to engage in daily physical activity during their hospital stay. I22 said *"With the rollator, I can go to the dining room by myself, but when I walk with the canes, I still need someone next to me."* I24 added *"I feel stronger, but I still have this fear of falling. I don't go out alone in the hallways because I'm afraid to lose my balance."* I32 reported *"I think I've made some progress, but I move considering my fatigue and my need for rest. I'm 87 years old, you know. I feel much less brave."*

Control group

At the end of the hospitalization, five patients (31%) reported positive improvement in their physical activity, which occurred mostly during family visits. C4 said *"In the morning, I take 30 minutes' walk outside. Sometimes, I just walk to the living room or the cafeteria to see the lunch menu and if there are some interesting cakes. In the afternoon, I have many visitors. Most of the time, I go to the park with them."* C1 mentioned *"Sometimes, I have visitors coming in the afternoon. When the weather was nice, we happen to go for a walk in the sun."* C15 stated *"I think I'm walking more now. I've been to all the places here: the garden, the library, the dining room and the corridors."*

Seven patients (44%) still reported pain and lack of confidence as barrier to being more active. C6 said *"I'm still feeling pain when I walk but I guess it's normal after surgery. Anyway, I do feel better now. I don't go in the hallways or outside because it's already difficult to walk and come back from the dining room. It's already stressful."* C3 stated *"I still have pain when I walk. Yesterday, my legs were fluttering".* C2 mentioned *"At one point, I wasn't feeling so well. Then, we did some exercises rather sitting down."*

C24 did not notice any changes in her habits. She appreciated being able to rest during her hospital stay. She shared *"I am hyperactive, so I enjoyed doing nothing here."*

5.4.2.3.Summary

At the end of the hospitalization, our intent was to understand older patients' acceptance and adoption of the gamified rehabilitation equipment and to figure out the behavior changes in both groups. Most of the patients (92%) from the intervention group, reported a positive feedback towards the serious game-based rehabilitation. The intervention was identified as fun, challenging, allowing self-learning, improving confidence, distracting but also exhausting. Positive change on the behavior at the hospital was observed, mainly from the intervention group. Patients felt more active and observed significant improvement in their general condition. They replaced some of their sedentary activities such as sleeping, sitting in the hallways or playing word games to going outside and breathing fresh air. Few of them engaged in social walking group. However, pain and lack of confidence were still identified as barrier to engage in daily physical activity, mainly from the control group.

5.5.Discussion

Our findings shed light on the older patients' attitudes towards rehabilitation process, their motivation for physical activity as well as their readiness for innovative technology. Acceptance and adoption of serious game-based rehabilitation and the resulting behavior change at the end of the hospital stay were also presented.

5.5.1. Opportunities and challenges

Opportunities and challenges for designing serious game-based rehabilitation interventions for elderly emerged from the discussion.

Bachmann et al (2010) demonstrated in their systematic review the potential of geriatric inpatient rehabilitation to improve functional capacity. However, understanding older patients' perception of the care process during hospitalization remains complex (Krevers et al 2002). Despite a few numbers of patients who viewed the hospital stay as a resting time, most of our patients perceived the rehabilitation process as long, repetitive and boring. This opens an opportunity for gamified rehabilitation designers to propose different solutions aiming at breaking this monotony and improving the recovery process. Using computer games to augment physical and cognitive rehabilitation allows a significant therapeutic benefit as well as a stronger engagement from the patient (Rego et al 2010). Furthermore, although some patients reported disinterest in technology mainly due to a lack of ability, the same proportion

showed enthusiasm and were in favor of the progress brought by the technological solution. In fact, inequalities between the younger and older generation, in terms of accessing the latest technological innovation (Fischer et al 2014; Hanson 2010), could make it more difficult for the elderly to use them properly. Nevertheless, as the benefits from using information technology arise, such as contacting family and friends or getting medical assistance, a potential long-term adoption of such tools is therefore conceivable (Macedo 2017; Vassli and Farshchian 2018). In our case, interest in trying a solution that could potentially help our patients' rehabilitation were observed.

Understanding the elderly patient's context and characteristics is necessary to design a successful rehabilitation intervention (Barbosa et al 2018). Most of our participants, reported pain and tiredness after the surgery as the main barrier for being active. Due to the complexity of their health problems, geriatric patients are especially vulnerable (Grimmer et al 2013) during their rehabilitation. Therefore, gamified solutions need to be customizable enough to adjust to the patient's conditions. In addition, our experiment exposed additional challenges associated to conducting study with older patients in a real-world hospital setting (Mackichan et al 2013). Subsequent drop-outs (around 30%) were observed from our study group. They were mainly caused by hospital transfer, loss of eligibility associated to a health conditions decline and a desire to stop participating in the trial. Listening to their doubts and worries as well as emphasizing the benefits of the intervention is therefore necessary to sustain elderly patients' involvements.

5.5.2. Intervention's impact

At the end of the hospitalization, most of the patients from the intervention group, reported a positive feedback towards the serious game-based rehabilitation. The intervention was identified as fun, challenging, allowing self-learning, improving confidence, distracting but also exhausting. In fact, previous research (Ling et al 2017; Idriss et al 2017; Brox et al 2017) suggests that serious games increase motivation in rehabilitation sessions which consist of repetitive exercises. Game characteristics which generally involve cognitive and motor skills represent an effective way to retain elderly's attention (Krichevets et al 1995) and distract them from pain and boredom (Burke et al 2009). Furthermore, the possibility to increase the game level tailored to the patient's ability gives a sense of challenge followed by a feeling of satisfaction when achieving the task (Lumsden et al 2016). Finally, in addition to the game components, our intervention was composed of an efficient trunk stabilization reassuring the patients while engaging in different movements such as "cleaning the window", "guiding an ambulance" or "flying a dragon" (Kozak et al 2017). They were therefore encouraged to perform upper limbs exercise in a safe manner.

Although not described in this paper, our serious game-based intervention demonstrated to be as effective as the standard rehabilitation in improving functional capacity of the elderly (Randriambelonoro et al). Additionally, positive change on the behavior at the hospital was observed in both groups but mostly in the group who received the gamified intervention. Patients felt more active and observed significant improvement in their general condition. They replaced some of their sedentary activities such as sleeping, sitting in the hallways or

playing word games to going outside and breathing fresh air. Few of them engaged in social walking groups. Similar results, namely, an improved mood and quality of life of patient with Parkinson's disease, were observed after a one-month usage of the Nintendo Wii (Herz et al 2013). Also, despite their relatively small sample size, Ma and Beckhoum (2008) shared the benefits of using serious games in movement therapy after stroke. Finally, Hall and al, concluded in their systematic review (Hall et all 2012) that most of the studies revealed significant positive effects on health outcomes associated with digital videogame play among older adults. Our findings reinforce the potential of serious game-based rehabilitation in engaging elderly patients to be more active.

5.5.3. Design recommendations

Based on our findings, we derived four recommended components for serious game-based rehabilitation designers: playfulness, personalization, performance feedback and safety.

5.5.3.1.Playfulness

Most of our participants enjoyed playfulness as it brought back family memory. Designers of serious game for elderly should consider involving games that reminds them of their childhood or the time spent with their family, especially their grandchildren. Some participants started to take part in walking groups or reported to go outside mainly with visitors. Fostering social interaction through the game by allowing two or more patients to compete or collaborate side by side, could be a vector of extrinsic motivation to exercise more. Finally, designing incentives related to their hobbies and everyday habits may engage them more in being physically active. As an example, when reaching a certain level through the game, patients could receive clues for their word games.

5.5.3.2.Personalization

We recommend a dynamic system, enabling a personalized goal setting, where the game level increases or decreases depending on the patient's improvement and condition. Our participants were eager to know and discover the next of steps in the game. Out of curiosity, some of them even targeted to try all the available games. Depending on the patient's evolution, the game should be easily adaptable and customizable. Additionally, the challenge should be focused on movement precision, especially in the rehabilitation domain. For example, if during the first week, the goal was to make a 90-degree angle with the left arm, the second week the patient could be asked to reach 120-degree angle. In case the patient is feeling pain due to any adverse events, the game level should be downgraded accordingly.

5.5.3.3.Performance feedback

Real time performance feedback is an important component of any gamified intervention. In this specific case, it is even more essential for patients while doing the exercise, to know if they are doing it correctly. Most of our patients reported to be very happy to discover that they have reached their goal. Performance feedback has the potential to bring self-confidence

to the player. Designers should consider implementing visible and easy to understand evaluation of the patient's rehabilitation exercise to increase their self-efficacy.

5.5.3.4.Safety

Safety represents a fundamental aspect to be covered while designing serious games for elderly patients' rehabilitation. All the intervention group patients, in our study, felt safe doing any movements thanks to a trunk stabilization preventing them for falling. However, this could limit the possible range of motion of the patient during the exercise. Designers should propose a safe rehabilitation solution considering the different movements required by the physiotherapist for treating a specific pathology.

5.5.4. Limitations

Our study presents different limitations. The sample size of the study was small, making it difficult to conclude firmly on the impact of serious game rehabilitation on elderly physical activity. The study duration was also relatively short. Nurse and physiotherapists inputs would also deserve further investigation. Effect after a long-term usage of the gaming components deserves further investigation. Results of the 3-weeks follow-up at home will be reported in another paper.

5.6.Conclusion

This paper presents a qualitative analysis of a randomized clinical trial to investigate how older patients are engaged and motivated through a gamified rehabilitation equipment during hospitalization. Before using the different devices, our intent was to understand older patients' attitudes towards rehabilitation process, their motivation for physical activity as well as their readiness for innovative technology. At the end of the hospital stay, our study showed that patients who received the intervention demonstrated stronger engagement in the rehabilitation process and exhibited a higher motivation to be active. Different recommendations on designing serious game for elderly rehabilitation are outlined, especially playfulness, personalization, performance feedback and safety. All in all, our findings suggest the potential short-term impact of serious game-based intervention on elderly motivation for rehabilitation program in a hospital setting.

5.7.Acknowledgements

This project has received funding from European Union's Horizon 2020 research and innovation program under grant agreement No 690425.

5.8.References

[1] United Nations. Department of Economic and Social Affairs. World Population Prospects 2019: Highlights. (2019). https://population.un.org/wpp/Publications/Files/WPP2019_10KeyFindings.pdf Accessed 02 September 2019

[2] World Health Organization. 10 facts on ageing and health. (2017). https://www.who.int/features/factfiles/ageing/en/ Accessed 02 September 2019

[3] Covinsky, K. E., Palmer, R. M., Fortinsky, R. H., Counsell, S. R., Stewart, A. L., Kresevic, D. ,et al. (2003). Loss of Independence in Activities of Daily Living in Older Adults Hospitalized with Medical Illnesses: Increased Vulnerability with Age. *Journal of the American Geriatrics Society*. 51: 451-458. https://doi.org/10.1046/j.1532-5415.2003.51152.x

[4] Kwok, C. L., Lee, C. K., Lo, W. T., & Yip, P. S. (2016). The Contribution of Ageing to Hospitalization Days in Hong Kong: A Decomposition Analysis. *International Journal of Health Policy and Management*, 6(3), 155–164. https://doi.org/10.15171/ijhpm.2016.108

[5] Ling, Y., Ter Meer, L.P., Yumak, Z., & Veltkamp, R.C. (2017). Usability Test of Exercise Games Designed for Rehabilitation of Elderly Patients After Hip Replacement Surgery: Pilot Study. *JMIR Serious Games*;5(4):e19. https://doi.org/10.2196/games.7969

[6] Lang, C.E., Macdonald, J.R., Reisman, D.S., Boyd, L., Jacobson Kimberley, T., Scindler-Ivens, S.M. et al. (2009). Observation of amounts of movement practice provided during stroke rehabilitation. *Arch Phys Med Rehabil*. 90:1692–1698. https://doi.org/10.1016/j.apmr.2009.04.005

[7] Burke, J.W., McNeill, M.D., Charles, D., Morrow, P., Crosbie, J.H., & McDonough, S.M. (2009). Optimizing engagement for stroke rehabilitation using serious games. *Visual Computer*, vol. 25, 1085-1099. https://doi.org/10.1007/s00371-009-0387-4

[8] Liddle, S., Baxter, D. & Gracey, J. (2007). Chronic low back pain: Patients' experiences, opinions and expectations for clinical management. *Disability and Rehabilitation*. 29:24, 1899-1909. https://doi.org/10.1080/09638280701189895

[9] Huotari, K., & Hamari, J. (2012) Defining Gamification – A Service Marketing Perspective. *Proceedings of the 16th International Academic MindTrek Conference 2012*. Tampere, Finland. 17-22. https://doi.org/10.1145/2393132.2393137

[10] Deterding, S., Dixon, D., Khaled, & R., Nacke, L. (2011). From game design elements to gamefulness: Defining "gamification". *Proceedings of the 15th International Academic MindTrek Conference 2011*. Tampere, Finland. 9-15. https://doi:org/10.1145/2181037.2181040

[11] Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does Gamification Work? A Literature Review of Empirical Studies on Gamification. *47th Hawaii International Conference on System Sciences (HICSS)*. Waikoloa. 3025-3034. https://doi.org/10.1109/HICSS.2014.377

[12] Johnson, D., Deterding, S., Kuhn, K., Staneva, A., Stoyanov, S., & Hides, L. (2016). Gamification for health and wellbeing: A systematic review of the literature. *Internet Interventions.* Vol 6, 89-106. ISSN 2214-7829, https://doi.org/10.1016/j.invent.2016.10.002

[13] Cugelman B. (2013). Gamification: what it is and why it matters to digital health behavior change developers. *JMIR serious games*, 1(1), e3. https://doi.org/10.2196/games.3139

[14] Lister, C., West, J.H., Cannon, B., Sax, T., & Brodegard, D. (2014). Just a Fad? Gamification in Health and Fitness Apps. *JMIR Serious Games*;2(2):e9. https://doi.org/10.2196/games.3413

[15] Skriloff, S., Gonzalez, D., Christensen, K., Bentley, L. & Mortensen, C. (2016). FitPlay Games: Increasing Exercise Motivation Through Asynchronous Social Gaming. *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems.* 164-167. https://doi.org/10.1145/2851581.2890367

[16] Lin, J.J., Mamykina, L., Lindtner, S., Delajoux, G., & Strub, H.B. (2006). Fish'n'Steps: Encouraging Physical Activity with an Interactive Computer Game. *Proceedings of the 8th International Conference on Ubiquitous Computing.* 4206. 261-278. https://doi.org/10.1007/11853565_16

[17] Consolvo, S., McDonald, D.W., Toscos, T., Chen, M.Y., Froehlich, J., Harrison, B., et al. (2008). Activity sensing in the wild: a field trial of UbiFit Garden. *In Proc. CHI 2008, ACM Press*. https://doi.org/10.1145/1357054.1357335

[18] Rice, M.D., Sun, M.W.T., Ling, J.N.S, Foo, M.H.N., Janell, K., Sheng, S.L.W., et al. (2011). Evaluating Gesture-based Games with Healthy Older Adults on a Large Screen Display. *CM SIGGRAPH Symposium on Video Games*. https://doi.org/10.1145/2018556.2018560

[19] Meekes, W., Stanmore, E.K. (2017). Motivational Determinants of Exergame Participation for Older People in Assisted Living Facilities: Mixed-Methods Study. *J Med Internet Res*;19(7):e238. https://doi.org/10.2196/jmir.6841

[20] El-Hilly, A.A., Iqbal, S.S., Ahmed, M., Sherwani, Y., Muntasir, M., Siddiqui, S., et al. (2016). Game On? Smoking Cessation Through the Gamification of mHealth: A Longitudinal Qualitative Study. *JMIR Serious Games*;4(2):e18. https://doi.org/10.2196/games.5678

[21] Randriambelonoro, M., Perrin, C., Blocquet, A., Kozak, D., Fernandez, J.T., Marfaing, T., et al. Computer-aided physical rehabilitation of older people: a pilot non-inferiority randomized clinical trial. To be submitted to the Archives of Physical Medicine and Rehabilitation

[22] Les, A., Niedzielska, E., Piotrowska, J., Staniszewski, M, Kozak, D. (2017). Analysis of the effectiveness of ActivLife training in people over 60 years of age: A pilot study. *Gerontechnology.* 16(3):189-19. https://doi.org/10.4017/gt.2017.16.3.008.00

[23] Cindy Ng, L.W., Jenkins, S. & Hill, K. (2012) Accuracy and responsiveness of the stepwatch activity monitor and ActivPAL in patients with COPD when walking with and without a rollator. *Disability and Rehabilitation*. 34:15, 1317-1322, https://doi.org/10.3109/09638288.2011.641666

[24] McCullagh, R., Dillon, C., O'Connell, A.M., Horgan, F., & Timmons, S. (2016). Step-Count Accuracy of 3 Motion Sensors for Older and Frail Medical Inpatients. *Archives of Physical Medicine and Rehabilitation*. Volume 98, Issue 2, 295-302. https://doi.org/10.1016/j.apmr.2016.08.476

[25] Olphert, W., Damodaran, L., Balatsoukas, P., & Parkinson, C. (2009). Process requirements for building sustainable digital assistive technology for older people. *Journal of Assistive Technologies*. 3(3):4-13. https://doi.org/10.1108/17549450200900019

[26] Bachmann, S., Finger, C., Huss, A., Egger, M., Stuck, A. E, Clough-Gorr, K.M. et al. (2010). Inpatient rehabilitation specifically designed for geriatric patients: systematic review and meta-analysis of randomised controlled trials. *BMJ*; 340 :c1718. https://doi.org/10.1136/bmj.c1718

[27] Krevers, B., Närväanen, A.L., & Öberg, B. (2002). Patient evaluation of the care and rehabilitation process in geriatric hospital care. *Disability and Rehabilitation.* 24:9, 482-491. https://doi.org/10.1080/09638280110105268

[28] Rego, P., Moreira, P.M., & Reis, L.P. (2010). Serious games for rehabilitation: A survey and a classification towards a taxonomy. *5th Iberian Conference on Information Systems and Technologies*. Santiago de Compostela. pp. 1-6.

[29] Fischer, S.H., David, D., Crotty, B.H., Dierks, M., & Safran, C. (2014). Acceptance and use of health information technology by community-dwelling elderly. *Int. J. Med. Inform.* 83(9), 624–635. https://doi.org/10.1016/j.ijmedinf.2014.06.005

[30] Hanson, V. (2010). Influencing technology adoption by older adults. *Interacting with Computers*. 22. 502-509. https://doi.org/10.1016/j.intcom.2010.09.001

[31] Macedo, I.M. (2017). Predicting the acceptance and use of information and communication technology by older adults: an empirical examination of the revised UTAUT2. *Comput. Hum. Behav*. 75, 935–948. https://doi.org/10.1016/j.chb.2017.06.013

[32] Vassli, L.T., Farshchian, B.A. (2018). Acceptance of health-related ICT among elderly people living in the community: a systematic review of qualitative evidence. *Int. J. Hum. Comput. Interact*. 34(2), 99–116. https://doi.org/10.1080/10447318.2017.1328024

[33] Barbosa, H., Castro, A., & Carrapatoso, E. (2018). Serious Games and Rehabilitation for Elderly Adults. *Global Scientific Journal (GSJ).* 6. 275-283.

[34] Grimmer, K., Beaton, K., Kumar, S., Hendry, K., Moss, J., Hillier, S., et al. (2013). Estimating the risk of functional decline in the elderly after discharge from an australian public tertiary hospital emergency department. *Australian Health Review: A Publication of the Australian Hospital Association*. 37(3),341–347. https://doi.org/10.1071/AH12034

[35] Mackichan, F., Adamson, J., & Gooberman-Hill, R. (2013). 'Living within your limits': Activity restriction in older people experiencing chronic pain. *Age and Ageing*, 42(6), 702–708. https://doi.org/10.1093/ageing/aft119

[36] Idriss, M., Tannous, H., Istrate, D., Perrochon, A., Salle, J.Y., Ho Ba Tho, M.C., et al. (2017). Rehabilitation-Oriented Serious Game Development and Evaluation Guidelines for Musculoskeletal Disorders. *JMIR Serious Games*;5(3):e14. https://doi.org/10.2196/games.7284

[37] Brox, E., Konstantinidis, S.T., & Evertsen, G. (2017). User-Centered Design of Serious Games for Older Adults Following 3 Years of Experience With Exergames for Seniors: A Study Design. *JMIR Serious Games*;5(1):e2. https://doi.org/10.2196/games.6254

[38] Krichevets, A.N., Sirotkina, E.B., Yevsevicheva, E.B., & Zeldin, L.M. (1995). Computer games as a means of movement rehabilitation. *Disability & Rehabilitation*, vol. 17, no. 2, pp. 100-105, 1995.

[39] Lumsden, J., Edwards, E.A., Lawrence, N.S., Coyle, D., & Munafò, M.R. (2016). Gamification of Cognitive Assessment and Cognitive Training: A Systematic Review of Applications and Efficacy. *JMIR Serious Games* ;4(2):e11. https://doi.org/10.2196/games.5888

[40] Kozak, D., Burgermeister, S., De Chassey, J.D.B., Naef, A., Maringue, A., & Dietrich, D. (2017). A functionality, safety and validity study of innovative REACH devices. *Gerontechnology*, 16(3), 181-188. https://doi.org/10.4017/gt.2017.16.3.007.00

[41] Herz, N.B., Mehta, S.H., Sethi, K.D., Jackson, P., Hall, P., & Morgan, J.C. (2013). Nintendo Wii rehabilitation ("Wii-hab") provides benefits in Parkinson's disease. *Parkinsonism & Related Disorders*. Volume 19, Issue 11. 1039-1042. https://doi.org/10.1016/j.parkreldis.2013.07.014

[42] Ma., M., and Bechkoum, K. Serious games for movement therapy after stroke. (2008). *IEEE International Conference on Systems, Man and Cybernetics*. Singapore. 1872-1877. https://doi.org/10.1109/ICSMC.2008.4811562

[43] Hall, A.K., Chavarria, E., Maneeratana, V., Chaney, B.H., & Bernhardt, J.M. (2012). Health Benefits of Digital Videogames for Older Adults: A Systematic Review of the Literature. *Games for Health Journal*, 1:6, 402-410. https://doi.org/10.1089/g4h.2012.0046

[44] Sardi, L., Idri, A., & Fernández-Alemán, J.L. (2017). A systematic review of gamification in e-Health. Journal of Biomedical Informatics. Volume 71, 31-48. https://doi.org/10.1016/j.jbi.2017.05.011

CHAPTER 6. COMPUTER-AIDED PHYSICAL REHABILITATION OF OLDER PEOPLE: A PILOT NON-INFERIORITY RANDOMIZED CLINICAL TRIAL

This chapter will be submitted as:

Randriambelonoro, M., Perrin, C., Carmona, G., Frangos, E., Hermann, F., Geissbuhler, A., & Graf, C. (2020). Computer-aided physical rehabilitation of older people: a pilot non-inferiority randomized clinical trial. *Archives of Physical Medicine and Rehabilitation*

6.1. Abstract

Background Developed countries are facing the challenge of ageing societies, lack of infrastructure for healthcare and high cost of care. Researchers have been attempting to answer these problems by using innovative technology to promote healthy ageing. The hospital-to-home transition is increasingly recognized as a critical period in the patient care, during which different incidents can occur and induce frequent re-hospitalization. Consequently, there is a growing interest in strengthening the physical and functional capacities of hospitalized older patients to prevent re-hospitalization.

Trial design Pilot multicenter non-inferiority randomized clinical trial.

Objectives To assess the efficacy of a personal mobility device compared with standard of care in the rehabilitation treatment of patient with musculoskeletal issues.

Methods 57 patients aged 67-95 years were randomly assigned to the intervention group (n=35) using the gamified rehabilitation equipment 3 times a week or to a control group (n=22) of usual standard care. Outcome measures included SPPB (Short Physical Performance Battery), IHGS (Isometric Hand Grip Strength), FIM (Functional Independence Measure) and number of steps.

Results There were no significant differences between the control and the intervention group for the primary outcome (SPPB) or any of the secondary outcomes (IHGS, FIM and steps), which demonstrates the non-inferiority effect of the serious game-based intervention. The analysis by mixed-effects regression on SPPB showed no indication of a group x time interaction (SPPB_I_t1=-0.76, p=0.26; SPPB_I_t2=0.21, p=0.74). Although not significant, a positive improvement of more than 2kg (Right: 2.52 kg; Left: 2.43 kg) for the patient from the intervention group was observed. In addition, patients who received the intervention exhibited stronger engagement in the rehabilitation process and a higher motivation to being more active.

Conclusions Serious game-based rehabilitation could be an effective alternative for older patients to regain their functional capacities.

Trial registration This trial has been registered in the trial register ClinicalTrials.gov (NCT03847454)

Funding This research has received funding from the European Union's Horizon 2020 research and innovation program under grant agreement No 690425.

6.2. Introduction

A globally growing geriatric population and rising incidence of non-communicable diseases (NCDs) emphasize the importance of providing a healthy ageing environment [1]. Rather than the absence of disease, “healthy ageing” is defined as a process that enables older people to continue to perform activity of daily living, maintain social contact and conserve dignity [2, 3, 4]. However, as people age, the incidence of chronic issues preventing them to be independent is also increasing. Due to their polymorbid conditions, elderly people are hospitalized longer and more frequently than younger ones, which is impairing their functional capacity [5]. Therefore, a rehabilitation phase is required to regain the independence of the geriatric patient before returning home.

Musculoskeletal disorders are one of the main reasons for geriatric hospitalization in Switzerland [6]. It includes conditions that affect joints, bones, muscles, and multiple body areas or system; which lead to the need of specific orthogeriatric rehabilitation. During their hospital stay, patients with musculoskeletal issues follow rehabilitation therapy to regain their body functions and perform daily tasks such as standing, walking, taking stairs or bathing independently. Functionality at discharge is inversely proportional with the risk of hospitalization [7]. After functional recovery, the hospital-to-home transition is increasingly recognized as a critical period in the geriatric patient care, notably to prevent further functional decline and re-hospitalizations [8]. Regular physical activity remains the central point to influence these two outcomes, but it needs motivation to be maintained over time [9].

Researchers have extensively studied the use of computer-aided physical rehabilitation to promote physical activity [10, 11]. Serious games coupled with monitoring devices such as Kinect have shown potential to positively impact patient’s motivation to do rehabilitation exercises [12, 13]. However, whether such devices would be as efficient as the standard care rehabilitation in the hospital and engage the elderly to remain active after discharge is still understudied.

The objective of the trial was to compare a gamified rehabilitation device with the standard of care in terms of efficacy in helping elderly patients to regain their functional capacities. Our hypothesis is that the patients in the intervention group would be as functional as the patients in the control group in terms of strength, speed and balance.

6.3. Methods

6.3.1. Subjects and design

The study consists in a two-arm multicenter non-inferiority randomized clinical trial examining the effectiveness of a gamified rehabilitation equipment for improving elderly functional capacities according to the physical performance at the Short Physical Performance Battery (SPPB). The study took place at two different sites in Switzerland: Loëx Hospital, a 104-beds geriatric rehabilitation hospital, and Joli-Mont, an 80-beds geriatric rehabilitation clinic. Both are part of the Geneva University Hospitals Group.

The eligibility criteria were stipulated as following: seniors (65+) hospitalized in one of the two study sites at Geneva University Hospitals, with musculoskeletal issues (pelvic or lower

limbs fracture, hip prosthesis, falls and low back pain), the ability to stand upright , and the ability to understand the instructions. Being able to interact with the equipment without any sensorial, physical or mental limitations was necessary. Patients considered too weak to interact with the device, requiring specific rehabilitation treatment (e.g. wrist fracture), or going to a care institution were excluded. Due to a limitation associated with the device's size, obese participants were also not eligible.

The hospital electronic medical records of all consecutive new hospitalized patients were accessed (from February to June 2019) to identify potential participants. All patients who fulfilled the eligibility criteria were screened by two medical doctors (in each of the sites) and one nurse. Two researchers then presented the project and provided written information about the study to the identified subjects. If the contacted patient agreed to participate, researchers asked the patient to sign an informed consent form. Participants were then allocated randomly to one of the two arms of the trial. Participants' baseline data were gathered during the study. The randomization was based on a single allocation ratio, with no block and no stratification. Due to the intervention type, allocation was not masked to participants in the intervention and control groups or to researchers who recruited patients.

The study has been approved by the Institution's Ethical Committee (number 2018-01516). The trial has been registered in trial register ClinicalTrials.gov (NCT03847454)

6.3.2. Intervention

ActivLife is an activation equipment coupled with a serious game. The games and the instructions are displayed on a screen in front the patient who is secured in the mechanical platform ActivLife. The screen has a Kinect sensor that allows the software to determine if the patient is doing the exercise correctly. The software allows the physiotherapist to program and schedule a specific treatment (series of games) for each patient.

Stepwatch is a wearable sensor that measures the activity of the patient in terms of number of steps, activity (low, medium, high), cadence and velocity. It is a wearable watch that does not display any information and can be worn on the ankle. Stepwatch demonstrated to capture small changes in step rate (99% accuracy [14,15]), thus it can be used to assess changes in physical activity in individuals who walk slowly or use a walking aid such as a rollator. Furthermore, it allowed us to collect the data locally which ensure the patient's privacy.

The intervention lasted 6 weeks: 3 weeks at the hospital and 3 weeks at the patient's domicile as illustrated in Figure 1. Figure 1 illustrates the research procedure.

During their hospital stay, the rehabilitation was performed under the supervision of two physiotherapists (one physiotherapist per site). Both groups were monitored constantly with the Stepwatch sensor and participated in training sessions of 30mn, 5 times a week. The intervention group used the ActivLife, 3 times a week, during these sessions, while the control group had all their sessions following the standard rehabilitation.

6.3.3. Primary and secondary Outcomes

Used as primary outcome, SPPB is an objective tool to assess the lower extremity functioning in older persons [16]. This test is associated with the risk of falls, the risk of decreased functionality and the risk of death [17, 18, 19]. The test consists of three parts: balance tests, a gait speed test and a chair stand test. In the balance tests, the participant must be able to stand unassisted without the use of a cane or walker. The participant is then asked to perform a side-by-side stand, a semi-tandem stand and a tandem stand for 10 seconds. In the gait speed test, the participant is asked to walk 4 meters at usual pace. In the chair stand test, the participant is prompted to stand up from a chair, 5 times straight, as quickly as he/she can, without using his/her arms. The SPPB test is based on a point system, with a maximal of 12 points, meaning an ability to function independently.

As secondary functional outcomes, we measured the handgrip strength (IHGS), the Functional Independence Measure (FIM), and the number of steps.

IHGS is a simple and cost-effective method for evaluating overall muscle strength [20]. It is associated with cardio-vascular mortality and is a main determinant of sarcopenia [21, 22]. The participant is asked to hold the dynamometer in the hand to be tested, with the arm on a right angle and the elbow next to the body. The base should rest on the palm of the hand, while the handle should rest on the middle of the four fingers. When the participant is ready, he/she is asked to tighten the dynamometer with maximum isometric effort, which is maintained for about 5 seconds. No other body movement is allowed. The participant is strongly encouraged to provide maximum effort. The score is expressed in kilograms.

The FIM serves as a basic indicator of the degree of functional ability/disability. This score is associated with the risk of re-hospitalization [23]. It is composed by 18 different items scored from 1 (complete assistance required) to 7 (complete independence). This independence assesses function in six areas including self-care, continence, mobility, transfers, communication, and social cognition [24]. The FIM is based on a point system, with a maximal of 126 points, meaning an ability to function independently.

Patient data were collected at three different times: at baseline (t0), after the intervention (t1 – end of hospitalization), and 3 weeks after the intervention ended (t2). SPPB test, IHGS test and semi-structured interview were conducted at time t0, t1 and t2. FIM was evaluated at time t0 and t1. A comorbidity score, the Cumulative Illness Rating Score (CIRS) [25] and a cognitive test, the Mini Mental State Evaluation (MMSE) [26] were used as cofounders to compare baseline data at time t0.

6.3.4. Statistics

The study is a non-inferiority trial, to test if the gamified rehabilitation concept is at least as efficient as the standard care with respect to the main outcome measured by SPPB. With a power calculation of 95%, a standard deviation of 8.8+/-1.2 and a non-inferiority limit of 0.4, the total sample size needed is 38. Adjusting for a dropout of 20%, the sample size needed increased to 46 patients in total, 23 in each group. To recruit this number of patients a 6-month inclusion period was anticipated.

Characteristics of subjects are presented as means (+/-SD) for continuous variables. The normality of the distribution of continuous variables was verified with Shapiro-Wilks tests. We used two-samples t-test and Fisher's exact test to compare baseline data. Mixed-effects multiple linear regression models were used to assess the group and time effects and their interaction, over the outcome, while taking into account the repeated measure design and adjusting for the presence of a physiotherapist at home and other variable such as the number of sessions. The difference-value was considered significant when $p < 0.05$; and highly significant when $p < 0.001$. Statistical analyses were performed with STATA version 16.0 (STATA Corporation, College Station, Tx, USA, 2019).

6.4. Results

6.4.1. Patients Flow Diagram

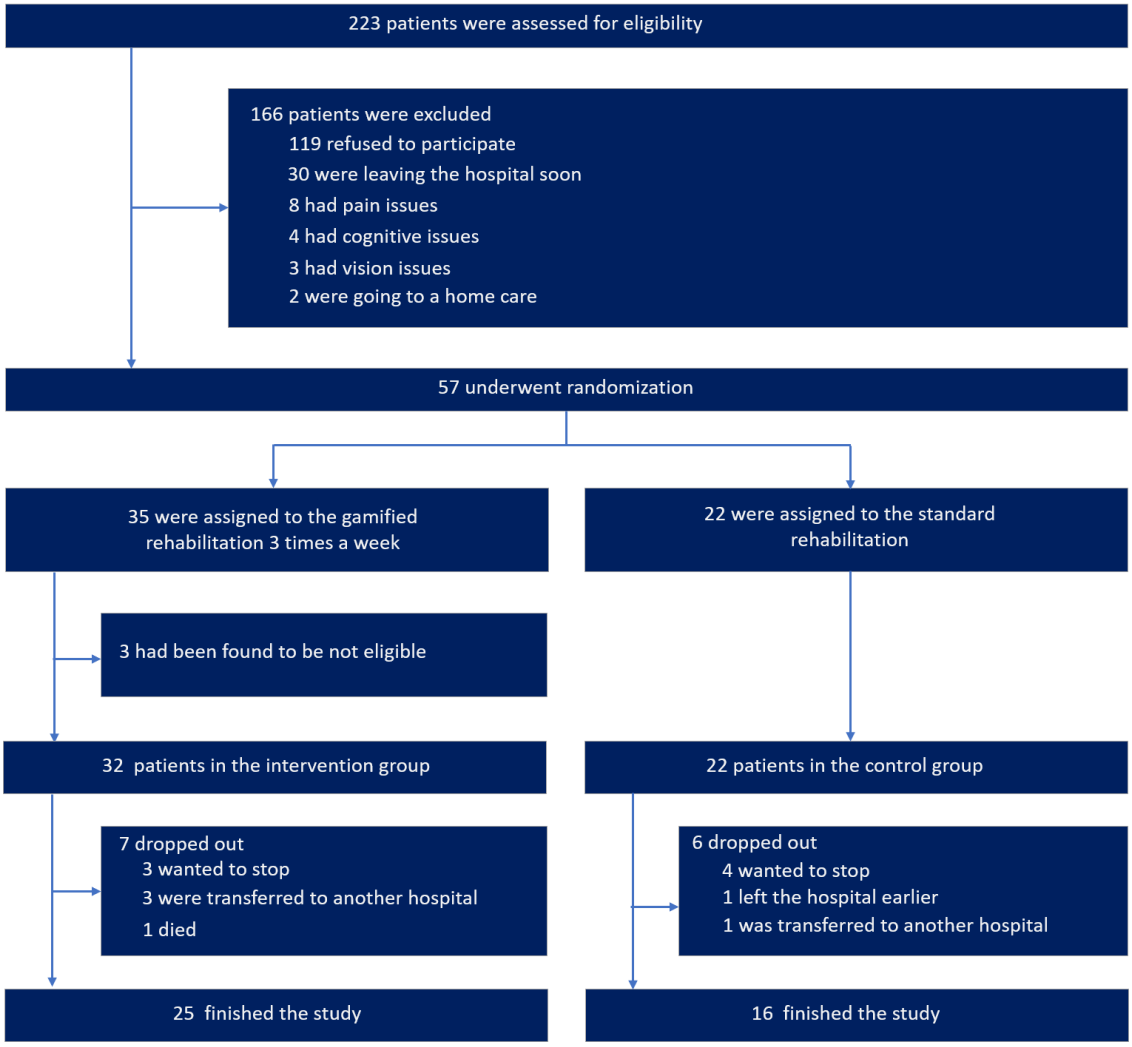


Figure 2. Patients' Flow Diagram

Patients Flow Diagram is described in Figure 2. A total of 223 individuals were screened for eligibility to participate. Of these, 166 were excluded from the study (119 refused to participate, 30 were leaving the hospital soon, 8 had pain issues, 4 had cognitive issues, 3 had vision issues, 2 were going to a care home). 57 patients underwent randomization to be

allocated to the intervention group (n=35) and the control group (n=22) and were included in the main analysis. At the follow-up, 10 patients dropped out from the intervention group (3 had been found not eligible, 3 wanted to stop, 3 were transferred to another hospital and 1 died) and 6 patients were lost from the control group (4 wanted to stop, 1 left the hospital earlier and 1 was transferred to another hospital).

6.4.2. Baseline data

Total participants mean age was 81.5 ± 6.8 years old with 68.4% female. The length of stay at hospital was 23.0 ± 11.6 days in average. CIRS scored 14.3 ± 6.4 on average. MMSE showed a mean value of 23.4 ± 5.1. FIM at admission was 97.4 ± 16.1 in average. The SPPB showed a mean of 6.36+/-2.78 at baseline. Subject's data at baseline are described in Table 1. Comparisons of the groups at baseline showed no evidence of differences between the groups in any of the measures.

6.4.3. Outcomes

There were no significant differences between the control group and the intervention group for the primary outcome (SPPB) or any of the secondary outcomes (IHGS, FIM and steps), which demonstrates the non-inferiority effect of the serious game-based intervention. Figure 3 illustrates the results of the four analyzed outcomes, namely SPPB, IHGS (right hand), FIM and Steps.

6.4.3.1. Primary outcome

SPPB: The analysis of mixed-effects ML regression on SPPB showed no indication of a group x time interaction (SPPB_group#Time1=-0.77, p=0.23; SPPB_group#Time2=0.21, p=0.75), which means that we observe no difference between the groups in SPPB improvement. Table 2 summarizes the SPPB outcomes.

Outcome variable	Coeff (95% CI)	p value	sig
SPPB (score)			
▪ Hospital	0.54 (-1.22, 2.30)	0.548	
▪ Group	-0.65 (-3.12,1.83)	0.609	
▪ Time 1	1.79 (0.81, 2.78)	0.000	1
▪ Time 2	1.60 (0.62, 2.59)	0.001	1
▪ Group#Time1	-0.77 (-2.03, 0.50)	0.234	
▪ Group#Time2	0.21 (-1.07, 1.48)	0.749	
▪ PhysioAtHome	2.43 (0.55, 4.31)	0.011	1
▪ Nb of sessions	0.17 (-0.12, 0.46)	0.248	

Table 2. Summary results: SPPB (Short Physical Performance Battery)

However, there is a main effect of time (SPPB_t1=1.79, p=0.000; SPPB_t2=1.60, p=0.001) reflecting overall improvement in SPPB score across the three measurement points. The main effect of having physiotherapy at home (SPPB= 2.43, p=0.01) indicated that remaining active

at home have positive effect on the SPPB score. While analyzing only the intervention patients who had more than 6 sessions on the machine, we observe better results. Group x time interaction was still absent (SPPB_group#Time1=0.76, p=0.23; SPPB_group#Time2=0.98, p=0.12) but positive improvement of SPPB at time t1 and t2 was noticed. Although not significant (SPPB_nb_of_sessions=0.17, p=0.25), the effect of the number of sessions on the machine tend to be positive.

6.4.3.2. Secondary outcomes

IHGS: The analysis of mixed-effects regression on IHGS showed no indication of a group x time interaction (Right: IHGS_group#Time1=-0.21, p=0.87; IHGS_group#Time2=2.33, p=0.13 / Left: IHGS_group#Time1=-0.40, p=0.69; IHGS_group#Time2=2.03, p=0.07), which means that we observe no difference between the groups in IHGS improvement. However, we observe a positive improvement of more than 2 kg (Right: 2.33 kg (not significant: 0.13), Left: 2.03 kg (not significant: 0.07) for the patient from the intervention group. Positive difference between the two hospitals were observed (Right: IHGS_hospital=6.44, p=0.05 / Left: IHGS_hospital=5.80, p=0.05) but still not significant. Table 3 summarizes the SPPB outcomes.

Outcome variable	Coeff (95% CI)	p value	sig
IHGS (score): Left side			
▪ Hospital	5.80 (-0.17, 11.78)	0.057	
▪ Group	-0.23 (-8.63, 2.16)	0.240	
▪ Time 1	1.24 (-0.31, 2.80)	0.118	
▪ Time 2	0.33 (-1.40, 2.05)	0.709	
▪ Group#Time1	-0.40 (-2.39, 1.58)	0.690	
▪ Group#Time2	2.03 (-0.18, 4.23)	0.072	
▪ PhysioAtHome	0.94 (-5.51, 7.40)	0.775	
IHGS (score): Right side			
▪ Hospital	6.44 (-0.10, 12.97)	0.054	
▪ Group	-2.62 (-8.31, 3.08)	0.368	
▪ Time 1	1.22 (-0.84, 3.28)	0.247	
▪ Time 2	-0.12 (-2.62, 2.37)	0.922	
▪ Group#Time1	-0.21 (-2.81, 2.39)	0.875	
▪ Group#Time2	2.33 (-0.72, 5.37)	0.135	
▪ PhysioAtHome	2.50 (-4.27, 9.27)	0.469	

Table 3. Summary results: IHGS (Isometric Hand Grip Strength)

FIM: The analysis of mixed-effects ML regression on FIM showed no indication of a group x time interaction (FIM_group#Time1=-5.14, p=0.29), which means that we observe no difference between the groups in FIM improvement. However, there is a main time effect (FIM_t1=10.8, p=0.004) reflecting overall improvement in FIM score between the two

measurement points. The main effect of group (FIM_group =3.38, p=0.45) was not significant. Table 4 summarizes the FIM and the Steps outcomes.

Outcome variable	Coeff (95% CI)	p value	sig
FIM (score)			
▪ Time 1	10.8 (3.40, 18.23)	0.004	1
▪ Group	3.38 (-5.40,12.17)	0.450	
▪ Group#Time1	-5.14 (-14.73, 4.44)	0.290	
Steps (nb of steps)			
▪ Group	120 (-576, 816)	0.740	
▪ Home	461 (3.70, 570)	0.006	1
▪ Group#Home	-164 (-453, 125)	0.270	
▪ Hospital	-478 (-1172, 215)	0.180	

Table 4. Summary results: FIM (Functional Independence Measure) and Steps

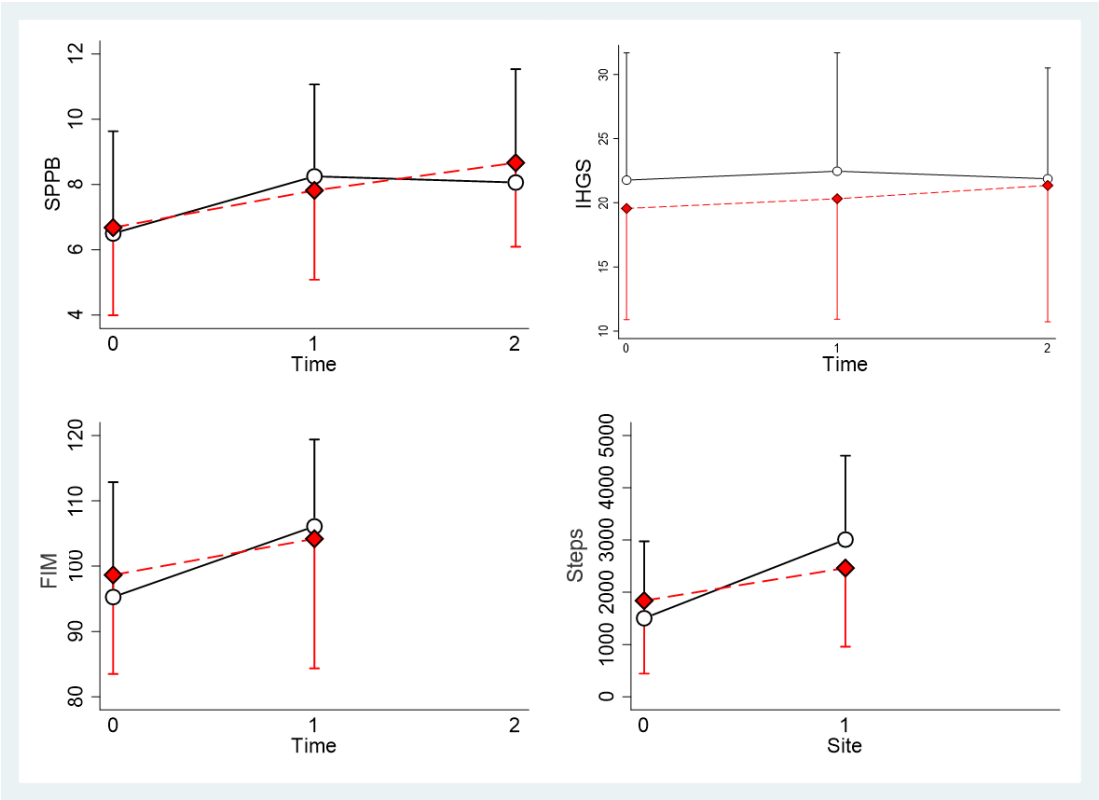


Figure 3. SPPB (Short Physical Performance Battery), IHGS (Isometric Hand Grip Strength)(right hand), FIM (Functional Independence Measure), Steps outcomes (IG (Intervention Group): red; CG (Control Group):black)

Steps: The analysis of mixed-effects regression on steps showed no indication of a group x site interaction (Steps_group#Home=-164, p=0.27), which means that we observe no difference between the groups in steps improvement. Nevertheless, there is a main effect of site (Steps_home=461, p=0.006) reflecting an improvement in the number of steps while

returning home. The main effect of group (Steps_group=120, p=0.74) was slightly positive but not significant. The mean number of steps (Steps_I=1839; Steps_C=1504) showed that participants from the intervention were somewhat more active at the hospital compared to the control group. However, at home, we failed to observe the same results (Steps_I=2463; Steps_C=3008). Negative difference between the two hospitals were observed (Steps_hospital=-478, p=0.18) but not significant.

6.5. Discussions

6.5.1. Principal findings

We evaluated the efficacy of a gamified rehabilitation equipment for improving functional capacities among older people with musculoskeletal issues. There were no significant differences between the control group and the intervention group for the primary outcome (SPPB: balance, gait speed, chair stand) or any of the secondary outcomes (IHGS, FIM and steps), which demonstrates the non-inferiority effect of the serious game-based intervention.

Literature demonstrates that serious games for older people overall health and specific disease management have been explored intensively. Parkinson disease [27, 28, 29] and stroke rehabilitation [30, 31, 32] have been a topic of interest for gamified intervention developers. However, a literature review on Kinect-based stroke rehabilitation systems [33] illustrates that previous studies were driven towards the feasibility and technical effectiveness of such systems rather than clinical effectiveness. Similar observation has been found on the use of gamification for cognitive assessment and cognitive training [34]. Although based on small sample size, our study has the benefit of investigating the clinical validity of a serious game-based rehabilitation in a real-world setting.

Besides, although not statistically significant, it is worth noting that, after 6 weeks, the handgrip strength test improved by 2kg in the intervention group compared to 0.3kg in the control group. This effect was likely due to the property of ActivLife encouraging safe upper limbs exercises. In addition to the game components, ActivLife is composed of an efficient trunk stabilization reassuring the patients while engaging in different movements such as “cleaning the window”, “guiding an ambulance” or “flying a dragon” [35]. Knowing that an improvement of 2kg is considered a minimal significant change in the handgrip test [36], this result demonstrates the potential of a gamified rehabilitation in improving even more elderly patients’ muscles strength.

Furthermore, although not reported here but in a different paper [37], motivation and engagement were observed in the participants who used the gamified rehabilitation equipment. Patients identified the intervention as playful, challenging, empowering and distracting and were more motivated to undergo their rehabilitation. More engaged and active, they showed positive behavior such as socializing with neighbor patients at the corridor, going to the cafeteria or enjoying fresh air at the hospital garden. This aligns with literature, where Bird and al [38], demonstrated that being exposed to exergames (exercise-based videogames) increases elderlies’ perception of the usefulness of technology for improving physical activity and could potentially induce motivation to be more active. A systematic review and meta-analysis [39] evaluating the efficacy of Active Video Games (AVGs) for

rehabilitation in respiratory conditions also reported no significant difference on physiological outcome measures (HR, SpO2 and dyspnea) between AVGs and traditional exercise. However, AVGs were found to be significantly more enjoyable than traditional exercise.

Additionally, if proven to be as effective as the standard care, the gamified rehabilitation reserves the benefit of being cost-effective by reducing the time spent by the physiotherapist with the person during a therapy session. We could imagine that; such tools would allow the physiotherapist to manage multiple patients at the same time requiring only a passive surveillance instead of actively monitoring each one of them. This would reduce considerably the burden of limited health professional associated with the increase of musculoskeletal disorders and the prevalence of ageing population [40]. However, as patients value patient-therapist interaction more than the amount or content of therapy during inpatient rehabilitation [41], further investigation is needed to understand the trade-off between those two components.

6.5.2. Limitations

Finally, our study presents different limitations. The sample size of the study was small, making it difficult to detect moderate effects (e.g., differences between groups), especially as we observed a lot of variability in the steps data. Due to the subsequent drop-outs, some data were missing. Besides, the limitation associated to the length of stay of the patients made it difficult to ensure that the intervention group had enough sessions on the machine. As suggest by Valimaki and al [42], gaming dose should be high enough to validate the efficacy of a gamified intervention. However, this trial reveals the complexities associated with adopting new technologies in clinical practice, including the need to consider elderly patients' context and all possible adverse events that could occur during hospitalization.

6.6. Conclusions

Our pilot study demonstrated the clinical efficacy of ActivLife, a gamified rehabilitation equipment, in the treatment of older patient with musculoskeletal issues. In addition, patients who received the intervention exhibited stronger engagement in the rehabilitation process and a higher motivation to being more active. Consequently, serious game-based rehabilitation could be a cost-effective alternative for elderly patients to regain their functional capacities.

6.7. Appendix

	Total (n=57)	Intervention group (n=35)	Control group (n=22)	P-value
Age, (mean years ± SD)	81.5 ± 6.8	82.2 ± 7.0	81.5 ± 6.8	0.34
Female, (n(%))	39 (68.42)	24 (68.57)	15 (68.18)	0.98
Length of stay (days ± SD)	23.0 ± 11.6	22.5 ± 11.1	23.8 ± 12.7	0.71
FIM (score ± SD)	97.4 ± 16.1	98.7 ± 15.2	95.3 ± 17.6	0.46
CIRS (score ± SD)	14.3 ± 6.4	14.4 ± 5.7	14.1 ± 7.5	0.88
MMSE (score ± SD)	23.4 ± 5.1	23.6 ± 6.1	23.1 ± 3.7	0.85
SPPB (score ± SD)	6.36+/-2.78	6.23+/-2.80	6.58+/-2.83	0.68

Table 1. Patients' baseline data

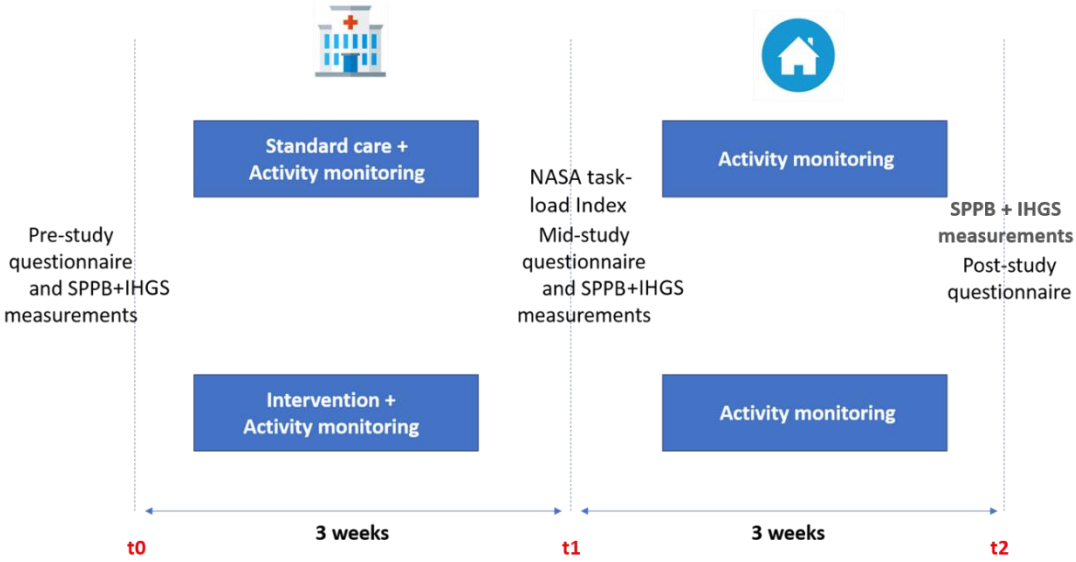


Figure 1. Research procedure

6.8. References

7. [1] World Health Organization. Tracking universal health coverage: first global monitoring report. (2015). ISBN 978 92 4 156497 7.

8. [2] World Health Organization. 10 facts on ageing and health. (2017). https://www.who.int/features/factfiles/ageing/en/ Accessed 02 September 2019.

9. [3] Wiles, J.L., Leibing, A., Guberman, N., Reeve, J., Allen, R.E. (2012). The meaning of "aging in place" to older people. *Gerontologist*. Jun;52(3):357-366. https://doi.org/10.1093/geront/gnr098

10. [4] Bacsu, J.R., Jeffery, B., Johnson, S., Martz, D., Novik, N., & Abonyi, S. (2012). Healthy aging in place: supporting rural seniors' health needs. *Online J Rural Nurs Health Care.* 12(2):77-87. https://doi.org/10.14574/ojrnhc.v12i2.52

11. [5] Mattison, M. Hospital management of older adults. (2019). https://www.uptodate.com/contents/hospital-management-of-older-adults/print Accessed 02 September 2019.

12. [6] Swiss Federal Statistical Office. (2016). Health - Pocket Statistics 2016. FSO number 1543-1600. https://www.bfs.admin.ch/bfs/en/home/news/whats-new.assetdetail.1442765.html Accessed 02 September 2019.

13. [7] Santos, L.J., Silveira, F.S., Müller, F.F., Araújo, H.D., Comerlato, J. B., Silva, M.C. da et al. (2017). Functional assessment of inpatients in the Intensive Care Unit of the University Hospital of Canoas. *Fisioterapia e Pesquisa*, 24(4), 437-443. https://doi.org/10.1590/1809-2950/17720924042017

14. [8] Mackichan, F., Adamson, J., & Gooberman-Hill, R. (2013). 'Living within your limits': Activity restriction in older people experiencing chronic pain. *Age and Ageing*, 42(6), 702–708. https://doi.org/10.1093/ageing/aft119

15. [9] Resnick, B. & Boltz, M. (2019). Optimizing Function and Physical Activity in Hospitalized Older Adults to Prevent Functional Decline and Falls. *Clinics in Geriatric Medicine*, 35(2), 237-251. https://doi.org/10.1016/j.cger.2019.01.003

16. [10] Taylor, L.M., Kerse, N., Frakking, T., & Maddison, R. (2018). Active Video Games for Improving Physical Performance Measures in Older People: A Meta-analysis. *Journal of geriatric physical therapy*, 108-123. https://doi.org/10.1519/JPT.0000000000000078

17. [11] Idriss, M., Tannous, H., Istrate, D., Perrochon, A., Salle, J.Y., & Ho Ba Tho, M.C. (2017). Rehabilitation-Oriented Serious Game Development and Evaluation Guidelines for Musculoskeletal Disorders. *JMIR Serious Games*. 5(3):e14. https://doi.org/10.2196/games.7284

18. [12] Lin, T., Hsieh, C. & Lee, J. (2013). A Kinect-Based System for Physical Rehabilitation: Utilizing Tai Chi Exercises to Improve Movement Disorders in Patients with Balance Ability. *7th Asia Modelling Symposium*. 149-153. https://doi.org/10.1109/AMS.2013.29

19. [13] Ling, Y., Ter Meer, L.P., Yumak, Z., & Veltkamp, R.C. (2017). Usability Test of Exercise Games Designed for Rehabilitation of Elderly Patients After Hip Replacement Surgery: Pilot Study. *JMIR Serious Games*. 5(4):e19. https://doi.org/10.2196/games.7969

20. [14] Cindy Ng, L.W., Jenkins, S. & Hill, K. (2012) Accuracy and responsiveness of the stepwatch activity monitor and ActivPAL in patients with COPD when walking with and without a rollator. *Disability and Rehabilitation*. 34:15, 1317-1322, https://doi.org/10.3109/09638288.2011.641666

21. [15] McCullagh, R., Dillon, C., O'Connell, A.M., Horgan, F., & Timmons, S. (2016). Step-Count Accuracy of 3 Motion Sensors for Older and Frail Medical Inpatients. *Archives of Physical Medicine and Rehabilitation*. Volume 98, Issue 2, 295-302. https://doi.org/10.1016/j.apmr.2016.08.476

22. [16] National Institute on Aging. Short Physical Performance Battery (SPPB): Assessing Physical Performance in the Older Patient. https://www.nia.nih.gov/research/labs/leps/short-physical-performance-battery-sppb Accessed 05 September 2019.

23. [17] Veronese, N., Bolzetta, F., Toffanello, E. D., Zambon, S., De Rui, M., Perissinotto, E., et al. (2014). Association between Short Physical Performance Battery and falls in older people: the Progetto Veneto Anziani Study. *Rejuvenation research*, 17(3), 276–284. https://doi.org/10.1089/rej.2013.1491

24. [18] Kim, J. C., Chon, J., Kim, H. S., Lee, J. H., Yoo, S. D., Kim, D. H., et al. (2017). The Association Between Fall History and Physical Performance Tests in the Community-Dwelling Elderly: A Cross-Sectional Analysis. *Annals of rehabilitation medicine*, 41(2), 239–247. https://doi.org/10.5535/arm.2017.41.2.239

25. [19] Pavasini, R., Guralnik, J., Brown, J. C., di Bari, M., Cesari, M., Landi, F., et al. (2016). Short Physical Performance Battery and all-cause mortality: systematic review and meta-analysis. *BMC medicine*, 14(1), 215. https://doi.org/10.1186/s12916-016-0763-7

26. [20] Amaral, C. A., Amaral, T., Monteiro, G., Vasconcellos, M., & Portela, M. C. (2019). Hand grip strength: Reference values for adults and elderly people of Rio Branco, Acre, Brazil. *PloS one*, 14(1), e0211452. https://doi.org/10.1371/journal.pone.0211452

27. [21] Celis-Morales, C. A., Welsh, P., Lyall, D. M., Steell, L., Petermann, F., Anderson, J., et al. (2018). Associations of grip strength with cardiovascular, respiratory, and cancer outcomes and all cause mortality: prospective cohort study of half a million UK Biobank participants. *BMJ (Clinical research ed.)*, 361, k1651. https://doi.org/10.1136/bmj.k1651

28. [22] Sousa-Santos, A. R., & Amaral, T. F. (2017). Differences in handgrip strength protocols to identify sarcopenia and frailty - a systematic review. *BMC geriatrics*, 17(1), 238. https://doi.org/10.1186/s12877-017-0625-y

29. [23] Hoyer, E.H., Needham, D.M., Atanelov, L., Knox, B., Friedman, M., & Brotman, D.J. (2014). FIM at Discharge and Rehospitalization. *J. Hosp. Med* ;5;277-282. https://doi.org/10.1002/jhm.2152

30. [24] Cech, D.J, and Martin, S.T. (2012). Functional Movement Development Across the Life Span. 3rd Edition. Elsevier. ISBN: 9781416049784.

31. [25] Salvi, F., Miller, M.D., Grilli, A., et al. (2008). A manual of guidelines to score the modified cumulative illness rating scale and its validation in acute hospitalized elderly patients. *J Am Geriatr Soc* ;56(10):1926-1931. https://doi.org/10.1111/j.1532-5415.2008.01935.x

32. [26] Folstein, M.F. et al. (1975). "Mini-mental state". A practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res*; 12: 189. https://doi.org/10.1016/0022-3956(75)90026-6

33. [27] Yu, W., Vuong, C., & Ingalls, T. (2011). An interactive multimedia system for Parkinson's patient rehabilitation. *In: Shumaker R. (eds) Virtual and Mixed Reality - Systems and Applications. Lecture Notes in Computer Science.* Springer, Berlin, Heidelberg. 6774:129-137. https://doi.org/10.1007/978-3-642-22024-1_15

34. [28] Assad, O., Hermann, R., Lilla, D., Mellies, B., Meyer, R., Shevach, L., et al. (2011). Motion-based games for Parkinson's disease patients. *Entertainment Computing – ICEC. Lecture Notes in Computer Science.* 6972:47-58. https://doi.org/10.1007/978-3-642-24500-8_6

35. [29] Paraskevopoulos, I.T., Tsekleves, E., Craig, C., Whyatt, C., & Cosmas, J. (2014). Design guidelines for developing customised serious games for Parkinson's disease rehabilitation using bespoke game sensors. *Entertain Comput*. 5(4):413-424. https://doi.org/10.1016/j.entcom.2014.10.006

36. [30] Cho, S., Ku, J., Cho, Y.K., Kim, I.Y., Kang, Y.J., Jang, D.P., et al. (2014). Development of virtual reality proprioceptive rehabilitation system for stroke patients. *Comput Methods Programs Biomed*. 113(1):258-265. https://doi.org/10.1016/j.cmpb.2013.09.006

37. [31] Ding, Q., Stevenson, I.H., Wang, N., Li, W., Sun, Y., Wang, Q., et al. (2013). Motion games improve balance control in stroke survivors: a preliminary study based on the principle of constraint-induced movement therapy. *Displays 2013*. 34(2):125-131. https://doi.org/10.1016/j.displa.2012.08.004

38. [32] Ibarra Zannatha, J.M., Tamayo, A.J., Sánchez, A.D., Delgado, J.E., Cheu, L.E., & Arévalo. W.A. (2013). Development of a system based on 3D vision, interactive virtual environments, ergonometric signals and a humanoid for stroke rehabilitation. *Comput Methods Programs Biomed*. 112(2):239-249. https://doi.org/10.1016/j.cmpb.2013.04.021

39. [33] Dimaguila, G.L., Gray, K., & Merolli, M. (2018). Person-Generated Health Data in Simulated Rehabilitation Using Kinect for Stroke: Literature Review. *JMIR Rehabil Assist Technol*. 5(1):e11. https://doi.org/10.2196/rehab.9123

40. [34] Lumsden, J., Edwards, E.A., Lawrence, N.S., Coyle, D., & Munafò, M.R. (2016). Gamification of Cognitive Assessment and Cognitive Training: A Systematic Review of Applications and Efficacy. *JMIR Serious Games* ;4(2):e11. https://doi.org/10.2196/games.5888

41. [35] Kozak, D., Burgermeister, S., De Chassey, J.D.B., Naef, A., Maringue, A., & Dietrich, D. (2017). A functionality, safety and validity study of innovative REACH devices. *Gerontechnology*, 16(3), 181-188. https://doi.org/10.4017/gt.2017.16.3.007.00

42. [36] Esain, I., Rodriguez-Larrad, A., Bidaurrazaga-Letona, I., & Gil, S. M. (2017). Health-related quality of life, handgrip strength and falls during detraining in elderly habitual exercisers. *Health and quality of life outcomes.* 15(1), 226. https://doi.org/10.1186/s12955-017-0800-z

43. [37] Randriambelonoro, M., Perrin, C., Blocquet, A., Kozak, D., Fernandez, J.T., Marfaing, T., et al. (2020). Hospital-to-Home Transition for Older Patients: Using Serious Games to Improve the Motivation for Rehabilitation – a Qualitative Study. *Population Ageing.* https://doi.org/10.1007/s12062-020-09274-7

44. [38] Bird, M.L., Clark, B., Millar, J., Whetton, S., & Smith, S. (2015). Exposure to "Exergames" Increases Older Adults' Perception of the Usefulness of Technology for Improving Health and Physical Activity: A Pilot Study. *JMIR Serious Games*. 3(2):e8. https://doi.org/10.2196/games.4275

45. [39] Simmich, J., Deacon, A.J., & Russell, T.G. (2019). Active Video Games for Rehabilitation in Respiratory Conditions: Systematic Review and Meta-Analysis. *JMIR Serious Games.* 7(1):e10116. https://doi.org/10.2196/10116

46. [40] Gheno, R., Cepparo, J. M., Rosca, C. E., & Cotten, A. (2012). Musculoskeletal disorders in the elderly. *Journal of clinical imaging science.* 2, 39. https://doi.org/10.4103/2156-7514.99151

47. [41] Peiris, C.L., Taylor, N.F., & Shields, N. (2012). Patients value patient-therapist interactions more than the amount or content of therapy during inpatient rehabilitation: a qualitative study. *Journal of Physiotherapy*. Volume 58, Issue 4. 261-268. https://doi.org/10.1016/S1836-9553(12)70128-5

48. [42] Välimäki, M., Mishina, K., Kaakinen, J.K., Holm, S.K., Vahlo, J., Kirjonen, M., et al. (2018). Digital Gaming for Improving the Functioning of People With Traumatic Brain Injury: Randomized Clinical Feasibility Study. *J Med Internet Res*. 20(3):e77. https://doi.org/10.2196/jmir.7618

CHAPTER 7. MIRANABOT: A CONVERSATIONAL AGENT TO PROMOTE HEALTHY EATING

7.1. Introduction

Diabetes, osteoporosis, stroke, cardiovascular disease and cancers, which are the most prevalent illnesses affecting elderly people are all diet-influenced [1,2]. Every year, 41 million people die due to these noncommunicable diseases (NCDs), which represent 71% of all the deaths in the world [3]. There is a significant emphasis on the role of micronutrients in health promotion and preventing NCDs. Micronutrients deficiencies are often common in older adults due to a number of factors such as lower food intake and lack of variety in the food they eat. A complete, clinically appropriate and cost-effective screening instrument to evaluate the elderly dietary status is therefore necessary [4].

Food recommender systems have received increasing attention to help people adopt healthier eating behavior. These systems focus on suggesting proper food items based on individuals' preferences and health conditions [5, 6]. Nevertheless, despite the extensive research and the large number of existing nutrition applications, food recommender systems are still facing many challenges in terms of nutrition habits tracking and delivery of the proper recommendations [7, 8]. People find it often difficult and time consuming to enter manually their food every day in the system. Many studies are also facing the uncertainty of the information given by the users as they may not know or tend to forget what they have eaten which makes it more challenging for the system to recommend the correct food item. Additionally, many studies have shown that the recommendation is not necessarily followed by a change in the behavior [9]. There is still a lack of understanding in how to incorporate efficiently behavior change techniques into a food recommender system.

With the recent development in natural language understanding, conversational agents have gained popularity over the past three years. In the nutrition domain, they allow to collect user data in an easy and user-friendly manner. Researchers at MIT developed a Web-based prototype of a speech-controlled nutrition-logging system which converts the entry spoken by the users into calories intake [10]. Researchers at the University of Applied Sciences in Western Switzerland worked on a chatbot that help people reduce their meat and increase fruits and vegetables consumption [11]. Users were able to set nutrition goals themselves and had a follow-up with the system every day. Although, only 11% could reach their objectives, more than half of the participants showed positive changes in their nutrition habits.

We developed a conversational agent called "MiranaBot" that helps the elderly to be aware of their eating habits in terms of variety and regularity. Rather than focusing on food quantity and nutritional value, the system targets the variety of the individuals' diet. From a regular description of the elderlies' meals during a certain period, "MiranaBot" is able to assess the quality of their nutrition, identify the foods they need to consume less and explain why and how to replace them. The system proposes personalized solutions tailored to the older adults' needs and context.

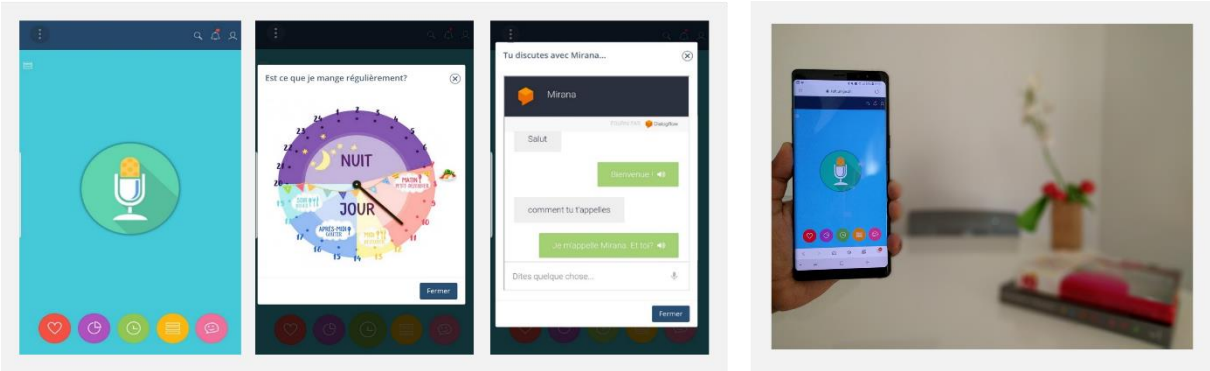


Figure X. Graphical User Interface – MiranaBot

7.2. Methodology

We conducted a qualitative study with 10 obese and diabetic patients, 2 nutritionists and 1 physician with a specific expertise on therapeutic education of chronically-ill patients from a large hospital in Switzerland. Observation sessions, focus group and semi-structured interviews were performed to get insights from the different participants. The prototype was developed through an iterative process, following the agile development principles [12]. A first prototype was developed. Changes were quickly implemented according to the participants feedback and in collaboration with them.

To understand patient's barriers for adopting a healthy nutrition, we first observed 4 consultation sessions led by a physician at the therapeutic education center. In addition, we followed a group of 10 patients for one-week period, while they came to the center for a workshop to learn about their disease. To inquire about their needs and their issues, we conducted a focus group with the patients, while presenting them the first prototype of the application. We conducted a semi-structured interview with one physician and two nutritionists, who helped us understand the real issues faced by the patients as well as the complexity of nutritional behavior change. The nutritionists participated in creating the nutritional database, we later use in the application.

We then selected the corresponding Behavior Change Techniques (BCTs) from the literature aiming at and answering and fulfilling the needs and requirements identified. These BCTs are then converted into a set of functionalities that we incorporated in MiranaBot.

7.3. Findings

Our observation followed by the discussion with the patients and the health professionals gave us insights on the barriers for patients to adopt a healthy nutrition as well as their needs and requirements for a successful behavior change. N1, N2 and P are used to refer to the nutritionists and the physician. Patients are addressed as P1 to P10.

7.3.1. Barriers to healthy nutrition

Lack of regularity, lack of variety, false belief and hunger unawareness were identified as the main barriers to healthy nutrition.

7.3.1.1. Lack of regularity

Nutritionists identified the lack of regularity in eating as the first issue commonly faced by their patients. N1 said *"Most of our patients are not eating three times a day. They are not taking breakfast, neither a lunch, which make them starve for a big quantity of food later during the day or at night."* P6 mentioned *"I don't really have time because of my job, so I'm just grabbing some food that I can easily find when I finish my round. It's easier like that for me."* P3 added *"I don't eat breakfast, I'm used to that. At lunch, I just grab some sandwiches, but I come back home, I think I deserve a big dinner. I'm usually very hungry at night."*

7.3.1.2. Lack of variety

Health professionals also mentioned the lack of food variety as a recurring issue observed in chronically-ill patients. N1 stated *"We often share the optimal plate during our workshop because most of our patients have difficulty to bring variety to their food. They are often eating the same aliment every day".* P4 said "I have to admit that I'm always eating pasta and pizza, because I like them, and you know, it's not expensive." P9 added *"I'm always buying the same thing when I do my groceries. I don't allow a lot of surprise about my food. I already have my personal habits."*

7.3.1.3. False belief

Following a consultation with a patient, the physician shared with us that one of the most difficult issue they face as health professionals is patient's false belief. P said, *"As one of my patients heard that olive oil was good for the heart, he started to put a big quantity of olive oil in everything he eats."* During the session with her doctor, P2 shared *"You know doctor, I only eat fresh cheese every day. Whatever is the menu, I always have my fresh cheese. It is fresh, so for me that's healthy."*

7.3.1.4. Hunger unawareness

Nutritionists and physician both mentioned the complexity of nutritional behavior change due to the different external and internal factors influencing people to eat. P shared with us *"Most of the time, when patients comes here, it is not only about food. It's about how they face the loss of their loved ones or the family problem they have. It's much more than a food addiction."* N2 stated *"It often happen that the patients do not eat because they feel hungry but rather to fill a void inside them."* P7 added *"You know, I just eat automatically because there is food there, I don't think too much about why."* P4 stated *"I'm not always feeling good after eating, that's how I somehow realize maybe that was too much."*

7.3.2. Needs and requirements

Monitoring, education, empowerment and practicality were identified as the main needs and requirements for nutrition behavior change.

7.3.2.1. Monitoring

Patients and health professionals both shared the need for a monitoring tool to raise self-awareness on patients' nutritional habits. P5 said *"It's true that I don't always remember the*

food I ate. So, I don't really realize how I eat. If a tool can help me know that, why not?" P7 added *"The nutritionist asked me to describe what I ate on a diary, so that she can know how to help me. I guess it's important to monitor what we eat but it is not always easy."* N2 stated *"Just being aware of what and how they eat would already represents a big advancement because a lot of my patients do not even think of how much and why they eat. That's the real issue. It is essential for them to monitor their eating habits."*

7.3.2.2. Education

The plethora of nutritional advice and beliefs accessible on the web make it often difficult for patients to identify trustworthy information. Patients appreciated the one-week workshop organized by the hospital as it taught them trustworthy and practical tips to change their habits. P6 said *"You know before coming here, I thought drinking smoothie should be fine as it is just fruits. I did not know that fruits also contain sugar and drinking a lot of smoothie is really not healthy."* N1 mentioned *"During our session here, we teach our patients to always eat proteins, carbs and vegetable. We share simple tricks to measure portion. Our patients need that."*

7.3.2.3. Empowerment

From the perspective of the health professionals, there is no one size fit all solution for everyone. Their role remains to give the necessary education. It is then up to the patients to build their own strategy and apply it for changing their behavior. P4 effectively confirmed *"I won't like it if you tell me what I have to eat. I should be free to choose what I like as long as I follow your general recommendation."*

7.3.2.4. Practicality

Most of our participants expressed the need for a practical and simple solution. P8 shared *"I used a nutrition application before to record my food, but it was too time consuming, so I gave up."* N2 mentioned *"Specifying the quantity of your pasta and beef is not so obvious. Patients need a hands-on tool that would motivate them and make their life easy."*

7.3.3. Summary

Lack of regularity, lack of variety, false belief and hunger unawareness were identified as the main barriers to healthy nutrition; whereas monitoring, education, empowerment and practicality were identified as the main needs and requirements for nutrition behavior change. Considering these findings, we decided on appropriate BCTs to include inside our systems.

7.4. MiranaBot: behavior change theory and techniques

Multiple researches have been conducted to identify motivational strategies for behavior change, especially in the nutrition and physical activity domain [13, 14]. Studies grounded in behavior theory appeared to show a positive impact on patient health behavior. Sawesi et al. [15] demonstrated a significant relationship between theory-based health behavior change intervention and patient engagement. Various systematic reviews and meta-analysis [16, 17, 18] showed that efficacy of dietary intervention is associated with well-defined behavior

change techniques, social support, increased contact frequency, goal setting, just-in-time feedback, review of goal progress and self-monitoring.

There are many models and theories on behavior change, some focusing on parts of the behavior change process [19], while others are more holistic, aiming to include all factors that can influence behavior change [20]. “MiranaBot” is based on the Self-regulation theory [21] and the COM-B (Capability, Opportunity, Motivation -Behaviour) model [19]. Self-monitoring, personalized visual feedback, goal setting, self-awareness and personalized education were the behavior change techniques incorporated in the tool.

7.4.1. Self-monitoring

“MiranaBot” allows the users to track their nutrition easily by asking them to briefly describe what they ate after each meal in terms of portions. The system will initially train the users on the portion size to help them describe their meals as best as possible.

7.4.2. Personalized visual feedback

“MiranaBot” provides visual feedback every day regarding nutrition variety and regularity. Food variety is illustrated on a plate showing the ratio of proteins, carbohydrates and vegetables consumed by the user versus the optimal plate suggested by the Swiss Society of Nutrition [22]. A visual watch resuming the elderlies eating times is intended to encourage self-reflection on their nutrition regularity.

7.4.3. Goal setting

Following a training period analysing the eating habits of the users, “MiranaBot” allows them to set personal goals, such as “eating three meals a day” or reducing “bread” and to define a timeline to reach their goals. Prompt reminder and useful information are pushed to help users reach their goal.

7.4.4. Self-awareness

Researchers have demonstratred that the reason to eat for most patients with chronic conditions are rather linked to psychological issues (emotions, stress) or automatic behavior than hunger itself [23, 24]. “MiranaBot” raises awareness of real physical hunger by asking users how they are feeling before and after each meal.

7.4.5. Personalized education

“MiranaBot” is developed to be continuously available to answer users’ questions during the process of change. Advice on food item alternatives, benefits of healthy nutrition behavior and promptly reminders towards their personal goals are frequently delivered.

7.5. MiranaBot: Components and functionalities

Figure Y illustrates the process and the different technical modules of the conversational agent. “Mirana” is composed of four main components: the speech to text module, the text analysis module, the natural conversation module and the graphical user interface.

Figure Y. Process and components of MiranaBot

7.5.1. Speech to text module

Users can easily describe, vocally, what they have eaten after each meal. This voice description is then converted in real time into text using DialogFlow, Google's human–computer interaction technologies based on natural language conversations. MiranaBot is now available in three languages, namely French, English and Danish. The first version of the application will be tested in the French speaking part of Switzerland.

7.5.2. Text analysis module

MiranaBot analyses users' every day food data to provide them with daily and weekly visual feedbacks. MiranaBot also detects frequencies, varieties, and quantities (in terms of portions) of a specific food (or category of food). With the help of nutritionists and dieticians, we were able to build a database of food classified by categories. We have set rules for each category which are then used by our algorithm to identify recurring patterns (e.g., eat too much bread = more than the normal consumption defined by the rules).

7.5.3. Natural conversation module

One important functionality of "Mirana" is its ability to discuss, have a conversation and answer the questions from the user. As mentioned previously, the conversational module is built upon DialogFlow developed by Google. Their module uses machine learning for small talk conversations to recognize user basic questions. In addition, we developed specific intents (or functions) related to our case to give "Mirana" the ability to answer questions about nutrition.

7.5.4. Graphical user interface

To maximize the output provided to the user, MiranaBot uses the screen of the host device (phone, tablet, computer) to display graphical feedback in addition to the vocal answers. The regularity of the nutrition and the variety of the food eaten by the users are displayed on the screen to induce better self-reflection. While describing their food, the user can also check, in real time, if what they said was transcribed correctly. In case of error, they have the possibility to edit the entry through the graphical user interface. The user interface is also used to share articles, tips or to send useful notifications to the user.

7.6. Conclusions

This chapter describes the development of a conversational agent to promote healthy nutrition for chronically-ill patients. We conducted a qualitative study with 13 patients and health professionals who helped us identify the main barriers as well as the needs and requirements for adopting a healthy nutrition behavior. Lack of regularity, lack of variety, false belief and hunger unawareness were identified as the main barriers to healthy nutrition; whereas monitoring, education, empowerment and practicality were identified as the main needs and requirements for nutrition behavior change. These findings allowed us to suggest the appropriate behavior change techniques to be used in our systems, which are self-monitoring, personalized visual feedback, goal setting, self-awareness and personalized education. Finally, these were translated into a set of functionalities that build up to construct our final solution: MiranaBot.

7.6. References

[1] Institute for Health Metrics and Evaluation. (2019). Data Visualizations. http://www.healthdata.org/results/data-visualizations Accessed 25 May 2019.

[2] World Health Organization. (2011). Global Status Report on Noncommunicable Diseases. http://www.who.int/nmh /publications/ncd_report2010/en Accessed 25 May 2019.

[3] Institute for Health Metrics and Evaluation. (2019). Switzerland. http://www.healthdata.org/Switzerland Accessed 25 May 2019

[4] Reber, E., Gomes, F., Vasiloglou, M. F., Schuetz, P., & Stanga, Z. (2019). Nutritional Risk Screening and Assessment. *Journal of clinical medicine*, 8(7), 1065. https://doi.org/10.3390/jcm8071065

[5] Burke, R., Felfernig, A., & Göker, M. H. (2011). Recommender Systems: An Overview. AI Magazine, 32(3), 13-18. https://doi.org/10.1609/aimag.v32i3.2361

[6] Shah, K., Salunke, A., Dongare, S., & Antala, K. (2017). Recommender systems: An overview of different approaches to recommendations. *International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS)*. 1-4. https://doi.org/10.1109/ICIIECS.2017.8276172

[7] Mika, S. (2011). Challenges for Nutrition Recommender Systems. *Proceedings of the 2nd Workshop on Context Awareness in Retrieval and Recommendation*. 786.

[8] Tran, T.N., Atas, M., Felfernig, A., & Stettinger, M. (2017). An overview of recommender systems in the healthy food domain. *Journal of Intelligent Information Systems*, 50, 501-526. https://doi.org/10.1007/s10844-017-0469-0

[9] Konstan, J.A & Riedl, J. (2012). Recommender systems: from algorithms to user experience. *User Modeling and User-Adapted Interaction*. 22, Issue 1–2, 101–123. https://doi.org/10.1007/s11257-011-9112-x

[10] Korpusik, M. & Glass, J. (2017). Spoken Language Understanding for a Nutrition Dialogue System . *IEEE Transactions on Audio, Speech, and Language Processing.* 1450 – 1461. https://doi.org/10.1109/TASLP.2017.2694699

[11] Casas, J., Mugellini, E., & Khaled, O.A. (2018). Food Diary Coaching Chatbot. *In Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (UbiComp '18).* 1676-1680. https://doi.org/10.1145/3267305.3274191

[12] Meso, P. & Jain, R. (2006). Agile Software Development: Adaptive Systems Principles and Best Practices. *Information Systems Management.* 23:3, 19-30, https://doi.org/10.1201/1078.10580530/46108.23.3.20060601/93704.3

[13] Duff, O., Walsh, D., Furlong, B., O'Connor, N., Moran, K. & Woods, C. (2017). Behavior Change Techniques in Physical Activity eHealth Interventions for People with Cardiovascular Disease: Systematic Review. *Journal of Medical Internet Research*, 19(8):e281. https://doi.org/10.2196/jmir.7782

[14] Schembre, S., Liao, Y., Robertson, M., Dunton, G., Kerr, J., Haffey, M. et al. (2018). Just-in-Time Feedback in Diet and Physical Activity Interventions: Systematic Review and Practical Design Framework. *Journal of Medical Internet Research*, 20(3):e106. https://doi.org/10.2196/jmir.8701

[15] Sawesi, S., Rashrash, M., Phalakornkule, K., Carpenter, J.S., Jones, J.F. (2016). The Impact of Information Technology on Patient Engagement and Health Behavior Change: A Systematic Review of the Literature. *JMIR Med Inform.* 4(1):e1. https://doi.org/10.2196/medinform.4514

[16] Greaves, C.J., Sheppard, K.E., Abraham, C., Hardemann, W., Rode, M., Evans, P.H., et al. (2011). Systematic review of reviews of intervention components associated with increased effectiveness in dietary and physical activity interventions. *BMC Public Health.* 11:119. https://doi.org/10.1186/1471-2458-11-119

[17] Michie, S., Abraham, C., Whittington, C., McAteer, J. & Gupta, S. (2009). Effective techniques in healthy eating and physical activity interventions: a meta-regression. *Health Psychol*, 28(6):690-701. https://doi.org/10.1037/a0016136

[18] Lara, J., Evans, E. H., O'Brien, N., Moynihan, P. J., Meyer, T. D., Adamson, A. J. et al. (2014). Association of behaviour change techniques with effectiveness of dietary interventions among adults of retirement age: a systematic review and meta-analysis of randomised controlled trials. *BMC medicine*, 12, 177. https://doi.org/10.1186/s12916-014-0177-3

[19] Prochaska, J. & Velicer, W. (1997). The Transtheoretical Model of Health Behavior Change. *American journal of health promotion.* 12. 38-48. https://doi.org/10.4278/0890-1171-12.1.38

[20] Klein M., Mogles N., van Wissen A. (2011). Why Won't You Do What's Good for You? Using Intelligent Support for Behavior Change. In: Salah A.A., Lepri B. (eds) Human Behavior Understanding. HBU 2011. *Lecture Notes in Computer Science*, vol 7065. https://doi.org/10.1007/978-3-642-25446-8_12

[21] Karoly P. (1993). Mechanisms of self-regulation: a systems view. *Annu Rev Psychol*, 44(1):23-52. https://doi.org/10.1146/annurev.psych.44.1.23

[22] Swiss Society of Nutrition. Balanced Diet- Optimum Plate. http://www.sge-ssn.ch/fr/toi-et-moi/boire-et-manger/equilibre-alimentaire/assiette-optimale/ Accessed on 13 september 2019.

[23] Cohen, D. & Farley, T. A. (2008). Eating as an automatic behavior. *Preventing chronic disease*, 5(1), A23.

[24] Yau, Y. H. & Potenza, M. N. (2013). Stress and eating behaviors. *Minerva endocrinologica*, 38(3), 255–267.

CHAPTER 8. GENERAL DISCUSSION

The present research examines self-monitoring technologies to promote healthy behavior in the long term for specific population, namely, chronically-ill patients and elderly. Although self-monitoring technologies shows great promises to promote behavior change, tracking health data is not enough to sustain the motivation and engagement, especially in the patient domain [1]. A large number of behavior change techniques are indeed incorporated in existing self-monitoring technologies. This offers an indication of these systems' ability to alter individuals' behaviors, but the long-term efficacy of these systems to alter exercise, sleep, and sedentary behaviors remains unknown [2]. Besides, these applications and the behavior theory behind are often generalized to the public and does not necessarily answer individuals' preferences and needs related to their specific pathology. Finally, the feasibility and the acceptability of these technologies across different specific population remain understudied.

8.1. Summary of main findings and contributions

The purpose of this thesis was to address this issue by examining how self-monitoring technologies are accepted and adopted by chronically-ill patients and elderly and what are the guidelines to design an effective intervention to promote long-term behavior change. By doing so, our study's findings provide several important contributions to the literature, advancing research for Human Computer Interaction, Digital Health, Conversational AI, Behavioral sciences, Nutrition and Physical Rehabilitation communities. In the following sections, we summarize the contributions of our work and share guidelines on developing self-monitoring technologies for these specific population.

8.1.1. Acceptance and adoption of self-monitoring technologies

RQ1: How self-monitoring technologies are accepted and adopted by chronically-ill patients and engage them in healthy lifestyle?

Qualitative studies with chronically-ill patients (18 participants, 7 months) using a wearable tracker were conducted.

Diabetic and obese users given a fitness tracker for seven months identified the device's playfulness, practicality, persuasiveness, personalization, and privacy as the top motivators for its use. These results provide insights into designing wearable health monitors that are engaging enough to effect long-term change. The healthcare community is already envisioning technology as a driver of behavioral change. A well-designed activity monitor can do much more than record a user's glucose level and weight; through personalization and pervasiveness, it can help users manage their lifestyle to incorporate healthier choices. Users' expectations of the device change with continued use. Thus, in addition to tailoring the device for the user and the user's environment and making it entertaining and easy to use, designers should consider ways that the device can adapt to changing expectations across periods of use.

RQ2: What are the opportunities and challenges for the elderly to be monitored and how would they integrate the system in their daily life?

Qualitative studies with elderly living at home (20 participants, 6 weeks), using a wearable tracker were conducted.

Before using the devices, our intent was to understand older adults' attitudes towards increasing physical activity as well as their readiness towards tracking technologies. The main barriers of being physically active were the absence of motivation, the incapacity due to health condition and the lack of perceived usefulness of physical exercise. Furthermore, although half of the participants were not technology-oriented, personal interest and enthusiasm driven by family towards technology were observed. After 6 weeks of usage, we discovered changes in behavior and usage intention which allowed us to identify opportunities and challenges for the older adults to adopt sensors and application for health and activity management. This study showed the potential of acceptance and adoption of simple and manageable technology for behavior change. As for usage intention, some older adults start to be eager to learn new tools as long as they are able to do it, which is more and more influenced by family from younger generation. However, designers and engineers should consider older adults' fear introducing novelty in their daily life as well as their need for social interaction and their need to remain in control of any system given to them.

8.1.2. Social incentives to motivate chronically-ill patients

RQ3: How does collaboration and competition affect the motivation of chronically-ill patients to exercise more?

A qualitative study with chronically-ill patients (16 participants, 8 weeks) was conducted.

Results show that participants exercised more with social incentives compared with their baseline. Collaborating with buddies to compete in a community was reported as motivating for dyads exercising with strong ties. Social interactions could be demotivating between dyads who did not know each other well. Finally, it is crucial to consider patients' technical literacy when designing behavior-changing technologies.

8.1.3. Gamification for elderly rehabilitation

A non-inferiority 6-week randomized clinical trial with 57 elderly patients with musculoskeletal issues was conducted.

RQ4: How does serious game-based rehabilitation impact elderly patients' motivation and activity at the hospital?

Before using the different devices, our intent was to understand older patients' attitudes towards rehabilitation process, their motivation for physical activity as well as their readiness for innovative technology. At the end of the hospital stay, our study showed that patients who received the intervention demonstrated stronger engagement in the rehabilitation process and exhibited a higher motivation to be active. Different recommendations on designing serious game for elderly rehabilitation are outlined, especially playfulness, personalization, performance feedback and safety. All in all, if applied on a larger population during a longer

period, our findings suggest the potential long-term impact of serious game-based intervention on elderly quality of life and independence while returning home.

RQ5: How efficient is the gamified rehabilitation compared to the standard rehabilitation treatment?

Findings show no significant differences were found between the control group and the intervention group for the primary outcome (SPPB: balance, gait speed, chair stand) or any of the secondary outcomes (IHGS, FIM and steps), which demonstrates the non-inferiority effect of the serious game-based intervention. In addition, patients who received the intervention demonstrated engagement in the rehabilitation process and motivation in being more active. Serious game-based rehabilitation could be a cost-effective alternative for elderly patients to regain their functional capacities.

8.1.4 Conversational agent for chronically-ill and elderly patients to promote healthy eating

RQ6: What are the needs and requirements for self-monitoring technology to promote healthy eating for chronically-ill patients?

A qualitative study with chronically-ill patients (10 patients, 2 nutritionists, 1 physician) were conducted.

Lack of regularity, lack of variety, false belief and hunger unawareness were identified as the main barriers to healthy nutrition; whereas monitoring, education, empowerment and practicality were identified as the main needs and requirements for nutrition behavior change. These findings allowed us to suggest the appropriate behavior change techniques to be used in our systems, which are self-monitoring, personalized visual feedback, goal setting, self-awareness and personalized education. Finally, these were translated into a set of functionalities that build up to construct our final solution: MiranaBot.

8.2. Intervention prescription

Based on our findings and contributions, we would like to propose a set of intervention prescription or “dose of intervention”, taking into account the target population and their specific conditions.

Population	Target	Intervention	Recommendation	Examples of tools / Duration
Healthy adult	Prevention	Education Persuasion Enablement	Playfulness, Practicality, Persuasiveness	Self-monitoring technologies, Conversational agent **Long-term** **Stability (60-100 days)**
Obese and diabetic adult	Treatment	Education Persuasion Incentivisation Environmental	Playfulness, Practicality, Persuasiveness, Personalization,	Conversational agent, Self-monitoring technologies, Social incentives-based technologies

		restructruring	Privacy, Collaboration within strong ties	**Min 6-month**
Elderly	Prevention	Persuasion Incentivisation Training Environmental restructruring	Reluctance introducing novelty, Need for social interaction, Need to remain in control	Self-monitoring technologies, Gamified technologies, **Long-term**
	Rehabilitation	Education Persuasion Incentivisation Training Enablement	Playfulness, Personalization, Performance Feedback, Safety	Self-monitoring technologies, Gamified technologies, Social incentives-based technologies **Min 3-month**

Table 1. Intervention prescription

8.3. Limitations and future work

Self-monitoring has been shown to have robust behavior change effects [3, 4, 5, 6]. Today's technological advancements provide a growing set of opportunities for individuals to self-monitor their behavior. Nevertheless, several limitations are associated with the engagement in using self-monitoring technologies for behavior change. The work presented in this dissertation has touched upon some of the open questions that still needs to be addressed in this domain, mainly the acceptance and adoption of these tools in clinical settings. However, deeper investigations, for a longer period, with a larger sample size are still needed by the research community.

Drawing from the findings elicited through the thesis, we developed a conversational agent for chronically-ill patients and elderly to promote healthy eating. The efficacy of this tool to identify users' nutritional habits and assist the user in the process of behavior change would requires further investigations. As we are proposing a mixed interface (visual and vocal) to the user, it would be of interest to understand the influences of these different interfaces in encouraging heathier behavior choice. A potential future work could be the integration of the neurological aspects in addition the psychological and behavioral aspects into development of technology for behavior change. Finally, adapting our findings to other countries and cultures considering the population contextual environment would be of interest.

Beyond medical domain, techniques used in behavioral economics such as the "nudge theory" are closely related to the topic we touched upon through this thesis. Popularized by Thaler and Sustein in 2008 [7], and based on Kahneman's work (System 1 and System 2) [8], nudging focuses on subtle encouragement and enablement through choice architecture. They define it as "any aspect of the choice architecture that alters people's behavior in a predictable way without forbidding any options or significantly changing their economic incentives." [7]. Although it was first intended to steer customer behavior in economy and marketing strategy, it can be adapted and applied much more widely for motivating change in people, namely in

taking healthier choice for a better quality of life. A potential future research would be combining the different techniques, we recommend in this thesis, with nudge theory and understanding the impact of considering choice architecture and the environmental context.

8.4. References

[1] Miyamoto, S., Henderson, S., Young, H., Pande, A. & Han, J. (2016). Tracking Health Data Is Not Enough: A Qualitative Exploration of the Role of Healthcare Partnerships and mHealth Technology to Promote Physical Activity and to Sustain Behavior Change. JMIR mHealth and uHealth. 4. e5. https://doi.org/10.2196/mhealth.4814

[2] Duncan, M., Murawski, B., Short, C.E., Rebar, A.L., Schoeppe, S., Alley, S. et al. (2017). Activity Trackers Implement Different Behavior Change Techniques for Activity, Sleep, and Sedentary Behaviors. Interact J Med Res. 6(2):e13. https://doi.org/10.2196/ijmr.6685

[3] Brickwood, K.J., Watson, G., O'Brien, J., Williams, A.D. (2019). Consumer-Based Wearable Activity Trackers Increase Physical Activity Participation: Systematic Review and Meta-Analysis. *JMIR Mhealth Uhealth.* 7(4):e11819. https://doi.org/10.2196/11819

[4] Lyons, E.J., Swartz, M.C., Lewis, Z.H., Martinez, E., & Jennings, K. (2017). Feasibility and Acceptability of a Wearable Technology Physical Activity Intervention With Telephone Counseling for Mid-Aged and Older Adults: A Randomized Controlled Pilot Trial. *JMIR Mhealth Uhealth.* 5(3):e28. https://doi.org/10.2196/mhealth.6967

[5] Carter, M.C., Burley, V.J., Cade, J.E. (2017). Weight Loss Associated With Different Patterns of Self-Monitoring Using the Mobile Phone App My Meal Mate. *JMIR Mhealth Uhealth.* 5(2):e8. https://doi.org/10.2196/mhealth.4520

[6] Patel, M.L., Hopkins, C.M., Brooks, T.L., & Bennett, G.G. (2019). Comparing Self-Monitoring Strategies for Weight Loss in a Smartphone App: Randomized Controlled Trial. *JMIR Mhealth Uhealth.* 7(2):e12209. https://doi.org/10.2196/12209

[7] Thaler, R.H., & Sunstein, C.R. (2008). Nudge: Improving Decisions About Health, Wealth, And Happiness. *New Haven : Yale University Press*.

[8] Kahneman, D. (2011). Thinking, Fast And Slow. *New York : Farrar, Straus And Giroux*. https://doi.org/10.1007/s00362-013-0533-y

APPENDIX

MiranaBot: **A conversational agent to promote healthy eating**

MIRANA
Your nutrition assistant

UNIVERSITÉ DE GENÈVE

HUG

Despite the extensive research and the large number of existing nutrition applications, food recommender systems are still facing many challenges in terms of nutrition habits tracking and delivery of the proper recommendations. «MiranaBot» is a conversational agent, with the goal to help the user to be aware of their eating habits in terms of variety and regularity. Rather than focusing on food quantity and nutritional value, the system targets the variety of the individuals' diet.

Nutrition tracking

"Mirana" allows the user to track their nutrition easily by asking them to briefly describe what they ate after each meal in terms of portions. The system will initially train the users on the portion size to help them describe their meal as best as possible.

Variety feedback

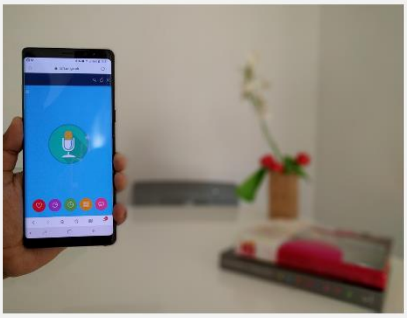
Every day, "Mirana" will provide a visual feedback to the users by showing them on a plate the ratio of proteins, carbs, and vegetables they have eaten thorough the day.

Regularity feedback

"Mirana" will also help the user to be aware of how regular they eat every day. The system will provide a visual feedback showing at what time during the day the user was eating.

Personalized recommendations

One of the major roles of "Mirana" will be to act as a nutritional assistant that can answer users' questions in the process of change. "Mirana" will be able to give advice on alternatives to the food the users have to consume less or more, explain in an easy way the benefit of some healthy nutrition behavior, give tips on certain food items and make reminders to help the users reach their goal.



Speech To Text module

Users can easily describe vocally what they have eaten after each meal. This voice description is then converted in real time into text, using the Google Speech API. As the first version of the application will be tested in the French speaking part of Switzerland, we choose the Google API for its 95% accuracy in recognizing French language.

Text Analysis module

Food data will be analyzed every day / every week in order to give a visual feedback to the user at the end of the day/week. The system will be able to detect frequencies, varieties, and quantities (in terms of portion) of a specific food (or category of food). With the help of nutritionist and dietician, we were able to build a database of food classified by categories. We have set rules for each category which will then be used by our algorithm to identify recurring patterns.

Natural conversation

The conversational module is built upon DialogFlow developed by Google. Their module use machine learning for small talk conversation, in order to recognize user basic questions. In addition, we developed specific intents related to our case (nutrition), to give "Mirana" the ability to answer question about nutrition.

Graphical User Interface

To maximize the output given to the user, we decided to use the screen of the device used (phone, tablet, computer), to give graphical feedback in addition to the vocal answer. The regularity of the nutrition and the variety of the food eaten by the users are displayed on the screen to induce better self-reflection. While describing their food, the user can also see in real time if what they said was transcribed correctly. In case of error, they have the possibility to edit the entry through the graphical interface. This interface is also used to share articles and tips or to push up useful notification to the user.

This project has received funding from European Union's Horizon 2020 research and innovation program under grant agreement No 690425.

REACH 2020

WWW.REACH2020.EU